

INCORPORATING CLIMATOLOGICAL INFORMATION INTO
ENSEMBLE DATA ASSIMILATION

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ABSTRACT

This thesis is concerned with the general question of how to improve data assimilation methods through the incorporation of climatological information. We examine two ways in which climatological information can augment the data assimilation process: through the analysis step by means of a weak constraint in the standard variational cost function, and through the forecast step by using a reduced stochastic forecast model which is valid on climatological time scales. We focus on how problems arising from finite ensemble size effects in ensemble data assimilation may be addressed by including climatological information in the assimilation procedure.

Firstly, we consider the problem of ensemble Kalman filtering in sparse observational networks. Incorporating climatological data is shown to control the phenomenon of covariance overestimation, which is particularly prevalent when filtering with small ensembles and sparse observational networks. We consider the situation where the observational space consists of variables that are directly observable with known observational error, as well as variables for which only their climatic variance and mean are given. To ensure that the variance of the unobserved ensemble members is no worse than the known climatological variance we develop a modification to the standard ensemble Kalman filter method called a variance limiting Kalman filter. We show that incorporating the information of the variance of some unobserved variables can improve the analysis skill and increase the stability of the data assimilation procedure, corroborating our findings with both analytical results and numerical simulations. We then apply this method to a problem where covariance overestimation arises due to model error and determine its range of optimal performance for a chaotic nonlinear system.

Secondly, we consider the problem of data assimilation in multiscale systems by stochastic parametrisation of subgrid-scale processes. To this end we consider a deterministic multiscale toy model in which a chaotic fast subsystem triggers rare transitions between slow metastable regimes, akin to weather or climate regimes. Using stochastic homogenisation techniques we derive a reduced stochastic parametrisation model for the slow dynamics only, and investigate how faithfully this model represents statistical properties such as the characteristic time scales of the slow dynamics. We find that while sensitive to uncertainties in the stochastic parametrisation, reduced stochastic models can actually perform better than a full deterministic model in the forecast step of an ensemble filtering algorithm. The improvement in skill is due to a controlled increase of the ensemble spread in stochastic forecast models, with an effect similar to covariance inflation or increased ensemble size. We corroborate this with numerical simulations.

PREFACE

This work was conducted between February 2008 and September 2011 at the University of Sydney under the supervision of Associate Professor Georg Gottwald. Some of the results of Chapters 3 and 5 have been published in

- Gottwald, G. A., L. Mitchell and S. Reich, 2011: Controlling overestimation of error covariance in ensemble Kalman filters with sparse observations: A variance limiting Kalman filter. *Monthly Weather Review*, **139** (8), 2650-2667, and
- Mitchell, L. and Gottwald, G. A., 2011: Data assimilation in slow-fast systems using homogenized climate models. *Journal of the Atmospheric Sciences* (to appear), doi : 10.1175/JAS-D-11-0145.1.

Results from Chapters 4 and 5 have been submitted for publication in the following articles:

- Mitchell, L. and G. A. Gottwald, 2011: Controlling model error of underdamped forecast models in sparse observational networks using a variance limiting Kalman filter. Submitted to *Quarterly Journal of the Royal Meteorological Society*.
- Mitchell, L. and G. A. Gottwald, 2011: On finite size Lyapunov exponents in multiscale systems. Submitted to *Chaos*.

The contents of this thesis is original work of which I am the sole author, unless explicitly stated otherwise. No part of this thesis has been submitted for the award of any other degree or diploma at this or any other university.

Sydney, August 2011

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INTRODUCTION

The aim of data assimilation in numerical weather prediction (NWP) is to produce an optimal estimate of the state of the atmosphere using two very different sources of information: a numerical model of the dynamics, and observations of the system (Kalnay, 2002). The data assimilation problem is of fundamental importance in NWP, as good estimates of the initial conditions of the atmosphere are essential for producing accurate forecasts. Indeed, improvements in data assimilation methods are a significant factor in increasing the accuracy of operational weather forecasts in recent years (Rabier, 2005). A major task for any data assimilation algorithm is to provide not only an optimal estimate of the atmospheric state – the *analysis* – but also an estimate of the associated analysis error, generally in the form of an *error covariance matrix*.

The models used in atmospheric applications are typically high-dimensional, non-linear, chaotic dynamical systems which aim to characterise the physical processes occurring at many different spatial and temporal scales. Observations are taken from the ground, air, or space-based sources at regular intervals and must be continually combined with model forecasts to produce the analysis. This is extremely difficult in practice, as the dimension of the model state to be estimated is typically orders of magnitude larger than the number of observations available, leading to a highly underdetermined problem. Furthermore, there are errors associated with every step of the procedure. Observations are contaminated with instrument and representativeness errors (see for example Liu and Rabier (2002)). The forecasts also contain errors due to a combination of model errors and errors arising from the inherent unpredictability of the dynamics given uncertain initial conditions (see for example Lorenz (1963, 1969), among many others). Data assimilation attempts to combine these sources of information

and their errors in a meaningful way to produce an analysis and its associated error covariance matrix.

Data assimilation has a long history, dating back to the realisation that objective state estimates were required to initialise the first NWP models (Charney, 1951). After early empirical methods for interpolating observations onto numerical grids (Gilchrist and Cressman, 1954; Bergthorsson and Döös, 1955; Cressman, 1959), *nudging* (Hoke and Anthes, 1976) was the first technique to use the forecast model in the analysis procedure. In this method forecasts are driven towards the observations by means of a forcing term added to the equations of motion. *Optimal interpolation* (OI) was the most widely used data assimilation method used in operational weather forecasting until the advent of more advanced methods in the mid-1980s (Lorenc, 1986). The closely related *3-dimensional variational assimilation* (3D-VAR) method is still used at the National Centre for Atmospheric Research (NCAR) in the USA (Barker et al., 2004). In OI a minimum variance error estimate is found, while in 3D-VAR a maximum likelihood estimate is found which maximises the joint probability density function for the state given both the forecast and the observations. The cost function that is minimised in these methods forms the basis for more advanced Kalman filtering techniques, which we describe in greater detail in Chapter 2. Today the most widely-used data assimilation method in operational weather forecasting is *4-dimensional variational assimilation* (4D-VAR) (Lewis and Derber, 1985; Le Dimet and Talagrand, 1986; Talagrand and Courtier, 1987) which is employed in the European Centre for Medium-Range Weather Forecasts (ECMWF) (Rabier et al., 2000) and the UK Met Office (Rawlins et al., 2007), among others. In 4D-VAR the analysis is the state which produces the best-fitting forecast to a set of observations distributed over a time interval. Operationally it involves a forward integration of the *tangent linear model* (TLM), followed by a backward integration of the adjoint or transpose TLM. For an overview of early NWP methods we direct the interested reader to the book by Lynch (2006), while a detailed exposition of these and other data assimilation methods can be found in the books by Daley (1993) and Kalnay (2002).

A limitation of the above methods is that they do not explicitly provide an estimate of the time-evolved error covariance matrices. In OI and 3D-VAR the error covariances are kept fixed; in 4D-VAR a quasi-climatological error covariance matrix is evolved using the TLM and adjoint, but is not explicitly available after performing an analysis. One

emerging approach to the data assimilation problem which gives an explicit estimate of the error covariances as evolved by the full nonlinear model after every analysis step is the *ensemble Kalman filter* (EnKF) (Evensen, 1994, 2006; Houtekamer and Mitchell, 1998), which provides a Monte Carlo estimate of the forecast error covariances using an ensemble of forecasts propagated by the full nonlinear model. Algorithmically, the EnKF consists of two stages: the first is the *forecast step*, where a *background* ensemble of state estimates is propagated forward in time to produce a forecast mean and error covariance estimate. The second step is the *analysis step*, where the forecast is combined with observations to produce an analysis mean and error covariance, which can be used to produce a new background ensemble from which the process can be restarted. The ensemble method is attractive because of its ease of implementation and relative computational affordability. Unlike techniques like 4D-VAR, the EnKF estimates states using the full nonlinear forecast model only; there is no need for backward integrations of the adjoint model or use of a tangent linear model. Also, with an EnKF the information contained in a covariance matrix can be stored in a matrix of ensemble perturbations, orders of magnitude smaller than the full-rank covariance matrix. This greatly reduces the computational burden in producing an analysis. Ensemble filters are presently used at the Canadian Meteorological Centre to produce global forecasts (Houtekamer and Mitchell, 2005).

Despite this computational efficiency, there are numerous external and internal sources of error involved when doing data assimilation with the EnKF. External error sources which are inherent in many forms of data assimilation include model error, observational error and deviations from Gaussianity. The primary source of internal errors specific to ensemble-based filters are sampling issues, which arise when using small ensembles to estimate the error covariances. Consequences of these finite size effects include covariance underestimation (Burgers et al., 1998; Anderson and Anderson, 1999), filter divergence (Houtekamer and Mitchell, 1998; Whitaker and Hamill, 2002) where the filter trusts its own forecast and ignores the information given by the observations, and spurious correlations between distant observation sites (Houtekamer and Mitchell, 1998). Common approaches to these problems include covariance inflation (Anderson and Anderson, 1999), localisation (Houtekamer and Mitchell, 2001; Hamill et al., 2001; Ott et al., 2004; Szunyogh et al., 2005) as well as multi-ensemble configurations (Houtekamer and Mitchell, 1998; Mitchell and Houtekamer, 2009). We will discuss these data assimilation challenges in further detail in Chapter 2.

Besides the model and the observations, there is a third source of useful information about the system which can be exploited, namely climatological or long-time-averaged statistical data. This information is so far relatively untapped in data assimilation, but may be valuable if it can be properly incorporated into algorithms. Climatological information exists in the form of historical observations or through climate reanalysis of the system. For example, the *Twentieth Century Reanalysis Project* (20CR) (Compo et al., 2011) recently produced a global atmospheric estimate for the entire 20th century using historical observational data. Such a dataset presents a valuable source of information which could be useful not only as a validation tool for data assimilation algorithms, but also as a secondary source of data to augment that of the existing model and observations.

This thesis focuses on ways in which climatological information might be used in data assimilation. Whereas most data assimilation algorithms place emphasis on the model and current observations, in this thesis we ask if it can also be beneficial to include climatological information in the data assimilation procedure. The question we address is how problems in ensemble filtering can be reduced by adding in information from climatology. We will investigate two particular ways in which climatological data can be incorporated in the data assimilation process, namely through the analysis step and through the forecast step of the EnKF algorithm.

Firstly, we will show that incorporating climatic data in the analysis step as *pseudo-observables* can counteract the overestimation of error covariance matrices, which can potentially spoil the analysis. Covariance overestimation receives far less attention in the literature than the more common problem of underestimation, however it has been shown to adversely affect the analysis in spatially sparse observational networks (Liu et al., 2008; Whitaker et al., 2009) where there are large data voids for the data assimilation algorithm to fill in. We will describe a new technique called a *variance limiting Kalman filter* (VLKF) to approach the problem of covariance overestimation in sparse observational networks in Chapter 3. Covariance overestimation can also occur as a result of model error, particularly when underdamped forecast models are used. Numerical models typically overestimate noise at the grid resolution, for which a common remedy is to introduce artificial model damping (Durran, 1999). This comes at the expense of destroying conservation laws in the system, which are particularly important in long-time simulations such as for climate models (Thuburn, 2008). In

Chapter 4 we propose an alternative to using artificial damping in forecast models for data assimilation. We will show that the VLKF method developed in Chapter 3 can be used to control the covariance overestimation which results when a forecast model with less damping is used.

Secondly, we will study ways in which climatological information can be incorporated in the forecast step, at the level of the forecast model itself. We study reduced stochastic climate models, which describe the long-time dynamics of the slow variables in a multiscale system in a statistical manner. In particular we use homogenisation to derive a reduced stochastic forecast model for a time-scale-separated system, which captures the effective slow dynamics of the full system. We ask if such a model can be beneficial in data assimilation when used for forecasting on long time scales. In Chapter 5 we employ a reduced stochastic model in the forecast step for ensemble filtering of a multiscale system. We will show that this strategy can be beneficial in reducing the impact of finite size effects associated with ensemble methods. Whereas the approach outlined in Chapters 3 and 4 is more appropriate for controlling covariance overestimation, we will see that reduced stochastic forecast models can be beneficial in addressing underestimation, the more familiar situation encountered in ensemble filtering. We find that reduced stochastic models can be used as a dynamically controlled alternative to covariance inflation, allowing for data assimilation using smaller ensembles.

We will address each of the above topics more specifically through detailed introductions at the beginning of each individual chapter. The outline of the remainder of this thesis is as follows: we will formulate the general data assimilation problem in a Bayesian framework and describe the EnKF in Chapter 2. In Chapter 3 we focus on how to incorporate climatological data into the analysis step of the EnKF and develop the variance limiting Kalman filter method, which deals with the problem of covariance overestimation in sparse observational networks. In Chapter 4 we apply the VLKF to a problem with model error, in particular in the situation where an underdamped model is used for forecasting. In Chapter 5 we turn to the problem of incorporating climatological information into ensemble filtering through the forecast step, by considering data assimilation in multiscale systems using reduced stochastic forecast models. We conclude with a discussion and outlook for further research in Chapter 6.

DATA ASSIMILATION AND ENSEMBLE KALMAN FILTERING

In this chapter we give a brief overview of data assimilation techniques and describe the *ensemble Kalman filter* (EnKF) method. We will focus on *filtering* methods which recursively combine an observation with a forecast, as opposed to *smoothing* methods such as 4D-VAR in which the analysis is the forecast trajectory which best fits a set of observations distributed over a time window. We will begin by formulating the general data assimilation problem in a Bayesian framework and deriving the analysis equations for the methods we will consider. We then describe a range of different methods for implementing the forecast step, one of which will be the EnKF algorithm. Finally, we will discuss some of the difficulties encountered in general data assimilation and for the EnKF specifically, approaches to which motivate the work in this thesis.

2.1 THE DATA ASSIMILATION PROBLEM

We assume an N -dimensional dynamical system

$$\dot{\mathbf{z}}_t = F(\mathbf{z}_t) \tag{2.1}$$

which describes the truth $\mathbf{z}_t$. The aim of data assimilation is to produce an optimal estimate of $\mathbf{z}_t$ at a given time given a typically chaotic, possibly incorrect model

$$\dot{\mathbf{z}} = f(\mathbf{z}) \tag{2.2}$$

and noisy observations $\mathbf{y}_o$ of the true state $\mathbf{z}_t$ taken at discrete times $t_i = i\Delta t_{\text{obs}}$. Observations $\mathbf{y}_o \in \mathbb{R}^n$ are expressed as a perturbed truth according to

$$\mathbf{y}_o(t_i) = \mathbf{H}\mathbf{z}_t(t_i) + \mathbf{r}_o(t_i), \quad (2.3)$$

where the observation operator $\mathbf{H} : \mathbb{R}^N \rightarrow \mathbb{R}^n$ maps from the whole space into observation space, and $\mathbf{r}_o \in \mathbb{R}^n$ is assumed to be independent and identically distributed observational Gaussian noise with associated error covariance matrix $\mathbf{R}_o$. At this stage we note that although we will use a linear observation operator $\mathbf{H}$ throughout this thesis, all of the arguments we will make can be extended to nonlinear observation operators $H(\mathbf{z})$ through linearisation of the observation operator $H(\mathbf{z} + \delta\mathbf{z}) = H(\mathbf{z}) + \mathbf{H}\delta\mathbf{z} + \mathcal{O}(\|\delta\mathbf{z}\|^2)$ (see Kalnay (2002) for further details). We apply the notation of Ide et al. (1997) unless explicitly stated otherwise.

Consider the forecast state vector $\mathbf{z}_f \in \mathbb{R}^N$ produced by an integration of the model (2.2) from the *background* state $\mathbf{z}_b \in \mathbb{R}^N$ with associated background error covariance $\mathbf{P}_b \in \mathbb{R}^{N \times N}$. The error covariance matrix associated with the forecast is $\mathbf{P}_f \in \mathbb{R}^{N \times N}$. We assume that the atmospheric state $\mathbf{z}$ conditioned on the forecast $\mathbf{z}_f$ is a realisation of a random process defined by a Gaussian probability density function (pdf), and so we can write the *prior* probability density function for the state $\mathbf{z}$ given the forecast $\mathbf{z}_f$ as

$$\rho(\mathbf{z}) = \frac{1}{\sqrt{\det(2\pi\mathbf{P}_f)}} \exp \left[-\frac{1}{2}(\mathbf{z}_f - \mathbf{z})^T \mathbf{P}_f^{-1}(\mathbf{z}_f - \mathbf{z}) \right]. \quad (2.4)$$

Similarly, we can define the likelihood function for $\mathbf{z}$ given the observations $\mathbf{y}_o$, or the conditional probability for $\mathbf{y}_o$ given the state $\mathbf{z}$ as

$$\rho(\mathbf{y}_o|\mathbf{z}) = \frac{1}{\sqrt{\det(2\pi\mathbf{R}_o)}} \exp \left[-\frac{1}{2}(\mathbf{y}_o - \mathbf{H}\mathbf{z})^T \mathbf{R}_o^{-1}(\mathbf{y}_o - \mathbf{H}\mathbf{z}) \right], \quad (2.5)$$

where we have again assumed Gaussianity. Bayes' theorem then states that the *posterior* pdf for the state $\mathbf{z}$ given the observations is

$$\rho(\mathbf{z}|\mathbf{y}_o) = \frac{\rho(\mathbf{z})\rho(\mathbf{y}_o|\mathbf{z})}{\rho(\mathbf{y}_o)}. \quad (2.6)$$

The pdf for the observations $\rho(\mathbf{y}_o)$ can be obtained by integrating over all possible states $\mathbf{z}$ so that $\rho(\mathbf{y}_o) = \int \rho(\mathbf{y}_o|\mathbf{z})\rho(\mathbf{z})d\mathbf{z}$. This distribution is the “climatological” distribution of the observations and does not depend on the state $\mathbf{z}$. Substituting (2.4) and (2.5) into Bayes’ theorem (2.6) we thus obtain

$$\rho(\mathbf{z}|\mathbf{y}_o) \propto \exp \left[-\frac{1}{2} \left((\mathbf{z}_f - \mathbf{z})^T \mathbf{P}_f^{-1} (\mathbf{z}_f - \mathbf{z}) + (\mathbf{y}_o - \mathbf{H}\mathbf{z})^T \mathbf{R}_o^{-1} (\mathbf{y}_o - \mathbf{H}\mathbf{z}) \right) \right].$$

The maximum likelihood state estimate is the value of $\mathbf{z}$ which maximises this pdf, which is equivalent to the minimum of the cost function

$$J(\mathbf{z}) = \frac{1}{2} (\mathbf{z}_f - \mathbf{z})^T \mathbf{P}_f^{-1} (\mathbf{z}_f - \mathbf{z}) + \frac{1}{2} (\mathbf{y}_o - \mathbf{H}\mathbf{z})^T \mathbf{R}_o^{-1} (\mathbf{y}_o - \mathbf{H}\mathbf{z}). \quad (2.7)$$

The *analysis* $\mathbf{z}_a$ is given as the minimum of this cost function, and can be found either explicitly or by employing a numerical minimisation algorithm on (2.7). The critical point $\mathbf{z} = \mathbf{z}_a$ can be calculated by solving $\nabla_{\mathbf{z}} J(\mathbf{z}) = 0$. We obtain

$$\left(\mathbf{P}_f^{-1} + \mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{H} \right) \mathbf{z}_a = \mathbf{P}_f^{-1} \mathbf{z}_f + \mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{y}_o, \quad (2.8)$$

which yields the analysis

$$\mathbf{z}_a = \mathbf{z}_f + \mathbf{K}_o [\mathbf{y}_o - \mathbf{H}\mathbf{z}_f], \quad (2.9)$$

where

$$\mathbf{K}_o = \mathbf{P}_f \mathbf{H}^T \left(\mathbf{H} \mathbf{P}_f \mathbf{H}^T + \mathbf{R}_o \right)^{-1} \quad (2.10)$$

is the *Kalman gain matrix* or optimal weight matrix for the observations. The analysis error covariance matrix can be calculated from $\mathbf{P}_a = \mathbb{E} [(\mathbf{z}_a - \mathbf{z}_t)(\mathbf{z}_a - \mathbf{z}_t)^T]$ where the expectation $\mathbb{E}$ is taken over many realisations. Assuming that the background and observational errors are uncorrelated and using (2.9), we obtain

$$\mathbf{P}_a = (\mathbf{I} - \mathbf{K}_o \mathbf{H}) \mathbf{P}_f. \quad (2.11)$$

Using the Shermann-Morrison-Woodbury formula (Golub and Loan, 1996) we may rewrite the Kalman gain matrix and the analysis covariance matrix as

$$\mathbf{K}_o = \mathbf{P}_a \mathbf{H}^T \mathbf{R}_o^{-1}$$

where

$$\mathbf{P}_a = \left(\mathbf{P}_f^{-1} + \mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{H} \right)^{-1}, \quad (2.12)$$

a reciprocal rule for the addition of the covariances. We remark that this form of $\mathbf{P}_a$ is computationally intractable for large models because it requires the inversion of the $N \times N$ matrix $\mathbf{P}_f$, however it will considerably simplify our calculations in Chapter 3.

After performing the analysis step the data assimilation process can be started again by setting the new background $\mathbf{z}_b$ equal to the analysis $\mathbf{z}_a$ and the background error covariance $\mathbf{P}_b$ equal to the analysis error covariance $\mathbf{P}_a$.

2.2 OVERVIEW OF FILTERING METHODS IN DATA ASSIMILATION

There are numerous data assimilation techniques which approach the Bayesian state estimation problem above by finding the minimum of the cost function (2.7). We will describe *optimal interpolation* (OI), *3-dimensional variational assimilation* (3D-VAR) and the linear, extended, and ensemble Kalman filters. These methods have in common that they use the same analysis step obtained by minimising the cost function (2.7). However, the methods are distinguished by the different ways in which the error covariances are estimated. We will give an overview of the different ways of calculating the forecast error covariances employed by these data assimilation techniques.

2.2.1 *Optimal interpolation and 3D-VAR*

Historically, until the emergence of variational methods in the mid-1980s, OI was the most widely used data assimilation method used in operational weather forecasting (Lorenc, 1986). 3D-VAR is still used at the National Centre for Atmospheric Research (NCAR) in the USA (Barker et al., 2004). The main difference between the two methods is that in OI the analysis $\mathbf{z}_a$ is calculated directly from (2.9) and (2.10) while in 3D-VAR the cost function (2.7) is minimised using a numerical minimisation algorithm.

This affords 3D-VAR a great computational advantage: for OI to be computationally tractable analyses must be calculated locally using small observation volumes (see for example [Lorenc \(1981\)](#)) whereas in 3D-VAR the analysis can be done globally. A further difference between the two methods is that in the presence of nonlinear observation operators 3D-VAR contains an “outer loop” to ensure that the analysis obtains the maximal likelihood state. It can be shown that formally the solutions obtained by either method are equivalent ([Lorenc, 1986](#)), however they differ in practice because of the different approximations made in the solution methods used. Operationally 3D-Var is found to be cheaper because it does not require the inversion of large matrices such as those required in calculating the Kalman gain matrix ([2.10](#)).

OI and 3D-VAR are arguably the simplest data assimilation methods because they consist of only the analysis step; neither method directly provides a means for estimating the forecast error covariance $\mathbf{P}_f$. In practice, the forecast error covariances are either kept constant in time, that is $\mathbf{P}_f = \mathbf{P}_b$, or estimated using a variety of methods based on horizontal and vertical correlations between errors ([Kalnay, 2002](#)). As such OI and 3D-VAR have largely been replaced in operations by more advanced methods such as 4D-VAR and the EnKF which propagate the error covariances (implicitly in the case of 4D-VAR) in time using the forecast model. However, we note that there has been a resurgence of OI in recent years in the field of oceanographic data assimilation, where the relatively slow-moving dynamics ensures that error covariances do not vary greatly between analysis cycles ([Oke et al., 2008](#)).

2.2.2 The linear Kalman filter

Proposed by [Kalman \(1960\)](#) for linear dynamics, the Kalman filter differs from OI or 3D-VAR in that error covariances are propagated forward in time by the forecast model. We assume that the forecast model ([2.2](#)) is a linear stochastic differential equation, and that the forecast $\mathbf{z}_f(t_i)$ at observation time t_i can be written as

$$\mathbf{z}_f(t_i) = \mathbf{M}(t_{i-1}, t_i)\mathbf{z}_b + \mathbf{q}(t_i), \quad (2.13)$$

where $\mathbf{z}_b$ is the background state estimate coming from the previous analysis step at time t_{i-1} , $\mathbf{M}(t_{i-1}, t_i)$ is the model transition matrix from time t_{i-1} to t_i and we assume $\mathbf{q}(t_i) \in \mathbb{R}^N$ to be a Gaussian random variable with mean zero and covariance matrix $\mathbf{Q}$ representing model error. Importantly, the background error covariance $\mathbf{P}_b$ coming from the analysis step at time t_{i-1} can now be propagated through to time t_i by means of the transition matrix $\mathbf{M}$, giving the forecast error covariance

$$\mathbf{P}_f(t_i) = \mathbf{M}(t_i, t_{i-1})\mathbf{P}_b(t_{i-1})\mathbf{M}(t_i, t_{i-1})^T + \mathbf{Q}. \quad (2.14)$$

An obvious limitation of the Kalman filter is that it assumes linear dynamics and as such cannot be used for filtering the complex nonlinear systems of interest in NWP applications. As we will see however, a simple modification makes the general idea of the linear Kalman filter applicable to nonlinear systems.

2.2.3 The extended Kalman filter (EKF)

An extension of the Kalman filter for nonlinear models is called the *extended Kalman filter* (EKF) (Sorensen, 1966; Jazwinski, 1970), which uses a linearisation of the full nonlinear model to propagate error covariance matrices forward in time. The EKF generalises the linear Kalman filter by replacing the forecast step (2.13) with a nonlinear forecast

$$\mathbf{z}_f(t_i) = \mathcal{M}_{i-1,i}(\mathbf{z}_b) + \mathbf{q}(t_i),$$

where now $\mathcal{M}_{i-1,i}$ is a nonlinear model which propagates the background $\mathbf{z}_b$ at time t_{i-1} to time t_i . Linearising this model about the trajectory between t_{i-1} and t_i , we obtain

$$\delta\mathbf{z}(t_i) = \mathbf{L}(t_{i-1}, t_i)\delta\mathbf{z}(t_{i-1}) + \mathcal{O}(\|\delta\mathbf{z}\|^2), \quad (2.15)$$

where the Jacobian matrix $\mathbf{L}(t_{i-1}, t_i) = D\mathcal{M}_{i-1,i}(\mathbf{z}_b)$ is the *tangent linear model*, which propagates a small perturbation $\delta\mathbf{z}$ to the state at time t_{i-1} to time t_i . The background error covariance matrix is propagated forward as in (2.14) to give an approximation to the forecast error covariance matrix

$$\mathbf{P}_f(t_i) = \mathbf{L}(t_i, t_{i-1})\mathbf{P}_b(t_{i-1})\mathbf{L}(t_i, t_{i-1})^T + \mathbf{Q}. \quad (2.16)$$

It can be shown that if the model $\mathcal{M}$ is perfect (ie if $\mathbf{Q} = \mathbf{0}$) and the background error covariances are the same then analyses produced by the EKF and 4D-VAR (as run without an outer loop) are identical (Lorenz, 1986). However, unlike 4D-VAR the EKF explicitly provides the forecast error covariance after propagating $\mathbf{P}_b$ forward over the forecast window.

Despite this, it is extremely impractical to use the EKF in operational data assimilation, due to the computational difficulty of storing the $N \times N$ tangent linear model $\mathbf{P}_f$ for large models. Propagating the error covariance as in (2.16) also requires evolving the tangent linear dynamics an additional $\mathcal{O}(N)$ times (Kalnay, 2002; Evensen, 2006). Furthermore, the linearisation performed in (2.15) is valid only for small perturbations $\delta\mathbf{z}$ or for small forecast intervals $\Delta t_{\text{obs}} = t_i - t_{i-1}$, which is not necessarily the case for realistic atmospheric systems. For large, highly nonlinear models the EKF is therefore too inaccurate and too computationally expensive to be useful. Our aim is to estimate the error covariances using the full nonlinear model, without requiring operations like propagating the tangent linear or adjoint models. One method for doing this is the ensemble Kalman filter, which will be the main focus of this work from this point onwards.

2.2.4 The ensemble Kalman filter (EnKF)

One of the emerging techniques for doing data assimilation using an ensemble of full nonlinear model forecasts of the atmospheric state is the *ensemble Kalman filter* (EnKF) (Evensen, 1994, 2006; Houtekamer and Mitchell, 1998). The defining feature of this method is that unlike previous methods it uses a Monte Carlo estimation of the forecast error covariance matrix $\mathbf{P}_f$. Furthermore, the information contained in $\mathbf{P}_f$ is encoded in the ensemble perturbations used in the estimation, which is far easier to store than the full error covariance matrix. In the EnKF the forecast error covariances are estimated from the sample covariance of the propagated ensemble, making it simpler to implement and less computationally expensive than previous methods. A schematic diagram of the EnKF algorithm is shown in Figure 2.2.4. An ensemble of guesses to the true state at time t_{i-1} with covariance $\mathbf{P}_a(t_{i-1})$ (dashed ellipse) is propagated forward to time t_i , when a new observation (cross) becomes available. The forecast mean $\mathbf{z}_f(t_i)$

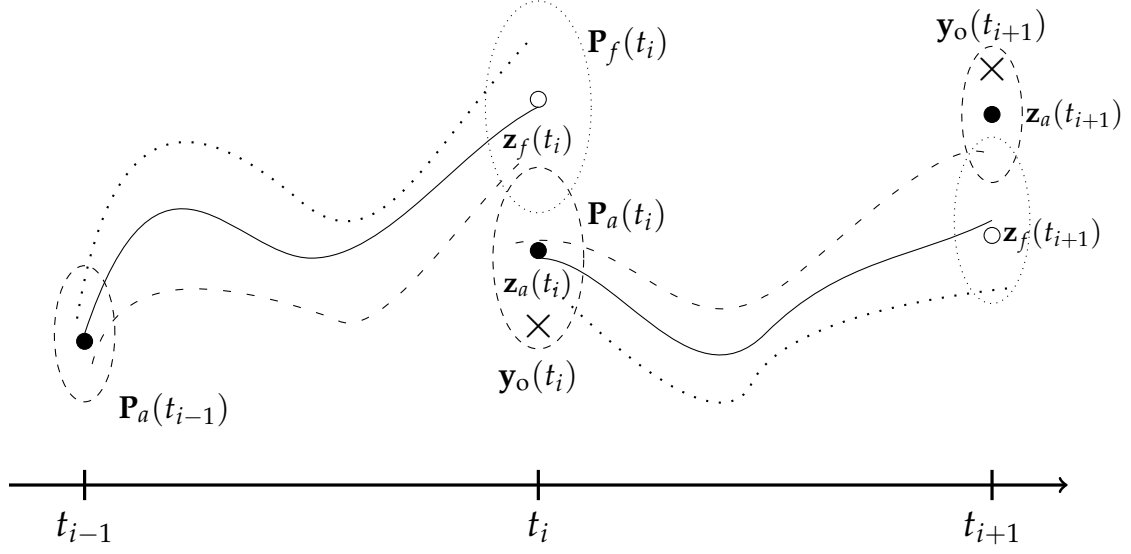


Figure 2.1.: Schematic diagram showing the forecast-analysis procedure for an EnKF.

(open circle) and covariance $\mathbf{P}_f(t_i)$ (dotted ellipse) are then estimated by calculating the sample mean and sample covariance of the forecast ensemble. This method for estimating $\mathbf{z}_f$ and $\mathbf{P}_f$ is the main feature of the EnKF algorithm. In the analysis step the forecast mean and covariance are used to create a new ensemble with mean $\mathbf{z}_a(t_i)$ (filled circle) and variance $\mathbf{P}_a(t_i)$ (dashed ellipse) by (2.9)-(2.11). The procedure is started again, propagating the ensemble to time t_{i+1} , and performing another analysis.

We now formalise this procedure. An ensemble with k members $\mathbf{z}_k$

$$\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_k] \in \mathbb{R}^{N \times k}$$

is propagated by the full nonlinear dynamics (2.2) according to

$$\dot{\mathbf{Z}} = \mathbf{f}(\mathbf{Z}), \quad \mathbf{f}(\mathbf{Z}) = [\mathbf{f}(\mathbf{z}_1), \mathbf{f}(\mathbf{z}_2), \dots, \mathbf{f}(\mathbf{z}_k)] \in \mathbb{R}^{N \times k} \quad (2.17)$$

to produce a forecast ensemble $\mathbf{Z}_f$. This ensemble is split into its mean

$$\mathbf{z}_f = \frac{1}{k} \sum_{i=1}^k \mathbf{z}_{f,i} = \mathbf{Z}_f \mathbf{w} \quad \text{with} \quad \mathbf{w} = \frac{1}{k} \mathbf{e} \in \mathbb{R}^k,$$

where $\mathbf{e} = [1, \dots, 1]^T \in \mathbb{R}^k$, and its ensemble deviation matrix

$$\mathbf{Z}'_f = \mathbf{Z}_f - \mathbf{z}_f \mathbf{e}^T = \mathbf{Z}_f \mathbf{T} ,$$

with the constant projection matrix

$$\mathbf{T} = \mathbf{I} - \mathbf{w} \mathbf{e}^T \in \mathbb{R}^{k \times k} .$$

The defining characteristic of the EnKF is that the sample covariance of the ensemble deviation matrix $\mathbf{Z}'_f$ is used to approximate the ensemble forecast covariance matrix

$$\mathbf{P}_f(t) = \frac{1}{k-1} \mathbf{Z}'_f(t) \left[\mathbf{Z}'_f(t) \right]^T \in \mathbb{R}^{N \times N} . \quad (2.18)$$

This shows the computational advantage of using the EnKF; instead of storing the $N \times N$ covariance matrix $\mathbf{P}_f$ one can instead store the $N \times k$ ensemble perturbation matrix $\mathbf{Z}'_f$, resulting in a huge computational saving for $k \ll N$. Note however that $\mathbf{P}_f(t)$ is rank-deficient for $k < N$; this is the typical situation in numerical weather prediction, where N is of the order of 10^9 and k is of the order of 100. This is what often leads to the problem of covariance underestimation, as the small number of ensemble members insufficiently capture the large number of degrees of freedom of the model. We will discuss this problem in greater detail in the next section.

To determine an ensemble $\mathbf{Z}_a$ which is consistent with the analysis error covariance matrix $\mathbf{P}_a$ as defined in (2.11) and (2.12), and satisfies

$$\mathbf{P}_a = \frac{1}{k-1} \mathbf{Z}'_a \left[\mathbf{Z}'_a \right]^T ,$$

where the prime denotes the deviations from the analysis mean, one can use the method of randomly perturbed observations (Evensen, 2006; Houtekamer and Mitchell, 1998; Burgers et al., 1998) or a deterministic method such as an ensemble square root filter (Simon, 2006). We will use the particular type of square root filter proposed in (Bishop et al., 2001; Tippett et al., 2003; Wang et al., 2004) – the *ensemble transform Kalman filter* (ETKF) – which seeks a transformation $\mathbf{S} \in \mathbb{R}^{k \times k}$ such that the analysis deviation ensemble $\mathbf{Z}'_a$ is given as a deterministic perturbation of the forecast ensemble $\mathbf{Z}_f$ via

$$\mathbf{Z}'_a = \mathbf{Z}'_f \mathbf{S} . \quad (2.19)$$

This method has the advantage of transforming the problem into ensemble space, greatly reducing the number of operations required when $k \ll N$ as is typically the case in operational data assimilation. Alternatively one could use the *ensemble adjustment Kalman filter* (Anderson, 2001) in which the ensemble deviation matrix $\mathbf{Z}'_f$ is pre-multiplied with an appropriately determined matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$. This is of course far more computationally expensive for large models due to the size of $\mathbf{A}$. Note that the matrix $\mathbf{S}$ (or $\mathbf{A}$) is not uniquely determined for $k < N$. The transformation matrix $\mathbf{S}$ can be obtained by (Wang et al., 2004)

$$\mathbf{S} = \bar{\mathbf{C}} (\mathbf{I}_k + \bar{\mathbf{\Gamma}})^{-\frac{1}{2}} \bar{\mathbf{C}}^T.$$

Here $\mathbf{C}\mathbf{\Gamma}\mathbf{C}^T$ is the singular value decomposition of

$$\mathbf{U} = \frac{1}{k-1} \mathbf{Z}'_f{}^T \mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{H} \mathbf{Z}'_f. \quad (2.20)$$

The matrix $\bar{\mathbf{C}} \in \mathbb{R}^{k \times (k-1)}$ is obtained by erasing the last zero column from $\mathbf{C} \in \mathbb{R}^{k \times k}$, and $\bar{\mathbf{\Gamma}} \in \mathbb{R}^{(k-1) \times (k-1)}$ is the upper left $(k-1) \times (k-1)$ block of the diagonal matrix $\mathbf{\Gamma} \in \mathbb{R}^{k \times k}$. The deletion of the 0 eigenvalue and the associated columns in $\mathbf{C}$ assure that $\mathbf{Z}'_a = \mathbf{Z}'_a \mathbf{S}$ and therefore that the analysis mean is given by $\mathbf{z}_a$. This method assures that the mean is preserved under the transformation (Wang et al., 2004) which is not necessarily true for general square root filters. We note that Bergemann et al. (2009) used the continuous Kalman-Bucy filter to calculate $\mathbf{Z}'_a$ without using any computations of matrix inverses, which may be advantageous in high-dimensional systems.

2.3 CHALLENGES IN ENSEMBLE KALMAN FILTERING

Despite its ease of implementation and flow-dependent estimation of error covariances there are several challenges in doing data assimilation with the EnKF. Some of these difficulties, such as model error and non-Gaussianity, are inherent in all forms of data assimilation based on the variational cost function approach described in Section 2.1. Others such as finite size effects and covariance over- and underestimation are unique to the EnKF (see Ehrendorfer (2007) for a recent review). We describe some of these challenges below.

Firstly, the assumption of Gaussian error statistics made in Section 2.1 is a major one, and one which may not be well justified for complex systems. In fact, it is known that nonlinear models often produce non-Gaussian statistics (Andersson et al., 2005). In particular, the analysis estimates produced by the various types of Kalman filter described are optimal only for linear systems. We remark that although the Kalman-Bucy theorem (Kalman and Bucy, 1961) assures that the Kalman filter is optimal for linear models, the constants involved may be so large that it is suboptimal for practical purposes (Harlim, 2011). Both the EKF and the EnKF have been shown to fail badly in the presence of a bimodal distribution (Miller et al., 1994, 1999). Although we will not consider non-Gaussian data assimilation in this work, we note that it is a major challenge and is currently an active area of research (Apte et al., 2007; Harlim and Hunt, 2007; Pires et al., 2010; Anderson, 2010). For further discussion on data assimilation in the presence of non-Gaussian statistics, the reader is directed to the review article by Bocquet et al. (2010).

Another problem inherent in all forms of data assimilation is that in any real-world application the dynamics of the truth (2.1) is unknown. As such there is always some model error associated with the model (2.2). How to parametrise this model error for use in data assimilation is a deep question to which there exists a plethora of different approaches. To take uncertainty in the model representation into account we mention here isotropic model error parametrisation (Mitchell and Houtekamer, 2000; Houtekamer et al., 2005), randomisation of existing deterministic parametrisation schemes (Buizza et al., 1999) and kinetic energy backscatter (Shutts, 2005). Recent comparisons between those methods are given by Houtekamer et al. (2009), Charron et al. (2010), as well as Hamill and Whitaker (2005). In this thesis we will consider two ways in which model error can arise: through the use of underdamped forecast models, and through stochastic parametrisations of the forecast model itself. In Chapter 4 we will develop a novel approach to deal with model error in the specific case where the forecast model is underdamped. In Chapter 5 we introduce model error into an EnKF through using a stochastic homogenised model in the forecast step, which we will show can in fact improve the analysis skill for a multiscale system.

Besides the inconsistencies between the filter assumptions and reality such as non-Gaussianity which render all Kalman filters suboptimal, and the problem of estimating model error which is inherent in all data assimilation methods, ensemble based Kalman

filters have the specific problem of sampling errors due to the insufficient size of the ensemble. This can manifest in various ways, including covariance underestimation, covariance overestimation, filter divergence and spurious non-local correlations.

The typical problem for EnKFs when using small ensembles is that error covariances can be greatly underestimated (Houtekamer and Mitchell, 1998; Burgers et al., 1998). In the worst case this can lead to filter divergence, when the filter trusts its own forecast and ignores the information given by the observations (Ott et al., 2004). The common approach to this problem employed in operational data assimilation is *inflation*, where the forecast error covariance is increased by an inflation factor δ^2 in order to simulate an ensemble with greater spread. There are a number of inflation techniques including multiplicative (Anderson and Anderson, 1999), additive (Corazza et al., 2003) and adaptive inflation (Anderson, 2007, 2009; Li et al., 2009) which estimates the inflation factor online from the observation innovations. In this thesis we will consider the simplest form of covariance inflation, global multiplicative inflation (Anderson and Anderson, 1999), where the forecast error covariance is inflated either by

$$\mathbf{P}_f \rightarrow \delta^2 \mathbf{P}_f \quad (2.21)$$

with $\delta > 1$, which inflates all components equally, or by inflating the forecast ensemble perturbations directly using $\mathbf{Z}'_f \rightarrow \delta \mathbf{Z}'_f$. We will use the latter form in this thesis. A drawback of global multiplicative inflation is that the inflation factor δ has to be determined empirically, which requires careful and expensive tuning. In Chapter 5 we will show how stochastic homogenised forecast models can be used as an alternative to covariance inflation, avoiding covariance underestimation by creating ensembles with sufficiently large spread during the forecast model integration itself.

Estimating covariances using small ensembles can also generate spurious correlations between remote observation sites (Houtekamer and Mitchell, 1998). The technique of *localisation* has been developed to overcome this problem, in which one only filters observations in a local region around each grid point (Houtekamer and Mitchell, 2001; Hamill et al., 2001; Ott et al., 2004; Szunyogh et al., 2005). The typical approach is to take the Schur product of the forecast error covariance matrix with a correlation function with local support to eliminate long-range correlations (Gaspari and Cohn, 1999).

Interestingly, finite size effects can also result in covariance overestimation, rather than underestimation. A theoretical understanding of this phenomenon is incomplete, however has been observed in a variety of situations. Overestimation of error covariances matrices has been shown to occur in spatially sparse observational networks (see for example [Liu et al. \(2008\)](#); [Whitaker et al. \(2009\)](#)), where there are large data voids for the data assimilation algorithm to fill in. This situation is particularly relevant for data assimilation in the southern hemisphere where there are fewer observations available, as well as in climate reanalysis, where sparse historical data is used to create analysed fields. When observations are not present to temper the growth of errors in data voids, the analysis can be spoiled by unrealistically high error covariances. This effect will of course be exacerbated when large covariance inflation is used; in data-poor regions, inflation can lead to unrealistically large ensemble covariances which have a detrimental effect upon the analysis (see for example [Whitaker et al. \(2004\)](#)). We remark that covariance overestimation could also be generated by localisation. It is known that localisation can be a significant source of imbalance in analysed fields (see for example [Houtekamer and Mitchell \(2005\)](#); [Kepert \(2009\)](#); [Houtekamer et al. \(2009\)](#)). This imbalance could generate spurious unwanted gravity wave activity, which may be another means by which overestimation of error covariances in sparse observational networks may occur.

As we will show, incorporating climatological information can be beneficial in alleviating many of the above problems in data assimilation. We will develop a new approach for incorporating climatological information into the analysis step of the ensemble Kalman filter procedure in the next chapter.

USING CLIMATOLOGICAL INFORMATION IN SPARSE OBSERVATIONAL NETWORKS: VARIANCE LIMITING KALMAN FILTERING

3.1 INTRODUCTION

In this chapter we will show that incorporating climatological information into the data assimilation scheme can help address the problem of covariance overestimation in sparse observational networks. We develop a framework for doing this by modifying the analysis step of the ensemble Kalman filter in the case of partial observations and limited information on the mean and variance of the unobserved regions. The challenge of dealing with sparse observational networks is a common problem in data assimilation, as the dimension of the forecast model is typically much larger than the number of observations available. Particular situations where large data voids can occur is in climate reanalysis in the pre-radiosonde era and mesospheric data assimilation. One problem which arises in such situations is covariance overestimation, as was described in Section 2.3. Our aim in this chapter is to develop a method which controls this overestimation by introducing climatological information into the filtering process.

When assimilating current weather data in numerical schemes for the troposphere, the main problem is covariance underestimation rather than overestimation. This is due to the availability of radiosonde data which assures wide observational coverage. However, in the pre-radiosonde era there were severe data voids, particularly in the southern hemisphere and in vertical resolution since most observations were taken

at surface level in the northern hemisphere. There is an increased interest in climate reanalysis (see for example (Bengtsson et al., 2007; Whitaker et al., 2004)), which has the challenge of dealing with large unobserved regions. In reanalysis historical atmospheric observations are analysed by a fixed modern forecast scheme to provide a global homogeneous data set covering troposphere and stratosphere for very long time periods. A remarkable effort is the international *Twentieth Century Reanalysis Project* (20CR) (Compo et al., 2011), which produced a global estimate of the atmosphere for the entire 20th century (1871 to the present) using only synoptic surface pressure reports and monthly sea-surface temperature and sea-ice distributions. Such a data set could help to analyse climate variations in the twentieth century or the multidecadal variations in the behaviour of the El-Niño Southern Oscillation (ENSO). An obstacle for reanalysis is the overestimation of error covariances if one chooses to employ ensemble filters to assimilate historical data (see for example Whitaker et al. (2004) where multiplicative covariance inflation is employed).

Overestimation of error covariances can also occur in modern numerical weather forecast schemes for which the upper lid of the vertical domain is constantly pushed towards higher levels. This is done to incorporate the mesosphere, with the aim of better resolving processes in the polar stratosphere (see for example Polavarapu et al. (2005); Sankey et al. (2007); Eckermann et al. (2009)). Unlike in the troposphere, the energy spectrum in the mesosphere is dominated by large-amplitude gravity waves. The high variability associated with these waves causes very large error covariances in the mesosphere which can be 2 orders of magnitude higher than at lower levels (Polavarapu et al., 2005). This renders ensemble filters very sensitive to small uncertainties in the forecast covariances. Being able to control the variances of mesospheric gravity waves is therefore a substantial challenge for data assimilation in the middle atmosphere.

In this chapter we develop a new method which controls covariance overestimation in sparse observational networks by incorporating climatological information into the analysis step of the ensemble Kalman filtering procedure. Our approach is to demand that the spread of the ensemble after the analysis step in the space spanned by the unobserved variables should not be greater than the climatic variance. This simple requirement leads to a modified set of ensemble Kalman filter equations which we call a *variance limiting Kalman filter* (VLKF). By penalising the distance of the analysis mean from the climatic mean (weighted by the error covariance that arises from the

above condition), we find that the analysis mean is controlled by driving it towards the climatological value. Through analytical results derived for a simple linear toy model and numerical simulations on a higher-dimensional nonlinear chaotic system we show that this method can improve on the analysis skill of standard ensemble Kalman filtering methods, as well as reduce the problem of filter divergence described in the previous chapter.

The remainder of this chapter is organised as follows. In Section 3.2 we introduce the dynamical setting and formulate the variance limiting Kalman filter in a variational setting. In Section 3.3 we illustrate the VLKF with a simple linear toy model for which the filter can be analysed analytically, and extract the parameter regimes where we expect VLKF to yield optimal performance. In Section 3.4 we apply the VLKF to the 40-dimensional Lorenz-96 system (Lorenz, 1996) and present numerical results illustrating the advantage of such a variance limiting filter. We conclude with a discussion in Section 3.5.

3.2 FORMULATION OF THE VARIANCE LIMITING KALMAN FILTER

Assume an N -dimensional dynamical system whose dynamics is given by (2.1), with the state variable $\mathbf{z} \in \mathbb{R}^N$. The exposition presented here is restricted to $\mathbb{R}^N$, but we note that the following formulation can be generalised for Hilbert spaces. We assume that the state space is decomposable according to $\mathbf{z} = (\mathbf{x}, \mathbf{y})$ with $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$ with $m < N$. We will take $n + m = N$ throughout, however we note that this assumption is not essential for our argument. Here $\mathbf{x}$ shall denote those variables for which direct observations are available, and $\mathbf{y}$ shall denote those variables for which only some integrated or statistical information is available. We coin the former variables *observables* and the latter variables *pseudo-observables*. We do not incorporate model error here and assume that the model (2.2) describes the truth. An application of the VLKF to a problem with model error will be discussed in Chapter 4.

As well as the usual observation operator $\mathbf{H}$ for the observables introduced in (2.3), we introduce an operator $\mathbf{h} : \mathbb{R}^N \rightarrow \mathbb{R}^m$ which maps from the whole space into the space of the pseudo-observables spanned by $\mathbf{y}$. We assume that the analysed pseudo-

observables have covariance $\mathbf{A}_{\text{clim}} \in \mathbb{R}^{m \times m}$ and constant mean $\mathbf{a}_{\text{clim}} \in \mathbb{R}^m$. This is the only information available for the pseudo-observables, and may be estimated, for example, from climatic measurements or climate reanalysis. The unknown error covariance of the pseudo-observations is denoted by $\mathbf{R}_w \in \mathbb{R}^{m \times m}$.

Data assimilation aims to find the best estimation of the current state given the forecast $\mathbf{z}_f$ with error covariance $\mathbf{P}_f$ and observations $\mathbf{y}_o$ of the designated variables¹ with error covariance $\mathbf{R}_o$. Pseudo-observations can be included following the standard Bayesian approach described in Chapter 2 once their mean $\mathbf{a}_{\text{clim}}$ and error covariance $\mathbf{R}_w$ are known. However, the error covariance $\mathbf{R}_w$ of a pseudo-observation is in general not equal to $\mathbf{A}_{\text{clim}}$. Here we show how to derive the error covariance $\mathbf{R}_w$ in order to ensure that the analysis does not exceed the prescribed variance $\mathbf{A}_{\text{clim}}$.

We formulate the Kalman filter as a minimisation problem analagous to the procedure in Chapter 2. The cost function for one analysis step as described in Chapter 2 with a given background mean $\mathbf{z}_f$ and associated error covariance $\mathbf{P}_f$ is written as

$$\begin{aligned} J(\mathbf{z}) = & \frac{1}{2}(\mathbf{z} - \mathbf{z}_f)^T \mathbf{P}_f^{-1}(\mathbf{z} - \mathbf{z}_f) + \frac{1}{2}(\mathbf{y}_o - \mathbf{H}\mathbf{z})^T \mathbf{R}_o^{-1}(\mathbf{y}_o - \mathbf{H}\mathbf{z}) \\ & + \frac{1}{2}(\mathbf{a}_{\text{clim}} - \mathbf{h}\mathbf{z})^T \mathbf{R}_w^{-1}(\mathbf{a}_{\text{clim}} - \mathbf{h}\mathbf{z}) , \end{aligned} \quad (3.1)$$

where $\mathbf{z}$ is the state variable at one observation time $t_i = i\Delta t_{\text{obs}}$. Note that the term involving the pseudo-observables corresponds to the notion of a *weak constraint* in variational data assimilation (Sasaki, 1970; Zupanski, 1997; Neef et al., 2006).

Proceeding as in Chapter 2 to find the analysis $\mathbf{z}_a$, we evaluate the equation for the critical point of (3.1) with $\nabla_{\mathbf{z}} J(\mathbf{z}) = 0$ to be

$$\left(\mathbf{P}_f^{-1} + \mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{H} + \mathbf{h}^T \mathbf{R}_w^{-1} \mathbf{h} \right) \mathbf{z}_a = \mathbf{P}_f^{-1} \mathbf{z}_f + \mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{y}_o + \mathbf{h}^T \mathbf{R}_w^{-1} \mathbf{a}_{\text{clim}} ,$$

giving the analysis mean

$$\mathbf{z}_a = \mathbf{z}_f + \mathbf{K}_o [\mathbf{y}_o - \mathbf{H}\mathbf{z}_f] + \mathbf{K}_w [\mathbf{a}_{\text{clim}} - \mathbf{h}\mathbf{z}_f] , \quad (3.2)$$

¹ We remark that the observations $\mathbf{y}_o \in \mathbb{R}^n$ are not to be confused with the pseudo-observables $\mathbf{y} \in \mathbb{R}^m$.

where the Kalman gain matrices are defined as

$$\begin{aligned}\mathbf{K}_o &= \mathbf{P}_a \mathbf{H}^T \mathbf{R}_o^{-1} \\ \mathbf{K}_w &= \mathbf{P}_a \mathbf{h}^T \mathbf{R}_w^{-1} .\end{aligned}\tag{3.3}$$

The analysis covariance $\mathbf{P}_a$ is given by the addition rule for variances, typical in linear Kalman filtering (Kalnay, 2002),

$$\mathbf{P}_a = \left(\mathbf{P}_f^{-1} + \mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{H} + \mathbf{h}^T \mathbf{R}_w^{-1} \mathbf{h} \right)^{-1} .\tag{3.4}$$

Note that (3.2) is the same as equation (2.9) for the analysis $\mathbf{z}_a$ except with the addition of a term driving the pseudo-observables towards the climatological mean, and (3.4) is the same as equation (2.12) for the analysis error covariance $\mathbf{P}_a$, with the addition of a term giving some weight to the error covariance of the pseudo-observables. To create a new ensemble using the ETKF we change the matrix $\mathbf{U}$ from equation (2.20) to

$$\mathbf{U} = \frac{1}{k-1} \mathbf{T}^T \mathbf{Z}_f^T \left(\mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{H} + \mathbf{h}^T \mathbf{R}_w^{-1} \mathbf{h} \right) \mathbf{Z}_f \mathbf{T}$$

and proceed as in Chapter 2.

We wish to derive a constraint on the variance of the pseudo-observables to be employed in the subspace of vectors $\mathbf{v}$ for which $|\mathbf{v}^T \mathbf{P}_a \mathbf{v}| > |\mathbf{v}^T \mathbf{A}_{\text{clim}} \mathbf{v}|$ in the equivalent EnKF. To control the variance of these unresolved pseudo-observables $\mathbf{h}\mathbf{z}$ we set

$$\mathbf{h} \mathbf{P}_a \mathbf{h}^T = \mathbf{A}_{\text{clim}} .\tag{3.5}$$

Introducing

$$\mathcal{P}^{-1} = \mathbf{P}_f^{-1} + \mathbf{H}^T \mathbf{R}_o^{-1} \mathbf{H} ,\tag{3.6}$$

the analysis error covariance of an EnKF without the weak constraint on the pseudo-observables (cf. (2.12)), and upon applying the Sherman-Morrison-Woodbury formula (see for example Golub and Loan (1996)) to $(\mathcal{P}^{-1} + \mathbf{h}^T \mathbf{R}_w^{-1} \mathbf{h})^{-1}$, equation (3.5) yields the desired equation for $\mathbf{R}_w$

$$\mathbf{R}_w^{-1} = \mathbf{A}_{\text{clim}}^{-1} - \left(\mathbf{h} \mathcal{P} \mathbf{h}^T \right)^{-1} ,\tag{3.7}$$

which is again a reciprocal addition formula for variances. Note that the naive expectation that $\mathbf{R}_w = \mathbf{A}_{\text{clim}}$ is true for $\mathcal{P} \rightarrow \infty$, but is not generally true. In fact, the expression for $\mathbf{R}_w$ which ensures that the variance of the analysed pseudo-observables is limited from above by $\mathbf{A}_{\text{clim}}$ involves all error covariances.

For sufficiently small background error covariance $\mathbf{P}_f$, the error covariance $\mathbf{R}_w$ as defined in (3.7) is not positive semi-definite. In this case the information given by the pseudo-observables has to be discarded. In the language of variational data assimilation the criterion of positive definiteness of $\mathbf{R}_w^{-1}$ determines whether the weak constraint is to be switched on or off. To determine those eigendirections for which statistical information available can be incorporated, we diagonalise $\mathbf{R}_w^{-1} = \mathbf{V}\mathbf{D}\mathbf{V}^T$ and define $\bar{\mathbf{D}}$ with $\bar{\mathbf{D}}_{ii} = \mathbf{D}_{ii}$ for $\mathbf{D}_{ii} \geq 0$ and $\bar{\mathbf{D}}_{ii} = 0$ for $\mathbf{D}_{ii} < 0$. The modified $\mathbf{R}_w^{-1} = \mathbf{V}\bar{\mathbf{D}}\mathbf{V}^T$ then uses climatological information on the pseudo-observables only in those directions which potentially improve the analysis. Noting that $\mathcal{P}$ denotes the analysis covariance of an EnKF (with $\mathbf{R}_w^{-1} = 0$), we see that equation (3.7) states that the variance constraint switches on for those eigendirections whose corresponding eigenvalues of $\mathbf{h}\mathcal{P}\mathbf{h}^T$ are larger than those of $\mathbf{A}_{\text{clim}}$. Hence the proposed VLKF as defined here controls covariance overestimation by incorporating climatic information about the unresolved variables; by restricting the posterior error covariance of the pseudo-observables to remain below their climatic variance we also drive the analysis mean towards the climatological mean.

3.3 ANALYTICAL LINEAR TOY MODEL

In order to understand under what circumstances the VLKF is switched on and when it produces better skill than the standard ETKF, we now apply it to a linear toy model which is analytically tractable. In this section we will find for this simple system the conditions under which the weak constraint provided by the VLKF is nonzero, and the conditions for which skill improvement over the ETKF occurs. The main results are that the weak constraint is non-zero more often when the observation interval or covariance inflation factor is large, and that the skill improvement is greater when either the observational noise or the time scale separation between observed and unobserved subsystems is increased.

We study the VLKF for the following coupled linear skew product system for two oscillators $\mathbf{x} \in \mathbb{R}^2, \mathbf{y} \in \mathbb{R}^2$

$$\begin{aligned} d\mathbf{x} &= \mathbf{A}\mathbf{x}dt - \mathbf{\Gamma}_x\mathbf{x}dt + \sigma_x d\mathbf{W}_{x,t} + \mathbf{\Lambda}\mathbf{y}dt \\ d\mathbf{y} &= \mathbf{B}\mathbf{y}dt - \mathbf{\Gamma}_y\mathbf{y}dt + \sigma_y d\mathbf{W}_{y,t}, \end{aligned}$$

where $\mathbf{A}, \mathbf{B}$ and $\mathbf{\Lambda}$ are all skew-symmetric, $\sigma_{x,y}$ and $\mathbf{\Gamma}_{x,y}$ are all symmetric, and $\mathbf{W}_{x,t}$ and $\mathbf{W}_{y,t}$ are independent two-dimensional Brownian processes². We assume here for simplicity that

$$\mathbf{\Gamma}_x = \gamma_x \mathbf{I}, \quad \mathbf{\Gamma}_y = \gamma_y \mathbf{I}, \quad \sigma_x = \sigma_x \mathbf{I}, \quad \sigma_y = \sigma_y \mathbf{I}, \quad \mathbf{R}_o = R_o \mathbf{I},$$

with the 2×2 identity matrix $\mathbf{I}$, and

$$\mathbf{A} = \omega_x \mathbf{J}, \quad \mathbf{B} = \omega_y \mathbf{J}, \quad \mathbf{\Lambda} = \lambda \mathbf{J},$$

with the skew-symmetric matrix

$$\mathbf{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Note that our particular choice for the structure of the matrices will imply that $\mathbf{R}_w = R_w \mathbf{I}$ after averaging over the Brownian motion paths.

This system models two noisy coupled oscillators $\mathbf{x}$ and $\mathbf{y}$. We assume that we have access to observations of the variable $\mathbf{x}$ at discrete observation times $t_i = i\Delta t_{\text{obs}}$, but have only statistical information about the variable $\mathbf{y}$. We assume knowledge of the climatic mean μ_{clim} and the climatic covariance σ_{clim}^2 of the unobserved variable $\mathbf{y}$. The noise is of Ornstein-Uhlenbeck type (Gardiner, 2003), and may represent either model error or parametrise highly chaotic nonlinear dynamics. Without loss of generality, the coupling is chosen such that the $\mathbf{y}$ -dynamics drives the $\mathbf{x}$ -dynamics but not vice versa. The form of the coupling is not essential for our argument, and it may be

² We will use bold font for matrices and vectors, and non-bold font for scalars here. It should be clear from the context whether bold fonts refer to a matrix or a vector.

oscillatory or damping with $\Lambda = \lambda \mathbf{I}$. We write this system in the more compact form for $\mathbf{z} = (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^4$ as

$$d\mathbf{z} = \mathbf{M}\mathbf{z} dt - \mathbf{\Gamma}\mathbf{z} dt + \sigma d\mathbf{W}_t + \mathbf{C}\mathbf{z} dt \quad (3.8)$$

with $\mathbf{W}_t = (\mathbf{W}_{x,t}, \mathbf{W}_{y,t})$ and

$$\begin{aligned} \mathbf{M} &= \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{pmatrix} & \mathbf{\Gamma} &= \begin{pmatrix} \mathbf{\Gamma}_x & \mathbf{0} \\ \mathbf{0} & \mathbf{\Gamma}_y \end{pmatrix} \\ \sigma &= \begin{pmatrix} \sigma_x & \mathbf{0} \\ \mathbf{0} & \sigma_y \end{pmatrix} & \mathbf{C} &= \begin{pmatrix} \mathbf{0} & \Lambda \\ \mathbf{0} & \mathbf{0} \end{pmatrix}. \end{aligned}$$

The solution of (3.8) can be obtained using Itô's formula and, introducing the propagator $\mathcal{L}(t) = \exp((\mathbf{M} - \mathbf{\Gamma} + \mathbf{C})t)$ which commutes with σ for our choice of the matrices, is given by

$$\mathbf{z}(t) = \mathcal{L}(t)\mathbf{z}_0 + \sigma \int_0^t \mathcal{L}(t-s) d\mathbf{W}_s,$$

with mean

$$\mathbf{m}(t) = \mathcal{L}(t)\mathbf{z}_0,$$

and covariance

$$\Sigma(t) = \sigma (2\mathbf{\Gamma} - \mathbf{C})^{-1} (\mathbf{I} - \exp(-(2\mathbf{\Gamma} - \mathbf{C})t)) \sigma^T, \quad (3.9)$$

where

$$\mathbf{C} = \begin{pmatrix} \mathbf{0} & \Lambda \\ -\Lambda & \mathbf{0} \end{pmatrix}.$$

The climatic mean $\mu_{\text{clim}} \in \mathbb{R}^4$ and covariance matrix $\Sigma_{\text{clim}} \in \mathbb{R}^{4 \times 4}$ are then obtained in the limit $t \rightarrow \infty$ as

$$\mathbf{m}_{\text{clim}} = \lim_{t \rightarrow \infty} \mathbf{m}(t) = \mathbf{0},$$

and

$$\Sigma_{\text{clim}} = \lim_{t \rightarrow \infty} \Sigma(t) = \sigma (2\Gamma - \mathcal{C})^{-1} \sigma^T .$$

In order for the stochastic process (3.8) to have a stationary density and for $\Sigma(t)$ to be a positive definite covariance matrix for all t , the coupling has to be sufficiently small with $\lambda^2 < 4\gamma_x\gamma_y$. Note that the skew product nature of the system (3.8) is not special in the sense that a non-skew product structure where $\mathbf{x}$ couples back to $\mathbf{y}$ would simply lead to a renormalisation of $\mathcal{C}$. However, it is pertinent to mention that although in the actual dynamics of the model (3.8) there is no back-coupling from $\mathbf{x}$ to $\mathbf{y}$, the Kalman filter generically introduces back-coupling of all variables through the inversion of the covariance matrices (cf. (3.4)).

We now investigate the variance limiting Kalman filter for this toy model. In particular we first analyse under what conditions $\mathbf{R}_w$ is positive definite and the variance constraint will be switched on, and second we analyse when the VLKF yields a skill improvement when compared to the standard ETKF.

We start with the positive definiteness of $\mathbf{R}_w$. When calculating the covariance of the forecast in an ensemble filter we need to interpret the solution of the linear toy model (3.8) as

$$\mathbf{z}_j(t_{i+1}) \stackrel{d}{=} \mathcal{L}(\Delta t_{\text{obs}}) \mathbf{z}_j(t_i) + \sigma \int_0^{\Delta t_{\text{obs}}} \mathcal{L}(\Delta t_{\text{obs}} - s) d\mathbf{W}_{j_s} \quad j = 1, 2, \dots, k ,$$

where $\mathbf{z}_j(t_{i+1})$ is the forecast of ensemble member j at time $t_{i+1} = t_i + \Delta t_{\text{obs}} = (i + 1)\Delta t_{\text{obs}}$ before the analysis propagated from its initial condition $\mathbf{z}_j(t_i) = \mathbf{z}_a(t_i) + \xi_j$ with $\xi_j \sim \mathcal{N}(0, \mathbf{P}_a(t_i))$ at the previous analysis³. The equality here is in distribution only; members of the ensemble are not equal in a pathwise sense as their driving Brownian motions will be different, but they will have the same mean and variance. The covariance of the forecast can then be obtained by averaging with respect to

³ Note that although we are mathematically interpreting the deterministic ETKF as a random filter here, this does not mean that in numerical experiments we use a random sample ensemble; the perturbations ξ_j are still generated deterministically as in the ETKF.

the ensemble and with respect to realisations of the Brownian motion, and is readily computed as

$$\mathbf{P}_f(t_{i+1}) = \mathcal{L}(\Delta t_{\text{obs}}) \mathbf{P}_a(t_i) \mathcal{L}^T(\Delta t_{\text{obs}}) + \mathbf{\Sigma}(\Delta t_{\text{obs}}), \quad (3.10)$$

where $\mathcal{L}^T(t) = \exp((- \mathbf{M} - \mathbf{\Gamma} + \mathbf{C}^T)t)$ denotes the transpose of $\mathcal{L}(t)$. The forecast error covariance of an ensemble with initial spread $\mathbf{P}_a$ is thus typically larger than the forecast covariance $\mathbf{\Sigma}$ of one trajectory with a non-random initial condition $\mathbf{z}_0$. The difference is most pronounced for small observation intervals when the covariance of the ensemble $\mathbf{P}_f$ will be close to the initial analysis covariance $\mathbf{P}_a$, whereas a single trajectory will have acquired little variance $\mathbf{\Sigma}$. In the long-time limit, $\mathcal{L}(t) \rightarrow 0$ so both $\mathbf{P}_f$ and $\mathbf{\Sigma}$ will approach the climatic covariance $\mathbf{\Sigma}_{\text{clim}}$ (cf. (3.9)).

In the following we restrict ourselves to the limit of small observation intervals $\Delta t_{\text{obs}} \ll 1$. In this limit, we can approximate $\mathbf{P}_a(t_i) \approx \mathbf{P}_f(t_{i+1})$ and explicitly solve (3.10) for the forecast covariance matrix $\mathbf{P}_f$. This assumption requires that the analysis is stationary in the sense that the filter has lost memory of the initial background covariance provided to start up the analysis. We have verified the validity of this assumption for small observation intervals and for a range of initial background variances. This assumption renders (3.10) a matrix equation for $\mathbf{P}_f$. To derive analytical expressions we further perform a Taylor expansion of the propagator $\mathcal{L}(\Delta t_{\text{obs}})$ and the covariance $\mathbf{\Sigma}(\Delta t_{\text{obs}})$ for small observation intervals Δt_{obs} . This is consistent with our stationarity assumption $\mathbf{P}_a(t_i) \approx \mathbf{P}_f(t_{i+1})$. The lengthy analytical expression for $\mathbf{P}_f(t_{i+1})$ can be obtained with the aid of *Mathematica* (*Mathematica Version 7.0, 2008*), and is given for completeness in equation A.1 of the Appendix.

As discussed in Chapter 2, one often uses variance inflation (*Anderson and Anderson, 1999*) in filtering to compensate for the loss of ensemble variance due to finite size effects, sampling errors and the effects of nonlinearities. We do this here by introducing an inflation factor $\delta > 1$ multiplying the forecast variance $\mathbf{P}_f$. Having determined the forecast covariance matrix $\mathbf{P}_f$ we are now able to write down an expression for the error covariance of the pseudo-observables $\mathbf{R}_w$. As previously we limit the variance and the mean of the pseudo-observable $\mathbf{y}$ to be $\mathbf{A}_{\text{clim}} = \mathbf{h}\mathbf{\Sigma}_{\text{clim}}\mathbf{h}^T = \sigma_{\text{clim}}^2$ and $\mathbf{a}_{\text{clim}} = \mathbf{h}\mathbf{m}_{\text{clim}} = \mu_{\text{clim}}$ respectively. Then, upon using the definitions (3.6) and (3.7), we find that the error covariance for the pseudo-observables $\mathbf{R}_w$ is positive definite

provided the observation interval Δt_{obs} is sufficiently large⁴. Particularly, in the limit of $\mathbf{R}_0 \rightarrow \infty$, we find that if

$$\Delta t_{\text{obs}}(\delta) > \frac{\delta \lambda^2 + 4\gamma_x \gamma_y (1 - \delta)}{2\gamma_x (1 + \gamma_y^2)}, \quad (3.11)$$

the variance constraint must be switched on. Note that for $\delta > 1$ the critical Δt_{obs} above which $\mathbf{R}_w$ is positive definite can be negative, implying that the variance constraint will be switched on for all (positive) values of Δt_{obs} . If no inflation is applied, i.e. if $\delta = 1$, (3.11) simplifies to

$$\Delta t_{\text{obs}} > \frac{\lambda^2}{2\gamma_x (1 + \gamma_y^2)} > 0. \quad (3.12)$$

Because $4\gamma_x \gamma_y - \lambda^2 > 0$ the critical observation interval Δt_{obs} is smaller for non-trivial inflation with $\delta > 1$ than if no variance inflation is incorporated. This is because the variance inflation will increase instances with $|\mathbf{h} \mathbf{P}_d \mathbf{h}^T| > |\sigma_{\text{clim}}^2|$. We have numerically verified that inflation increases the number of instances where the variance constraint is switched on. It is also beneficial to increase the observation interval for switching on the variance constraint. The derivative of R_w^{-1} with respect to Δt_{obs} at the critical observation interval Δt_{obs} is

$$\frac{\partial R_w^{-1}}{\partial \Delta t_{\text{obs}}} = \frac{\gamma_y^2 + 1}{(\delta - 2)^2 \delta \sigma_y^2} > 0$$

in the limit of small coupling λ , indicating that the frequency of occurrence when the variance constraint is switched on increases monotonically with the observation interval Δt_{obs} . This has been verified numerically with the application of VLKF for (3.8) and is illustrated in Figure 3.1. It is pertinent to mention that for sufficiently large coupling strength λ or sufficiently small values of γ_x , equation (3.12) may not be consistent with the assumption of small observation intervals $\Delta t_{\text{obs}} \ll 1$.

Finite ensemble size effects are significant in ensemble filtering, and as such the analytical results obtained earlier for the ideal case with $k \rightarrow \infty$ must be treated with care. As discussed in Section 3.1, finite ensemble sizes cause the overestimation

⁴ We actually compute $\mathbf{R}_w^{-1}$, however, since $\mathbf{R}_w$ is diagonal for our choice of the matrices, positive definiteness of $\mathbf{R}_w^{-1}$ implies positive definiteness of $\mathbf{R}_w$.

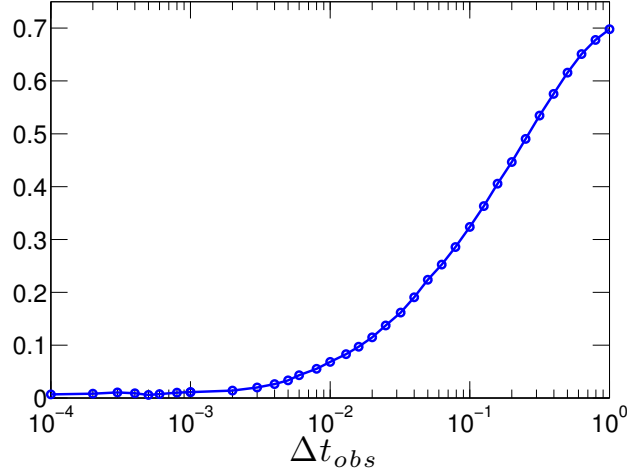


Figure 3.1.: Proportion of incidences when the variance constraint is switched on and $\mathbf{R}_w$ is positive definite as a function of the observation interval Δt_{obs} for the stochastic linear toy model (3.8). We used $\gamma_x = 1$, $\gamma_y = 1$, $\sigma_x = 1$, $\sigma_y = 1$, $\lambda = 0.2$. We used $k = 20$ ensemble members, 100 realisations and $\mathbf{R}_o = \mathbf{H}\Sigma_{clim}\mathbf{H}^T$ and no inflation with $\delta = 1$. The analytically calculated critical observation interval according to equation (3.12) is $\Delta t_{obs} = 10^{-2}$.

of error covariances in sparse observation networks (Liu et al., 2008; Whitaker et al., 2009), implying that $\mathbf{R}_w$ is positive definite and the variance limiting constraint will be switched on. This finite size effect is illustrated in Figure 3.2, where the maximal singular value of $\mathbf{h}\mathbf{P}_a\mathbf{h}^T$, averaged over 50 realisations, is shown for the ETKF as a function of ensemble size k for different observational noise variances. Here we used no inflation, i.e. $\delta = 1$, in order to focus on the effect of finite ensemble sizes. It is clearly seen that the projected covariance decreases for sufficiently large ensembles. The covariance will asymptote from above to $\mathbf{h}\Sigma_{clim}\mathbf{h}^T$ in the limit $k \rightarrow \infty$. For sufficiently small observational noise the filter corrects large forecast error covariances by incorporating the observations into the analysis, leading to a decrease in the analysis error covariance.

We remark that for large observation intervals $\Delta t_{obs} \rightarrow \infty$ and large observational noise $\mathbf{R}_o \rightarrow \infty$, we have $\mathbf{P}_f \rightarrow \Sigma_{clim}$ and our analytical formulae indicate that the variance constraint should not be switched on (cf. (3.6) and (3.7)). However, in numerical simulations we observe the curious result that increasing the ensemble size k causes the variance constraint to be switched on for almost every analysis cycle. The reason is that

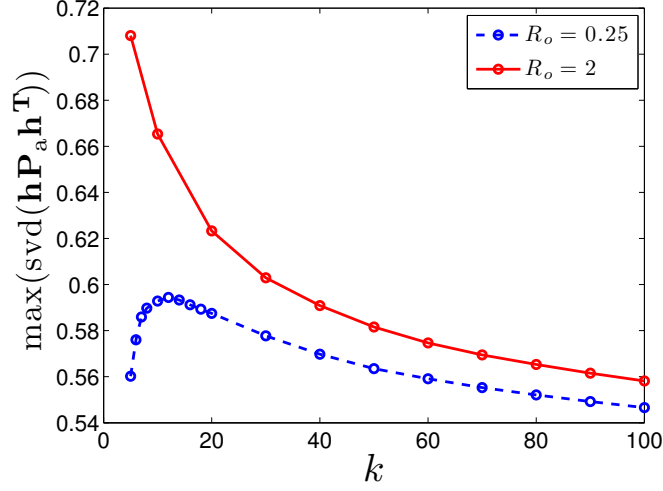


Figure 3.2.: Average maximal singular value of $\mathbf{hP}_a \mathbf{h}^T$ as a function of ensemble size k for the stochastic linear toy model (3.8) using standard ETKF without inflation, with $R_o = 0.25$ (dashed curve) and $R_o = 2$ (solid curve). Other parameters are $\sigma_x = \sigma_y = \gamma_x = \gamma_y = 1$, $\lambda = 0.2$, $\Delta t_{\text{obs}} = 1$, for which the climatic variance is $\mathbf{h}\Sigma_{\text{clim}} \mathbf{h}^T \approx 0.505$. We used 50 realisations for the averaging.

numerical finite size fluctuations in the covariance matrices produce more instances of the variance constraint being switched on. This is shown in Figure 3.3, where we plot the maximal eigenvalue of $\mathbf{hP}_f \mathbf{h}^T$ as a function of time for $k = 5$ (left) and $k = 200$ (right). The figure shows that the average maximal eigenvalue of $\mathbf{hP}_f \mathbf{h}^T$ approaches $\mathbf{A}_{\text{clim}}$ from above. We therefore write $\mathbf{hP}_f \mathbf{h}^T = a \mathbf{A}_{\text{clim}}$, where $a > 1$. Then in the limit as $R_o \rightarrow \infty$, $\mathcal{P} \rightarrow \mathbf{P}_f$, and so equation (3.7) becomes

$$\mathbf{R}_w^{-1} = \left(1 - \frac{1}{a}\right) \mathbf{A}_{\text{clim}}^{-1} > 0,$$

meaning the variance constraint is switched on for $a > 1$. We note here that there are less fluctuations of the maximal eigenvalue of $\mathbf{hP}_f \mathbf{h}^T$ for $k = 200$ than $k = 5$, but more instances with the maximal eigenvalue of $\mathbf{hP}_f \mathbf{h}^T$ greater than $\mathbf{A}_{\text{clim}}$. The variance constraint is therefore switched on more often for large ensembles, even though $\mathbf{P}_f \rightarrow \Sigma_{\text{clim}}$ as $k \rightarrow \infty$.

The fact that the variance constraint is switched on does not necessarily imply that the variance limiting filter will perform better than the standard ETKF. In particular,

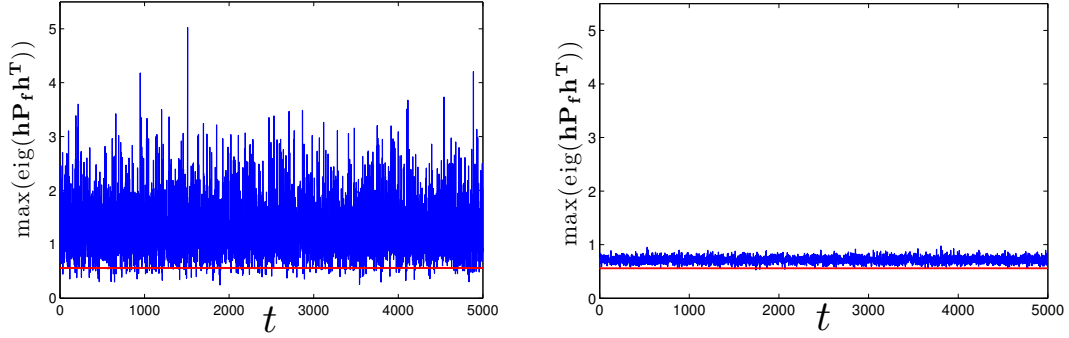


Figure 3.3.: Time series of the maximal eigenvalues of $\mathbf{hP}_f \mathbf{h}^T$ for EnKF (blue) with $k = 5$ (left) and $k = 200$ (right). The climatic variance is shown in red. We used $\mathbf{R}_o = 0.25$, other parameters as in Figure 3.2.

for very large observation intervals Δt_{obs} when the ensemble will have acquired the climatic mean and covariances, $\mathbf{R}_w^{-1}$ will be equal to zero by (3.7), and so the VLKF and ETKF will have equal skill. We now turn to the question under what conditions VLKF is expected to yield improved analysis skill compared to the standard ETKF. To this end we introduce the (squared) RMS error

$$\mathcal{E} = \mathbb{E}^{t,W} \|\mathbf{z}_a(t_i) - \mathbf{z}_t(t_i)\|_{\mathbf{G}}^2, \quad (3.13)$$

between the truth $\mathbf{z}_t$ and the analysis mean $\mathbf{z}_a$ (the square root is left out here for convenience of exposition). Here $\mathbb{E}^t$ denotes the temporal average over analysis cycles, and $\mathbb{E}^W$ denotes averaging over different realisations of the Brownian paths W . We introduce the norm $\|\mathbf{a}\mathbf{b}\|_{\mathbf{G}} = \mathbf{a}^T \mathbf{G} \mathbf{b}$ to investigate the overall RMS error $\mathcal{E}$ using $\mathbf{G} = \mathbf{I}$, the RMS error for the observed variables $\mathcal{E}_o$ using $\mathbf{G} = \mathbf{H}^T \mathbf{H}$ and the RMS error for the unobserved variables $\mathcal{E}_{uo}$ using $\mathbf{G} = \mathbf{h}^T \mathbf{h}$. Using the Kalman filter equation (3.2) for the analysis mean with $\mathbf{K}_w = 0$, we obtain for the ETKF

$$\mathcal{E}^{\text{ETKF}} = \mathbb{E}^{t,W} \|(\mathbf{I} - \mathbf{K}_o \mathbf{H})(\mathbf{z}_f(t_i) - \mathbf{z}_t(t_i)) + \mathbf{K}_o \mathbf{r}_o(t_i)\|_{\mathbf{G}}^2,$$

where $\mathbf{r}_o(t_i)$ is a random realisation of the observational error. Solving the linear toy model (3.8) for each member of the ensemble and then performing an ensemble average, we obtain

$$\mathbf{z}_f(t_i) = \mathcal{L}(\Delta t_{\text{obs}}) \mathbf{z}_a(t_{i-1}). \quad (3.14)$$

Substituting a particular realisation of the truth $\mathbf{z}_t(t)$, and performing an average over many realisations, we finally arrive at

$$\mathcal{E}^{\text{ETKF}} = \mathbb{E}^t \|(\mathbf{I} - \mathbf{K}_o \mathbf{H}) \mathcal{L}(\Delta t_{\text{obs}}) \boldsymbol{\xi}_{t_{i-1}}\|_{\mathbf{G}}^2 + \mathbb{E}^t \|(\mathbf{I} - \mathbf{K}_o \mathbf{H}) \boldsymbol{\eta}_{t_i}\|_{\mathbf{G}}^2 + \mathbb{E}^t \|\mathbf{K}_o \mathbf{r}_o\|_{\mathbf{G}}^2, \quad (3.15)$$

with the mutually independent normally distributed random variables

$$\begin{aligned} \boldsymbol{\xi}_{t_i} &= \mathbf{z}_a(t_i) - \mathbf{z}_t(t_i) \sim \mathcal{N}(0, \mathbf{P}_a(t_i)) \\ \boldsymbol{\eta}_{t_i} &= \sigma \int_{t_{i-1}}^{t_i} \mathcal{L}(\Delta t_{\text{obs}} - s) d\mathbf{W}_s \sim \mathcal{N}(0, \boldsymbol{\Sigma}(\Delta t_{\text{obs}})) \\ \mathbf{r}_o &\sim \mathcal{N}(0, \mathbf{R}_o). \end{aligned} \quad (3.16)$$

We have numerically verified the validity of our assumptions of the statistics of $\boldsymbol{\xi}_{t_i}$ and $\boldsymbol{\eta}_{t_i}$. Note that for $\boldsymbol{\xi}_{t_i}$ to have mean zero and variance $\mathbf{P}_a(t_i)$ filter divergence has to be excluded.

To obtain an expression for the squared error of a VLKF analysis for the cases where the weak constraint is switched on, we substitute the Kalman filter equation (3.2) - this time with the $\mathbf{K}_w$ term retained - again into (3.13) and perform an average over many realisations. Proceeding in a similar fashion as for the ETKF analysis error, we then obtain for the VLKF

$$\begin{aligned} \mathcal{E}^{\text{VLKF}} &= \mathbb{E}^t \|(\mathbf{I} - \mathbf{K}_o \mathbf{H}) \mathcal{L}(\Delta t_{\text{obs}}) \boldsymbol{\xi}_{t_{i-1}}\|_{\mathbf{G}}^2 + \mathbb{E}^t \|(\mathbf{I} - \mathbf{K}_o \mathbf{H}) \boldsymbol{\eta}_{t_i}\|_{\mathbf{G}}^2 \\ &\quad + \mathbb{E}^t \|\mathbf{K}_o \mathbf{r}_o\|_{\mathbf{G}}^2 + \mathbb{E}^t \|\mathbf{K}_w \mathbf{h} \boldsymbol{\zeta}_{t_i}\|_{\mathbf{G}}^2 \\ &\quad + 2 \mathbb{E}^t [(\mathbf{I} - \mathbf{K}_o \mathbf{H}) (\mathcal{L}(\Delta t_{\text{obs}}) \boldsymbol{\xi}_{t_{i-1}})^T \mathbf{G} (\mathbf{K}_w \mathbf{h} \boldsymbol{\zeta}_{t_i})], \end{aligned} \quad (3.17)$$

with the normally distributed random variable

$$\boldsymbol{\zeta}_{t_i} = \mathbf{z}_f(t_i) \sim \mathcal{N}(0, \frac{1}{k} \mathbf{P}_f(t_i)), \quad (3.18)$$

where we used $\mathbf{a}_{\text{clim}} = 0$ during the expansion of (3.13). Note that using our stationarity assumption to calculate $\mathbf{P}_f$ we have $\boldsymbol{\zeta}_{t_i} \stackrel{d}{\sim} (1/k) \boldsymbol{\xi}_{t_{i-1}}$. We have again numerically verified the statistics for $\boldsymbol{\zeta}_{t_i}$. The expression for the RMS error of the VLKF (3.17) can be considerably simplified. Since for large ensemble sizes $k \rightarrow \infty$ the random variable $\boldsymbol{\zeta}_{t_i}$

becomes a deterministic variable with mean zero, we can neglect all terms containing ζ_{t_i} . We therefore summarise to

$$\mathcal{E}^{\text{VLKF}} = \mathbb{E}^t \|(\mathbf{I} - \mathbf{K}_o \mathbf{H}) \mathcal{L}(\Delta t_{\text{obs}}) \boldsymbol{\xi}_{t_{i-1}}\|_{\mathbf{G}}^2 + \mathbb{E}^t \|(\mathbf{I} - \mathbf{K}_o \mathbf{H}) \boldsymbol{\eta}_{t_i}\|_{\mathbf{G}}^2 + \mathbb{E}^t \|\mathbf{K}_o \mathbf{r}_o\|_{\mathbf{G}}^2. \quad (3.19)$$

For convenience we have omitted superscripts for $\mathbf{K}_o$ and $\boldsymbol{\xi}_{t_{i-1}}$ in (3.15) and (3.19) to denote whether they have been evaluated for the ETKF and VLKF. Note that although the expressions (3.15) and (3.19) are formally the same, generally $\mathcal{E}^{\text{ETKF}} \neq \mathcal{E}^{\text{VLKF}}$ because the analysis covariance matrices $\mathbf{P}_a$ are different for both methods, leading to different gain matrices $\mathbf{K}_o$ and different statistics of $\boldsymbol{\xi}_t$ in (3.15) and (3.19). Note that we assumed here that the weak constraint provided by the VLKF is always switched on, whereas in practice this is not always the case.

We can now estimate the proportional skill improvement for the VLKF over the ETKF defined as

$$\mathcal{S} = \mathcal{E}^{\text{ETKF}} / \mathcal{E}^{\text{VLKF}} \quad (3.20)$$

with values of $\mathcal{S} > 1$ indicating skill improvement of VLKF over ETKF. Henceforth we will choose $\mathbf{G} = \mathbf{h}^T \mathbf{h}$ in order to concentrate on the skill improvement for the unobserved variables only, but drop the subscript from $\mathcal{E}_{uo}$ for convenience of exposition. Recalling that $\mathcal{E}^{\text{ETKF}} \approx \mathcal{E}^{\text{VLKF}}$ for large observation intervals Δt_{obs} , we expect skill improvement for small Δt_{obs} . We again perform a Taylor expansion in small Δt_{obs} of the skill $\mathcal{S}$. The resulting analytical expressions obtained using *Mathematica* are very lengthy and cumbersome, and do not in themselves provide much useful information. For completeness we give a (truncated) expression for the skill $\mathcal{S}$ in equation (A.2) of the Appendix, as well expressions for $\mathcal{S}$ in the limits as $\mathbf{R}_o \rightarrow \infty$ and $\mathbf{R}_o \rightarrow 0$.

We find that $\mathcal{S} \rightarrow \infty$ in the limit of the y -dynamics becoming infinitely faster than the x -dynamics, that is as $\gamma_y \rightarrow \infty$ or $\gamma_x \rightarrow 0$. Furthermore, in the limit of large noise we find that the skill increases with increasing time scale separation, as

$$\frac{\partial \mathcal{S}}{\partial \gamma_y} = \frac{\delta \lambda^2}{4 \gamma_x \gamma_y^2} + \Delta t_{\text{obs}} \left(\frac{\delta \lambda^4}{8 \gamma_x^2 \gamma_y^2} - 2(\delta - 1) \right)$$

is positive for sufficiently small observation time Δt_{obs} or inflation δ . This suggests that the skill is controlled by the ratio of the time scales of the observed and the unobserved

variables. If the time scale of the pseudo-observables is much larger than that of the observables, the VLKF will exhibit superior performance over the ETKF. This can be intuitively understood since $1/(2\gamma_y)$ is the time scale on which the equilibrium climatic is reached for the pseudo-observables $\mathbf{y}$. If the pseudo-observables have relaxed towards equilibrium within the observation interval Δt_{obs} , and their variance has acquired the climatic covariance $\mathbf{h}\mathbf{P}_a\mathbf{h}^T = \sigma_{\text{clim}}^2$, we expect the variance limiting to be beneficial.

Furthermore, we find analytically that the skill improvement increases with increasing observational noise R_o (at least in the small observation interval approximation). In particular we find that

$$\frac{\partial \mathcal{S}}{\partial R_o} = \frac{\gamma_x \lambda^2 (4\gamma_x \gamma_y - \lambda^2) \sigma_y^2}{2 (\gamma_y^2 \sigma_x^2 + \gamma_x \gamma_y \sigma_x^2 + \lambda^2 \sigma_y^2)^2}$$

at $R_o = 0$, which is positive as $4\gamma_x \gamma_y - \lambda^2 > 0$. The increase of skill with increasing observational noise can be understood phenomenologically in the following way. For $R_o = 0$ the filter trusts the observations, which as a time series carry the climatic covariance. This implies that there is a realisation of the Wiener process such that the analysis can be reproduced by a model with the true values of $\gamma_{x,y}$ and $\sigma_{x,y}$. This is also the case in the other extreme $R_o \rightarrow \infty$, where the filter trusts the model and disregards the information given by the observations. For $0 \ll R_o \ll \infty$ the analysis reproducing system would have a larger covariance σ_x than the true value. This slowed down relaxation towards equilibrium of the observed variables can be interpreted as an effective decrease of the damping coefficient γ_x . This effectively increases the time scale separation between the observed and the unobserved variables, which was conjectured above to be beneficial for skill improvement.

As expected, the skill improves with increasing inflation factor $\delta > 1$. The improvement is exactly linear for $\Delta t_{\text{obs}} \rightarrow 0$, as shown in equation (A.3) of the Appendix. This is due to the variance inflation leading to an increase of instances with $\mathbf{h}\mathbf{P}_a\mathbf{h}^T > \sigma_{\text{clim}}^2$, for which the variance constraint will be switched on.

In Figure 3.4 we present a comparison of the analytical results (3.15) and (3.19) with results from a numerical implementation of ETKF and VLKF for varying damping coefficient γ_y . Since γ_y controls the time-scale of the $\mathbf{y}$ -process, we are forced to vary Δt_{obs} with γ_y in order not to violate the small observation interval approximations used

in our analytical expressions. We choose Δt_{obs} as a function of γ_y such that the singular values of the first-order approximation of the forecast variance is a good approximation to the full forecast variance for this Δt_{obs} . For Figure 3.4 we have $\Delta t_{\text{obs}} \in (0.005, 0.01)$ to preserve the validity of the Taylor expansion. Besides the increase of the skill with γ_y , Figure 3.4 shows that the value of $\mathcal{S}$ increases significantly for larger values of the inflation factor $\delta > 1$.

The linear toy model (3.8) presented in this section is intended only as an illustrative example; the skill improvements obtained here are far too small to be of practical significance. However, as we will see in the next section, the results obtained here become far more significant for more complex higher-dimensional models, where the dynamic Brownian driving noise is replaced by nonlinear chaotic dynamics.

3.4 NUMERICAL RESULTS FOR THE LORENZ-96 SYSTEM

We illustrate the VLKF method using the Lorenz-96 system (Lorenz, 1996) and show its use in sparse observational networks at both improving the analysis skill and stabilising the filter. In the first subsection we demonstrate that the VLKF improves analysis skill over the standard ETKF for small observation intervals and in sparse observation networks, with greater skill for increasing observational noise. This is analogous to our analytical results of Section 3.3 for the linear toy model. In the second subsection we investigate the phenomenon of filter blow-up and find parameter regions where the VLKF reduces the instances of blow-up compared to the ETKF.

Lorenz (1996) proposed the following model for the atmosphere

$$\dot{z}_m = z_{m-1}(z_{m+1} - z_{m-2}) - z_m + F \quad m = 1, \dots, D \quad (3.21)$$

with $\mathbf{z} = (z_1, \dots, z_D)$ and periodic $z_{m+D} = z_m$. This system is a toy-model for midlatitude atmospheric dynamics, incorporating linear damping, forcing and nonlinear transport. The dynamical properties of the Lorenz-96 system have been investigated, for example, in (Lorenz and Emanuel, 1998; Orrell and Smith, 2003; Gottwald and Melbourne, 2005), and in the context of data assimilation it was investigated in, for

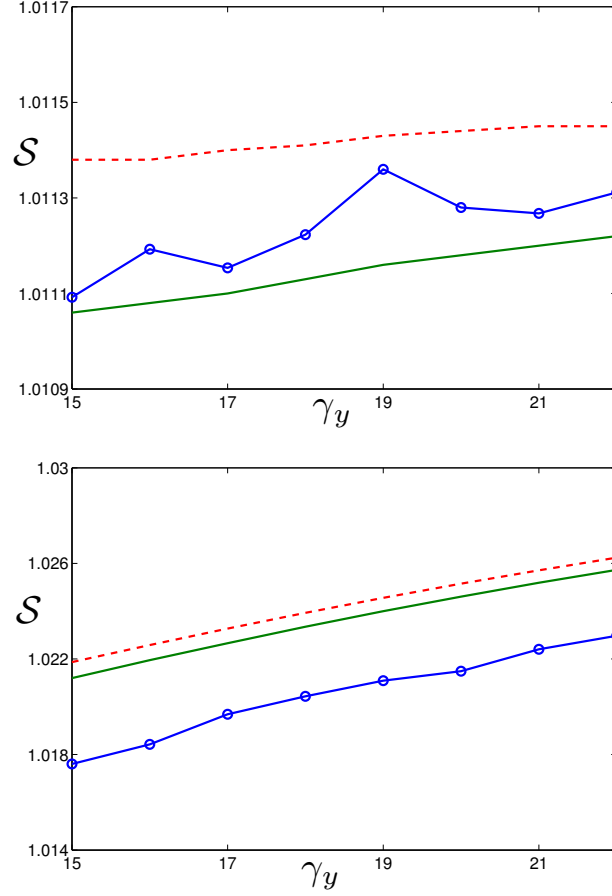


Figure 3.4.: Dependence of the skill improvement $\mathcal{S}$ of VLKF over ETKF on the damping coefficient γ_y of the pseudo-observable. We show a comparison of direct numerical simulations (open circles) with analytical results using (3.17) (continuous curve) and the approximation of large ensemble size (3.19) (dashed curve). Parameters are $\gamma_x = 1$, $\lambda = 2$, $\sigma_x = \sigma_y = 1$, $R_o = 0.25$. We used an ensemble size of $k = 20$ and averaged over 1000 realisations. Top: No inflation with $\delta = 1$. Bottom: Inflation with $\delta = 1.02^2$.

example, (Ott et al., 2004; Fisher et al., 2005; Harlim and Majda, 2010), among many others. We use $D = 40$ modes and set the forcing to $F = 8$, parameters corresponding to a strongly chaotic regime (Lorenz, 1996). For these parameters one unit of time corresponds to 5 days in the Earth’s atmosphere as calculated by calibrating the e -folding time of the asymptotic growth rate of the most unstable Lyapunov mode with a time scale of 2.1 days (Lorenz, 1996). Assuming the length of a midlatitude belt to be approximately 30,000 km, the discretisation of the Earth’s circumference into $D = 40$ grid points corresponds to a spacing between adjacent grid points z_m of approximately 750 km, which roughly equals the Rossby radius of deformation at midlatitudes. We estimated the advection velocity to be approximately 10.4 ms^{-1} from simulations, which compares well with typical wind velocities in the midlatitudes.

We will investigate the effect of using the VLKF on improving the analysis skill when compared to a standard ETKF, and on stabilising the filter and avoiding blow-up as discussed in (Ott et al., 2004; Kepert, 2004; Harlim and Majda, 2010). We perform twin experiments using a $k = 41$ -member ETKF and VLKF with the same truth time series, the same set of observations and the same initial ensemble. We have chosen an ensemble with $k > D$ in order to eliminate the effect that a finite-size ensemble can only fit as many observations as the number of its ensemble members (Lorenz, 2003). Here we want to focus on the effect of limiting the variance.

The system (3.21) is integrated using the implicit mid-point rule (see for example Leimkuhler and Reich (2005)) to a time $T = 30$ with a time step $dt = 1/240$. The total integration time corresponds to an equivalent of 150 days, and the integration timestep dt corresponds to half an hour. We measured the approximate climatic mean μ_{clim} and variance σ_{clim}^2 via a long time integration over a time interval of $T = 2000$, which corresponds to approximately 27.5 years. Because of the symmetry of the system (3.21), the mean and the standard deviation are the same for all variables z_m , and are measured to be $\mu_{\text{clim}} = 2.34$ and $\sigma_{\text{clim}} = 3.63$ respectively.

We study the performance of the filter and its dependence on the observation interval Δt_{obs} as well as the proportion of the system observed $1/N_{\text{obs}}$. Here $N_{\text{obs}} = 2$ means only every second variable is observed, $N_{\text{obs}} = 4$ only every fourth, and so on. To

quantify the improvement of the VLKF over the ETKF we measure the site-averaged RMS error

$$\mathcal{E} = \sqrt{\left\langle \frac{1}{LD_G} \sum_{i=1}^L \|\mathbf{z}_a(t_i) - \mathbf{z}_t(t_i)\|_{\mathbf{G}}^2 \right\rangle} \quad (3.22)$$

between the truth $\mathbf{z}_t$ and the analysis mean $\mathbf{z}_a$, where the average is taken over 500 independent realisations of $L = \lfloor T/\Delta t_{\text{obs}} \rfloor$ filter cycles using different truths with different initial conditions and observations and D_G denotes the number of variables involved. Similarly to the linear toy model, we introduce the norm $\|\mathbf{a}\|_{\mathbf{G}}^2 = \mathbf{a}^T \mathbf{G} \mathbf{a}$ to investigate the error over all variables $\mathcal{E}$ using $\mathbf{G} = \mathbf{I}$, the error $\mathcal{E}_o$ distributed over the observed variables only using $\mathbf{G} = \mathbf{H}^T \mathbf{H}$ and the error $\mathcal{E}_{uo}$ distributed over the unobserved variables using $\mathbf{G} = \mathbf{h}^T \mathbf{h}$.

The initial ensemble at $t = 0$ is drawn from an ensemble with standard deviation σ_{clim} ; the filter is then subsequently spun up for 200 analysis cycles to ensure statistical stationarity. Figure 3.5 demonstrates that the filter has relaxed sufficiently after this many cycles; shown are time series of the maximal singular values of $\mathbf{H}\mathbf{P}_a\mathbf{H}^T$ and $\mathbf{H}\mathbf{P}_f\mathbf{H}^T$ (top) and of $\mathbf{h}\mathbf{P}_a\mathbf{h}^T$ and $\mathbf{h}\mathbf{P}_f\mathbf{h}^T$ (bottom) for a long run of the ETKF, initialised by an ensemble with standard deviation σ_{clim} . Apart from occasional spikes of the singular values which persist indefinitely the filtered ensemble relaxes within less than 100 cycles.

We have used a constant global multiplicative inflation factor as in (2.21) of $\delta = 1.05$ for both filters. We note that the optimal inflation factor at which the RMS error $\mathcal{E}$ is minimal is different for the VLKF and ETKF. For $\Delta t_{\text{obs}} = 5$ hours and $N_{\text{obs}} = 4$ we find that $\delta = 1.06$ produces minimal RMS errors for the VLKF and $\delta = 1.04$ produces minimal RMS errors for the ETKF. For $\delta < 1.04$ filter divergence occurs with the ETKF, so we choose $\delta = 1.05$ as a compromise between controlling filter divergence and minimising the RMS errors of the analysis. This is illustrated in Figures 3.6 which were obtained by averaging over 30 realisations. We remark that in an operational meteorological data assimilation application it would be necessary to perform a δ -optimisation independently for different observational networks and forecast intervals, and our approach would not be sufficient. However, we wish to focus only on the comparing the inclusion of the weak constraint in VLKF so will not tune the inflation factor further.

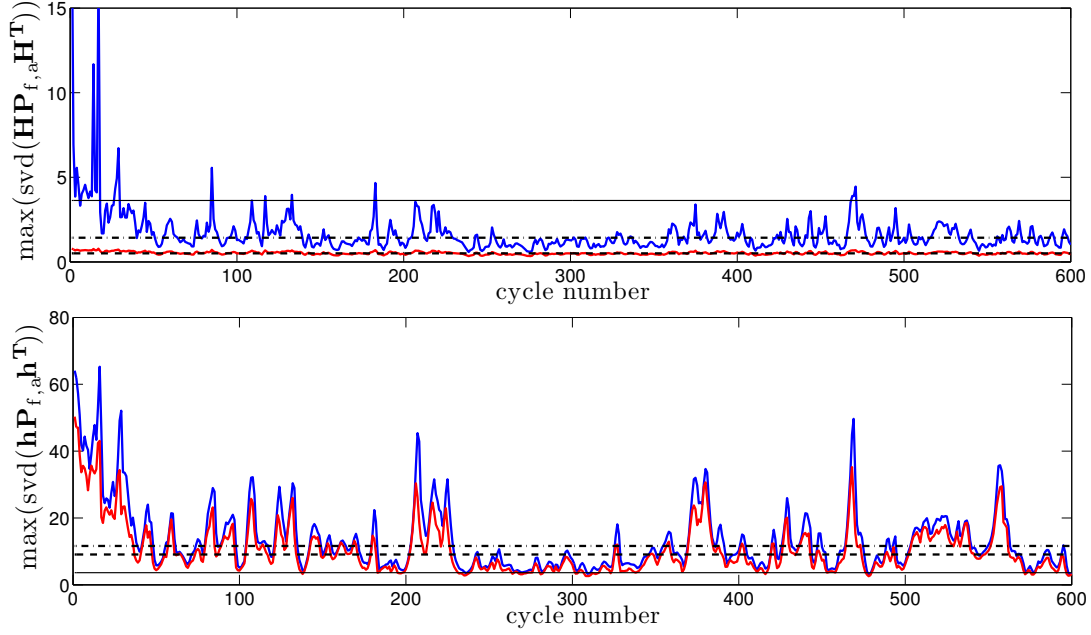


Figure 3.5.: Time series of the maximal svds of $\mathbf{P}_f$ (blue) and $\mathbf{P}_a$ (red) as projected onto the observables (top) and pseudo-observables (bottom), for $N_{\text{obs}} = 4$ and $\Delta t_{\text{obs}} = 6$ hours. The filter was initialised with an ensemble of standard deviation σ_{clim} (solid black line). The dashed lines represent the time-averaged mean of $\max(\text{svd}(\mathbf{H}\mathbf{P}_a\mathbf{H}^T))$ and $\max(\text{svd}(\mathbf{h}\mathbf{P}_a\mathbf{h}^T))$, the dot-dashed lines represent the time-averaged mean of $\max(\text{svd}(\mathbf{H}\mathbf{P}_f\mathbf{H}^T))$ and $\max(\text{svd}(\mathbf{h}\mathbf{P}_f\mathbf{h}^T))$.

We assume Gaussian observational noise of the order of 25% of the climatological standard deviation σ_{clim} , and set the observational error covariance matrix $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$. We find that for larger observational noise levels the variance limiting correction (3.7) is used more frequently, in accordance with our finding in the previous section for the toy model.

3.4.1 Improving analysis skill

We will study under what circumstances the VLKF improves on the analysis when compared with the ETKF, and how large this improvement can be. However, before doing so we remark that the VLKF algorithm as presented in Section 3.2 is not restricted

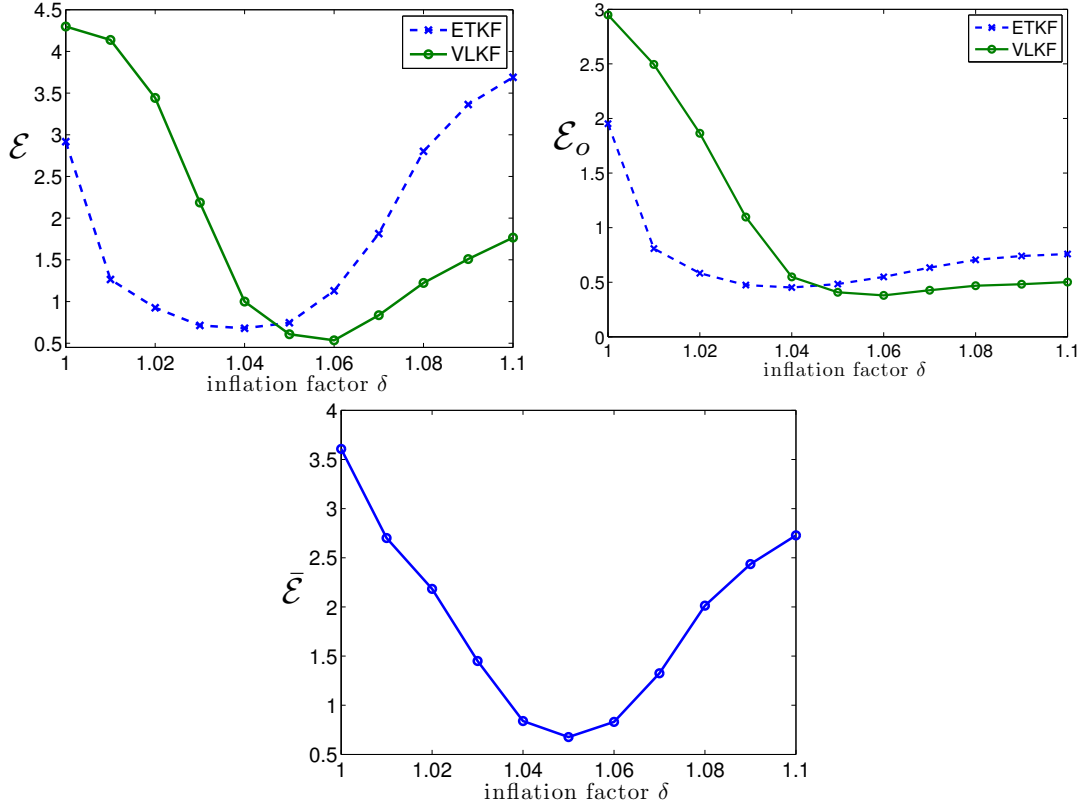


Figure 3.6.: RMS error for ETKF (dashed, crosses) and VLKF (solid, circles) as a function of the inflation factor δ . Top left: Calculated over all variables. Top right: Calculated over the observed variables only. Bottom: Combined average error $\bar{\mathcal{E}} = (\mathcal{E}^{\text{ETKF}} + \mathcal{E}^{\text{VLKF}})/2$ of ETKF and VLKF, over observables and pseudo-observables. We used $\Delta t_{\text{obs}} = 6$ hours and $N_{\text{obs}} = 4$.

to ensemble Kalman filters but can also be used to modify the extended Kalman filter (EKF). Unfortunately, we find that due to the strong nonlinearities in the Lorenz-96 model the EKF does not yield satisfactory results, even in the variance limiting formulation. This is demonstrated in Figure 3.7 which shows a sample analysis using the VLKF with forecast error covariances propagated as in the EKF with $N_{\text{obs}} = 5$, $\Delta t_{\text{obs}} = 18$ hours and $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$ for an arbitrary unobserved and arbitrary observed component of the Lorenz-96 model. Even using the VLKF, the analysis tracks the truth poorly. Using the standard EKF, the filter suffers from filter divergence, an issue which is discussed for the EnKF in the next subsection.

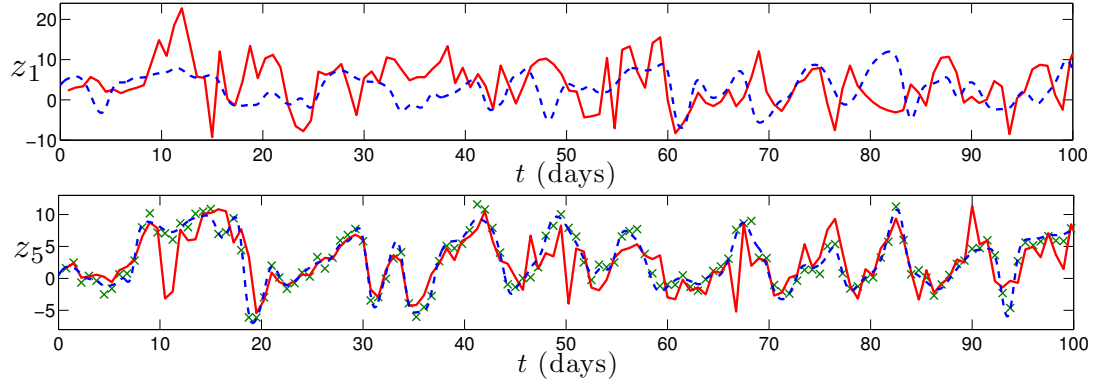


Figure 3.7.: Sample VLKF analysis (continuous red line) for observed z_5 (bottom panel) and unobserved z_1 (top panel) component. The blue dashed line is the truth, the green crosses are observations. Parameters used were $N_{\text{obs}} = 5$, $\Delta t_{\text{obs}} = 18$ hours and $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$.

Henceforth we will concentrate on the analysis skill for the VLKF as compared to the ETKF. Figure 3.8 shows a sample analysis using the ETKF with $N_{\text{obs}} = 5$, $\Delta t_{\text{obs}} = 18$ hours and $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$ for an arbitrary unobserved component (top panel) and an arbitrary observed component (bottom panel) of the Lorenz-96 model. While the figure shows that the analysis tracks the truth reasonably well for the observed component, the analysis is quite poor for the unobserved component. Substantial improvements are seen for the VLKF when we incorporate information about the variance of the unobserved pseudo-observables, as can be seen in Figure 3.9. We set the mean and the variance of the pseudo-observables to be the climatic mean and variance, $\mathbf{a}_{\text{clim}} = \mu_{\text{clim}} \mathbf{e}$ and $\mathbf{A}_{\text{clim}} = \sigma_{\text{clim}}^2 \mathbf{I}$ to filter the same truth with the same observations as used to produce Figure 3.8. For these parameters (and in this realisation) the quality of the analysis in both the observed and unobserved components is improved.

As for the linear toy model (3.8), finite ensemble sizes exacerbate the overestimation of error covariances. In Figure 3.10 the maximal singular value of $\mathbf{hP}_a \mathbf{h}^T$, averaged over 150 realisations, is shown for the ETKF as a function of ensemble size k . Again we use no inflation, i.e. $\delta = 1$, in order to focus on the effect of finite ensemble sizes. The projected covariance clearly decreases for larger ensemble sizes. However, here in the limit as $k \rightarrow \infty$ the maximal singular value of $\mathbf{hP}_a \mathbf{h}^T$ underestimates the climatic variance $\sigma_{\text{clim}}^2 = 13.18$.

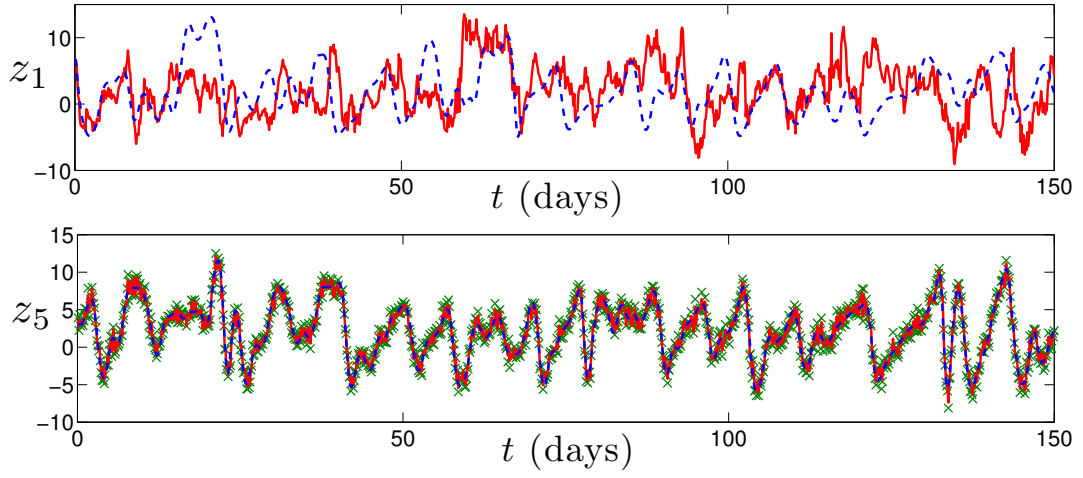


Figure 3.8.: Sample ETKF analysis (continuous red line) for observed z_5 (bottom panel) and unobserved z_1 (top panel) component. The blue dashed line is the truth, the green crosses are observations. Parameters used were $N_{\text{obs}} = 5$, $\Delta t_{\text{obs}} = 18$ hours and $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$.

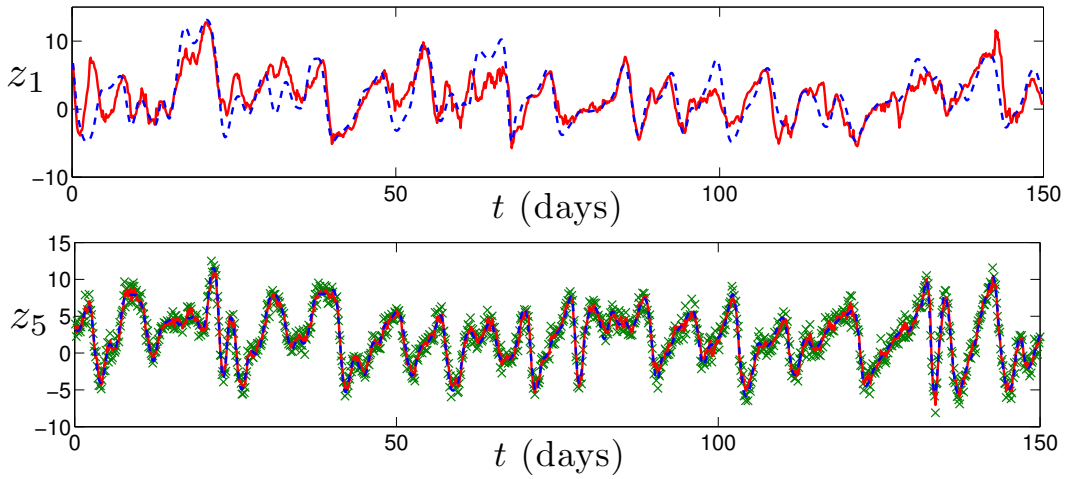


Figure 3.9.: Sample VLKF analysis (continuous green line) for observed z_5 (bottom panel) and unobserved z_1 (top panel) component. The blue dashed line is the truth, the red crosses are observations. Parameters as in Figure 3.8.

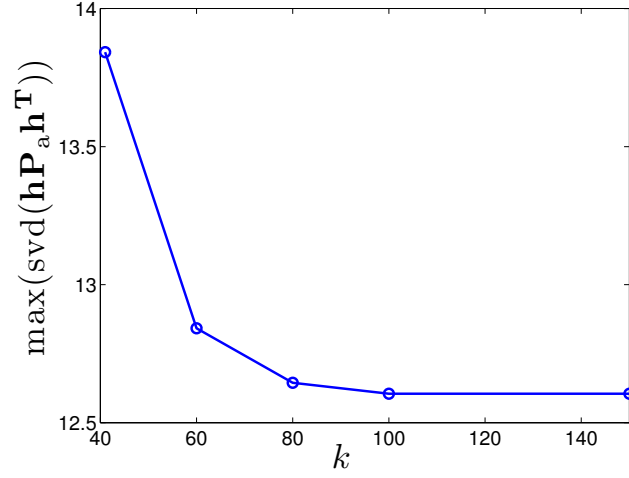


Figure 3.10.: Average maximal singular value of $\mathbf{hP}_a \mathbf{h}^T$ as a function of ensemble size k for the Lorenz-96 model (3.21), using standard ETKF without inflation. All other parameters are as in Figure 4. We used 150 realisations for the averaging.

In Tables 3.1 we display $\mathcal{E}$ for the ETKF and VLKF respectively, as a function of N_{obs} and Δt_{obs} . We performed 500 independent realisations to ensure statistical significance of the results. The increased RMS error for larger observation intervals Δt_{obs} can be linked to the increased variance of the chaotic nonlinear dynamics generated during longer integration times between analyses. Figure 3.13 shows the average proportional improvement of the VLKF over ETKF, obtained from the values of Tables 3.1. The figure shows that the skill improvement is greatest when the system is observed frequently. For large observation intervals Δt_{obs} the ETKF and VLKF yield very similar RMS errors. We have checked that for large observation intervals Δt_{obs} both filters still produce analyses which track the truth reasonably well, as illustrated in Figures 3.11 and 3.12.

Note that the observation intervals Δt_{obs} considered here are all much smaller than the e -folding time of 2.1 days. The most significant improvement occurs for small observation intervals Δt_{obs} , and when the system is observed on a sparse observational network with $N_{\text{obs}} > 2$. An interesting feature is that there is generally less skill improvement for $N_{\text{obs}} = 6$ than for $N_{\text{obs}} = 5$, indicating that as the availability of observations decreases the ETKF analysis becomes more like a random variable drawn from the climatology and so the climatological weak constraint provides less useful

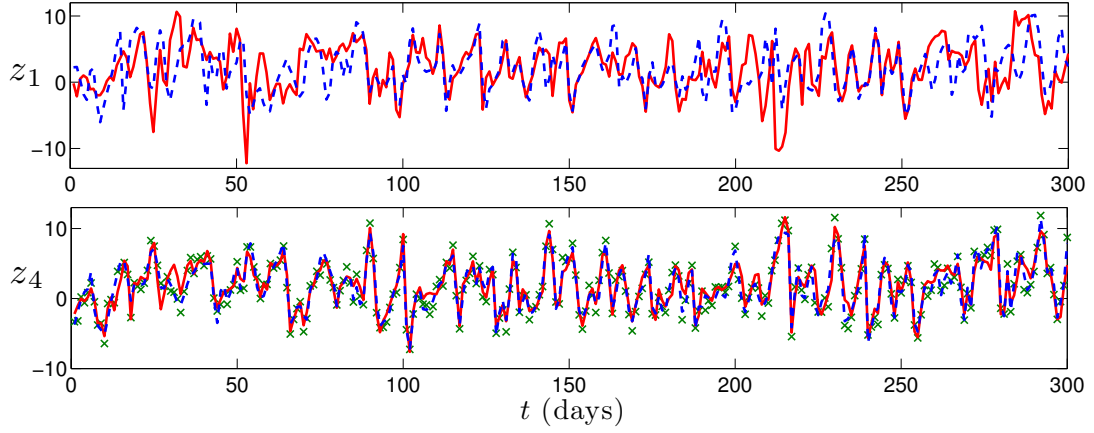


Figure 3.11.: Sample ETKF analysis (continuous red line) for using a long observation interval $\Delta t_{\text{obs}} = 24$ hours for an observed component z_4 (bottom panel) and unobserved component z_1 (top panel). The blue dashed line is the truth, the green crosses are observations. Parameters used were $N_{\text{obs}} = 4$ and $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$.

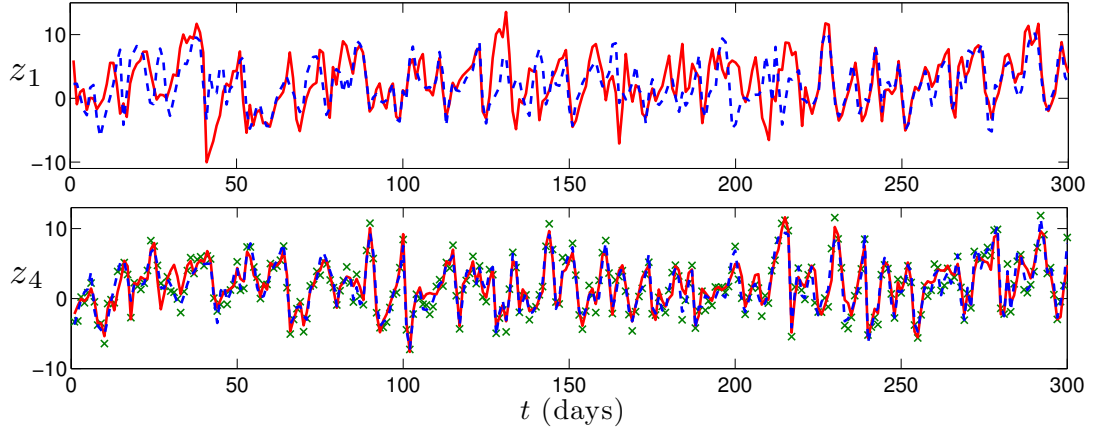


Figure 3.12.: Sample VLKF analysis for using a long observation interval $\Delta t_{\text{obs}} = 24$ hours for an observed component z_4 (bottom panel) and unobserved component z_1 (top panel). Colours and parameters are as in Figure 3.11.

information. The dependence of the skill $\mathcal{S}$ upon the observation interval is consistent with our analytical findings in Section 3.3.

Table 3.1.: RMS errors for ETKF (upper table) and VLKF (bottom table), averaged over 500 realisations, and with $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$ as observational noise.

N_o	6	4.40	3.64	3.42	3.32	3.29	3.30	3.30	3.28	3.26	3.26
	5	4.08	2.88	2.70	2.83	3.02	3.07	3.17	3.21	3.19	3.20
	4	2.42	1.17	1.35	1.72	2.18	2.37	2.62	2.84	2.98	3.06
	3	0.49	0.51	0.60	0.71	0.89	1.11	1.38	1.68	2.02	2.25
	2	0.31	0.34	0.38	0.43	0.49	0.55	0.66	0.75	0.90	1.13
	1	0.19	0.21	0.24	0.26	0.29	0.31	0.33	0.36	0.39	0.44
		0.025	0.05	0.075	0.1	0.125	0.15	0.175	0.2	0.225	0.25
		3 h	6 h	9 h	12 h	15 h	18 h	21 h	24 h	27 h	30 h
		Δt_{obs}									
N_o	6	3.20	3.09	3.10	3.15	3.20	3.22	3.27	3.27	3.26	3.27
	5	2.73	2.28	2.51	2.70	2.89	3.03	3.07	3.14	3.15	3.15
	4	1.30	1.03	1.28	1.66	2.04	2.29	2.55	2.70	2.88	2.96
	3	0.48	0.51	0.59	0.70	0.87	1.07	1.39	1.71	1.95	2.21
	2	0.31	0.34	0.38	0.44	0.50	0.56	0.64	0.77	0.95	1.14
	1	0.19	0.21	0.24	0.26	0.29	0.31	0.33	0.36	0.39	0.44
		0.025	0.05	0.075	0.1	0.125	0.15	0.175	0.2	0.225	0.25
		3 h	6 h	9 h	12 h	15 h	18 h	21 h	24 h	27 h	30 h
		Δt_{obs}									

We have tested that the increase in skill depicted in Figure 3.13 is insensitive to incomplete knowledge of the statistical properties of the pseudo-observables by simultaneously perturbing $\mathbf{A}_{\text{clim}}$ and $\mathbf{a}_{\text{clim}}$ and then monitoring the change in RMS error. We performed experiments where we drew $\mathbf{A}_{\text{clim}}$ and $\mathbf{a}_{\text{clim}}$ independently from uniform distributions $(0.9 \mathbf{A}_{\text{clim}}, 1.1 \mathbf{A}_{\text{clim}})$ and $(0.9 \mathbf{a}_{\text{clim}}, 1.1 \mathbf{a}_{\text{clim}})$. We found that for parameters $N_{\text{obs}} = 2, 4, 6$, $\eta = 0.05, 0.25, 0.5$ (with η measuring the amount of the climatic variance used through $\mathbf{R}_o = (\eta \sigma_{\text{clim}})^2 \mathbf{I}$), and $\Delta t_{\text{obs}} = 3, 6$ and 30 hours over 100 realisations there was on average no more than 7% difference in the analysis mean and the singular values of the covariance matrices to those for a control run where $\mathbf{A}_{\text{clim}} = \sigma_{\text{clim}}^2 \mathbf{I}$ and $\mathbf{a}_{\text{clim}} = \mu_{\text{clim}} \mathbf{e}$ were used.

An interesting question is how the proportional skill improvement is distributed over the observed and unobserved variables, as illustrated in Figures 3.14 and 3.15. In

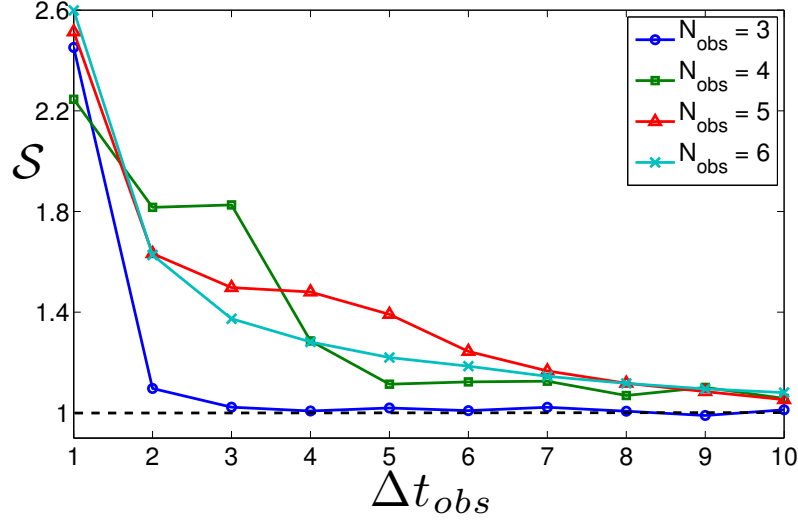


Figure 3.13.: Proportional skill improvement of VLKF over ETKF as a function of the observation interval Δt_{obs} for different values of N_{obs} , with observational noise $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$. A total of 500 realisations were used to perform the ensemble average in the RMS errors $\mathcal{E}$ using (3.22) for ETKF and VLKF. Δt_{obs} is measured in hours.

Figure 3.14 we show the proportional skill improvement $\mathcal{S}$ for the observed variables and the pseudo-observables, respectively. In Figure 3.15 we show the RMS errors $\mathcal{E}_o$ and $\mathcal{E}_{uo}$ for the ETKF and VLKF for observables and pseudo-observables respectively. Figure 3.14 shows that the skill improvement is larger for the pseudo-observables than for the observables. This is to be expected, as it is specifically the variance of the pseudo-observables that the VLKF constrains. It is shown that the skill improvement is better for the pseudo-observables than for the observables for all observation intervals Δt_{obs} . In contrast, VLKF exhibits an improved skill for the observed variables either for small observation intervals for all values of N_{obs} or for all (sufficiently small) observation intervals when $N_{\text{obs}} > 4$. We have checked that the analysis still tracks the truth reasonably well for these sparse observational networks, and the discrepancy with the ETKF is not due to the analysis not tracking the truth anymore. As expected, the RMS error asymptotes for large observation intervals Δt_{obs} (not shown) to the standard deviation of the observational noise $0.25\sigma_{\text{clim}} \approx 0.91$ for the observables, and to the climatic standard deviation $\sigma_{\text{clim}} = 3.63$ for the pseudo-observables (not shown), albeit slightly reduced for small values of N_{obs} due to the impact of the surrounding observed variables (see Figure 3.16).

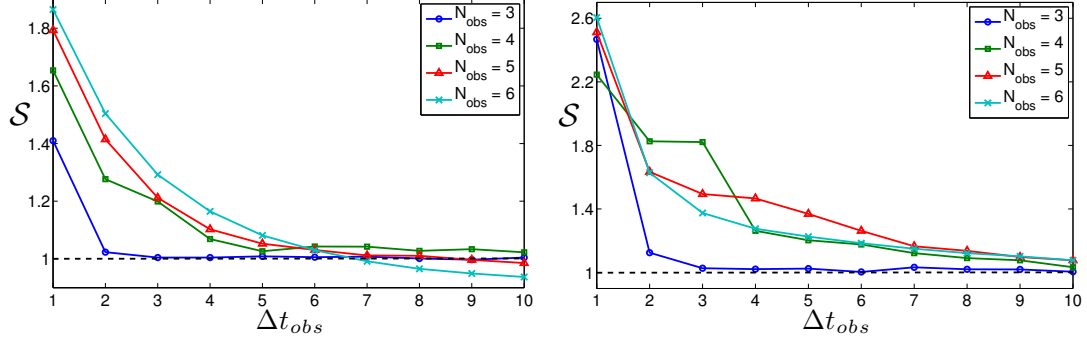


Figure 3.14.: Proportional skill improvement of VLKF over ETKF as a function of the observation interval Δt_{obs} for different values of N_{obs} . The RMS error $\mathcal{E}$ is calculated using only the observed variables (left) or only the pseudo-observables (right). Δt_{obs} is measured in hours. Parameters as in Figure 3.13.

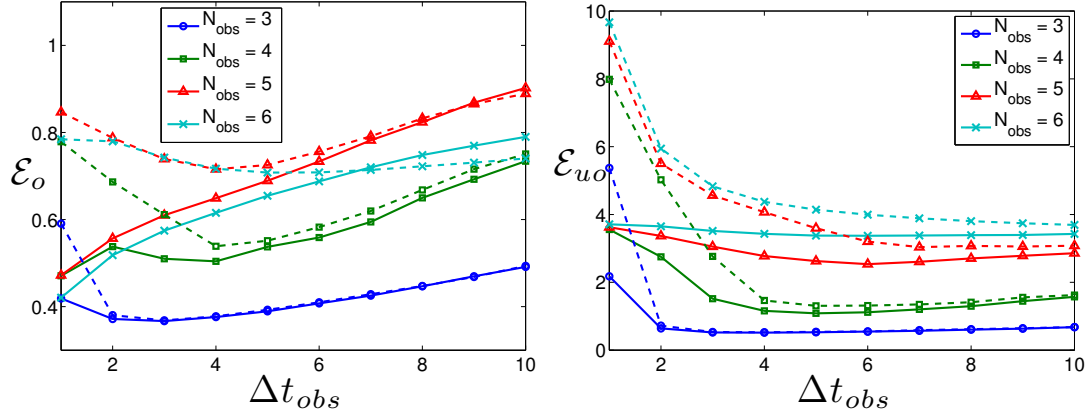


Figure 3.15.: Observed and unobserved RMS errors $\mathcal{E}_o$ (left) and $\mathcal{E}_{uo}$ (right) for the VLKF (solid lines) and ETKF (dashed lines) with $\mathbf{R}_o = (0.25\sigma_{clim})^2 \mathbf{I}$. Δt_{obs} is measured in hours. Parameters as in Figure 3.13.

Note that there is an order of magnitude difference between the RMS errors for the observables and the pseudo-observables for large N_{obs} (cf. Figures 3.15). This suggests that the information provided by the observations does not travel far away from the observational sites. However, the nonlinear coupling in the Lorenz-96 system (3.21) allows for information of the observed components to influence the error statistics of the unobserved components. Therefore the RMS error for the pseudo-observables adjacent to observables are better than those far away from observables. Moreover, the specific structure of the nonlinearity introduces a translational symmetry-breaking (one may think of the nonlinearity as a finite difference approximation of an advection term $\mathbf{z}\mathbf{z}_x$), which causes those pseudo-observables to the right of an observable to have a more reduced RMS error than those to the left of an observable. This is illustrated in Figure 3.16 where the RMS error is shown for each site when just one variable is observed. The advective time scale of the Lorenz-96 system is much larger than Δt_{obs} , which explains why the skill is not equally distributed over the sites, and why for large values of N_{obs} we observe a large difference between the site-averaged skills of the observed and unobserved variables. We note here that the RMS error of the pseudo-observables is larger than the climatic standard deviation σ_{clim} . This is because we have used a large $N_{\text{obs}} = 39$; as can be seen in Figure 3.15 (right), when using the ETKF with $N_{\text{obs}} > 5$ the RMS error for the pseudo-observables approaches σ_{clim} from above as $\Delta t_{\text{obs}} \rightarrow \infty$, and more slowly for larger N_{obs} . As such, for $N_{\text{obs}} = 39$ we would need to use a very large Δt_{obs} to have $\mathcal{E} \rightarrow \sigma_{\text{clim}}$ for the pseudo-observables. We have checked that $\mathcal{E}$ decreases for $\Delta t_{\text{obs}} > 10$ hours, and expect that $\mathcal{E} \rightarrow \sigma_{\text{clim}}$ as $\Delta t_{\text{obs}} \rightarrow \infty$ as in Figure 3.15.

Finally, in Figure 3.17 we show how the RMS error behaves as a function of the observational noise level. We see that for $N_{\text{obs}} = 4$ the VLKF always has a smaller RMS error than the ETKF. This again confirms the results from our analysis of the toy model in Section 3.3, that the VLKF yields best performance for small observation intervals Δt_{obs} and for large noise levels. For large observation intervals the ETKF and VLKF perform equally well, since the chaotic model dynamics will cause the ensemble to acquire the climatic variance during the forecast interval.

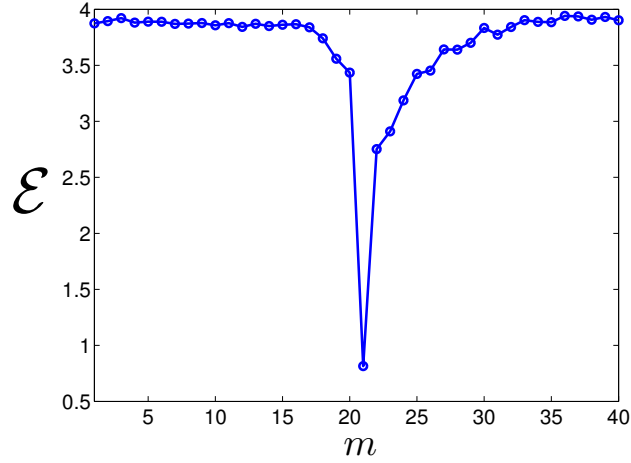


Figure 3.16.: RMS error $\mathcal{E}$ for each variable z_m as a function of the lattice site m . Only one observable was used at $m = 21$ in a standard ETKF. We used $\Delta t_{\text{obs}} = 10$ hours and $\mathbf{R}_o = (0.25 \sigma_{\text{clim}}^2) \mathbf{I}$. The results are averaged over 100 realisations.

3.4.2 Controlling filter divergence

Ott et al. (2004) observed that if not all variables z_m are observed, error covariances become rank-deficient and the analysis mean provided by the EnKF does not track the truth well. The authors suggested that the sparsity of observations leads to an inhomogeneous background error, which causes an underestimation of the error covariance. Harlim and Majda (2010) reported a more severe phenomenon of divergence of solutions towards infinity in finite time, which they called *catastrophic* filter divergence. We study here this catastrophic divergence (as opposed to the filter divergence of Ott et al. (2004) where the analysis mean simply does not track the truth well, taking large excursions away from the climatological mean but still remaining bounded for all times) and its dependence on the observation interval Δt_{obs} and proportion of the system observed $1/N_{\text{obs}}$. We note that blow-up divergence occurs only in the case of sufficiently small observational noise and moderate values of Δt_{obs} . Once Δt_{obs} is sufficiently large (in fact, larger than the Lyapunov e -folding time of 2.1 days) we notice that no catastrophic divergence occurs, independent of N_{obs} . This is likely because for large observation intervals the ensemble always acquires large variance through the nonlinear ensemble propagation. We prescribe Gaussian observational noise of the

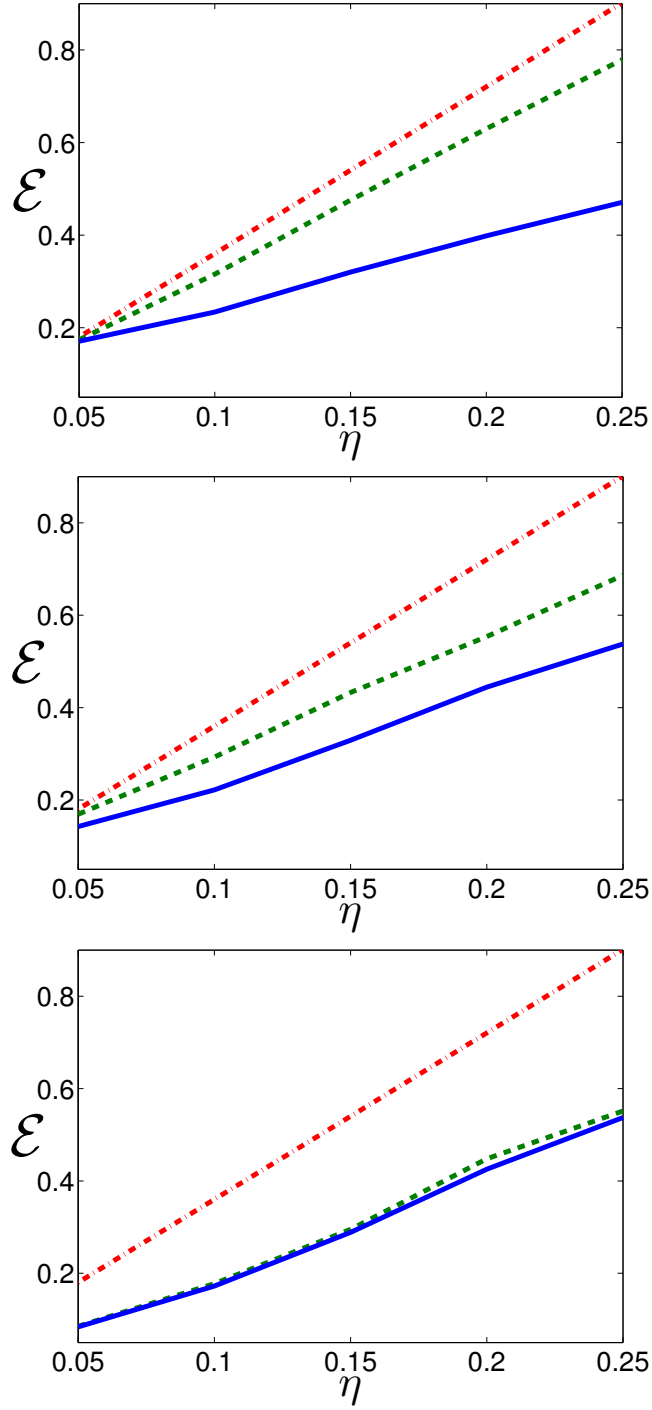


Figure 3.17.: RMS error $\mathcal{E}$ of VLKF (blue solid lines) and ETKF (green dashed lines), as a function of the observational noise, measured here by η defined via $\mathbf{R}_o = (\eta \sigma_{\text{clim}})^2 \mathbf{I}$. The red dotted-dashed line indicates the RMS error if only observations were used. We show results from top to bottom for observation intervals $\Delta t_{\text{obs}} = 1$ hour, $\Delta t_{\text{obs}} = 2$ hours and $\Delta t_{\text{obs}} = 5$ hours. $N_{\text{obs}} = 4$ was used and 1000 realisations were carried out.

order of 5% of the climatological standard deviation σ_{clim} , and set the observational error covariance matrix to $\mathbf{R}_o = (0.05 \sigma_{\text{clim}})^2 \mathbf{I}$. The initial ensemble at $t = 0$ is again drawn from an ensemble with variance σ_{clim}^2 .

To study the performance of the VLKF when blow-up occurs in the ETKF we count the number N_b of blow-ups that occur before a total of 100 realisations have terminated without blow-up. The proportions of blow-ups for the respective filters is then given by $N_b / (N_b + 100)$. We tabulate this proportion in Tables 3.2 for the ETKF and VLKF respectively and the proportional improvement in Table 3.3. The ‘ $\times$ ’ symbols in the table represent cases where no successful realisations could be obtained due to blow-up.

Both filters suffer from severe filter instability for $N_{\text{obs}} = 6$, i.e. for very sparse observational networks, at small observation intervals Δt_{obs} . No blow-up occurs for either filter when every variable is observed. Note the reduction in occurrences of blow-ups for large observation intervals Δt_{obs} as discussed above. We have checked that for all N_{obs} there is no blow-up for the ETKF (and VLKF) for sufficiently large Δt_{obs} (not shown); the larger N_{obs} is the smaller the upper bound of Δt_{obs} such that no blow-ups occur. Collapse is most prominent for the ETKF (and for the VLKF, but to a much lesser extent) for larger values of N_{obs} and at intermediate observation intervals, which depend on N_{obs} . Tables 3.2 and 3.3 clearly show that incorporating information on the pseudo-observables strongly increases the stability of the filter and suppresses blow-up.

Figure 3.18 illustrates how this blow-up occurs in the ETKF for small observational noise. The blow-up is caused by an alignment of the ensemble, as illustrated for one realisation in the bottom left panel which shows the condition number $s_{\text{max}}/s_{\text{min}}$ of the forecast error covariance $\mathbf{P}_f$, where s_{max} and s_{min} are the maximum and minimum singular values of $\mathbf{P}_f$ respectively. Preceding the catastrophic ensemble divergence which occurs here after 20 analysis cycles is an ensemble collapse which occurs after 5 analysis cycles. Interestingly, the ensemble does not blow up immediately after this alignment, but instead retains small variance for almost 10 analysis cycles before catastrophically diverging. This is illustrated in the bottom right panel of Figure 3.18, which shows the maximal singular value s_{max} of $\mathbf{P}_f$. However, we note that after the ensemble collapse the forecast ensemble no longer tracks the truth, as shown in the top panel. Numerical simulations show that the VLKF does not prevent the ensemble

Table 3.2.: Proportion of catastrophically diverging realisations with the ETKF (upper table) and VLKF (lower table). Observational noise with $\mathbf{R}_o = (0.05\sigma_{\text{clim}})^2 \mathbf{I}$ was used.

N_o	6	0.14	×	×	0.98	0.96	0.76	0.32	0.05	0.02	0.01
	5	0.02	0.40	0.67	0.73	0.84	0.89	0.94	0.82	0.49	0.19
	4	0	0.04	0.22	0.29	0.49	0.64	0.77	0.83	0.89	0.82
	3	0	0	0	0.03	0.04	0.11	0.15	0.44	0.58	0.67
	2	0	0	0	0	0	0.01	0	0.01	0.05	0.15
		0.025 3 h	0.05 6 h	0.075 9 h	0.1 12 h	0.125 15 h	0.15 18 h	0.175 21 h	0.2 24 h	0.225 27 h	0.25 30 h
		Δt_{obs}									
N_o	6	0.01	0.42	0.11	0.01	0	0	0	0	0	0
	5	0	0.24	0.36	0.10	0.01	0	0	0	0	0
	4	0	0.03	0.22	0.12	0.06	0.02	0	0	0	0
	3	0	0	0	0.02	0	0.01	0.01	0.01	0	0
	2	0	0	0	0	0	0	0	0	0	0.01
		0.025 3 h	0.05 6 h	0.075 9 h	0.1 12 h	0.125 15 h	0.15 18 h	0.175 21 h	0.2 24 h	0.225 27 h	0.25 30 h
		Δt_{obs}									

Table 3.3.: Proportional improvement of the VLKF over ETKF as calculated as the ratio of the values from Tables 3.2.

N_o	6	14	×	×	98.00	∞	∞	∞	∞	∞	∞
	5	∞	1.67	1.86	7.30	84.00	∞	∞	∞	∞	∞
	4	1	1.33	1.00	2.42	8.17	32.00	∞	∞	∞	∞
	3	1	1	1	1.5	∞	11.00	15.00	44.00	∞	∞
	2	1	1	1	1	1	∞	1	∞	∞	15.00
		0.025 3 h	0.05 6 h	0.075 9 h	0.1 12 h	0.125 15 h	0.15 18 h	0.175 21 h	0.2 24 h	0.225 27 h	0.25 30 h
		Δt_{obs}									

collapse seen here after 5 analysis cycles, but rather controls the later exponential divergence of the ensemble by limiting the ensemble spread. We note however that despite the gain in stability the VLKF has poor skill in the cases when blow-up occurs for the ETKF, because the analyses no longer track the truth well.

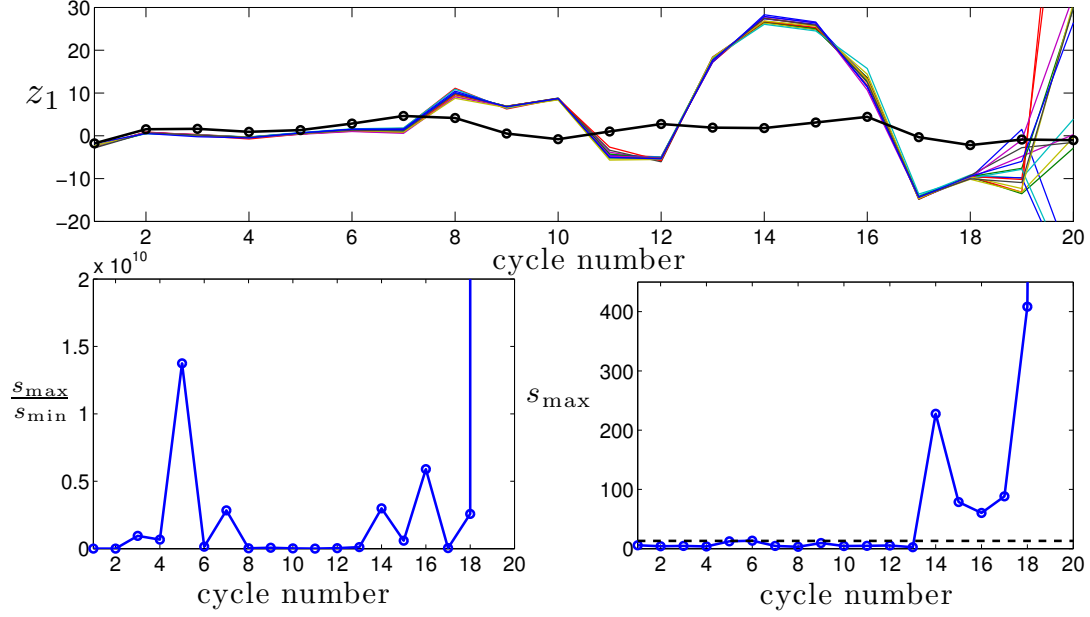


Figure 3.18.: An example of catastrophic filter divergence in the ETKF, for $N_{\text{obs}} = 5$, $\Delta t_{\text{obs}} = 18$ hours and $\eta = 0.15$. Top: Arbitrary unobserved component of the truth $\mathbf{z}_1$ (black, circles) and forecast ensemble (colours), showing filter divergence after 20 analysis cycles. Bottom left: Ratio of maximal to minimal singular values of $\mathbf{P}_f$, showing ensemble collapse after 5 cycles. Bottom right: Maximal singular value $s_{\max}$ of $\mathbf{P}_f$, showing ensemble divergence after 20 analysis cycles. The dashed line shows σ_{clim}^2 .

The fact that incorporating information about the variance of the unobserved variables improves the stability of the filter is in accordance with the interpretation of filter divergence of sparse observational networks provided in (Ott et al., 2004). In this situation the VLKF is clearly acting in a similar fashion to a model error weak constraint and avoiding filter divergence by adding more climatic information in to the error covariances. This aspect of the VLKF and its application to problems with model error will be explored in detail in Chapter 4.

3.5 DISCUSSION

In this chapter we have developed a framework to include information about the variance of unobserved variables in a sparse observational network. The filter is designed to control the overestimation of error covariances encountered in sparse observational networks, and limits the posterior analysis covariance of the unresolved variables to stay below their climatic variance. We have done so in a variational setting and found a relationship between the error covariance of the variance constraint $\mathbf{R}_w$ and the assumed target variance of the unobserved pseudo-observables $\mathbf{A}_{\text{clim}}$.

We have illustrated the beneficial effects of the variance limiting filter in improving the analysis skill when compared to a standard ensemble square root Kalman filter. We expect the variance limiting constraint to improve data assimilation for ensemble Kalman filters when finite size effects of small ensembles overestimate the error covariances, in particular in sparse observational networks. We found that the skill will improve for small observation intervals Δt_{obs} and sufficiently large observational noise. Substantial skill improvement was found for both observables and pseudo-observables. These effects can be understood with a simple linear toy model which allows for an analytical treatment, as well as numerically with a higher-dimensional nonlinear chaotic system. We established numerically that the VLKF reduces the probability of catastrophic filter divergence and improves the stability of the filter when compared to the standard ensemble transform Kalman filter.

The effect of the variance limiting filter to control the unrealistically large error covariances of poorly resolved variables due to finite ensemble sizes may find useful applications. We mention here that the variance constraint is able to adaptively damp the unrealistic excitation of ensemble spread in underresolved spatial regions due to inappropriate uniform inflation. This may be an alternative to recently developed spatially adaptive inflation schemes (Anderson, 2007; Li et al., 2009). In addition, it is known that localisation of covariance matrices in EnKFs leads to imbalance in the analysed fields (see, e.g., Houtekamer and Mitchell (2005); Kepert (2009) for a recent study). Filter localisation typically excites unwanted gravity waves which when uncontrolled can substantially degrade filter performance. One may construct balance constraints as pseudo-observations and thereby potentially reduce this undesired aspect

of covariance localisation. As more specific applications for the VLKF, we mention climate reanalysis and data assimilation for the mesosphere. It would be interesting to see how the proposed variance limiting filter can be used in climate reanalysis schemes to deal with the vertical sparsity of observational data and the sparse observational network in the southern hemisphere in the pre-radiosonde era (see [Whitaker et al. \(2004\)](#)). One would need to establish though whether the historical observation intervals Δt_{obs} are sufficiently small as to allow for skill improvements. Similarly, the VLKF may help to control the dynamically dominant gravity wave activity in the mesosphere as the upper lid is pushed further and further upwards (see for example [Polavarapu et al. \(2005\)](#)). However, a word of caution is required here. In some atmospheric data assimilation problems, it is not at all uncommon to have an ensemble prior variance for certain variables that is significantly larger than the climatological variance, when the atmosphere is locally far away from equilibrium. One relevant example would be in the vicinity of strong fronts over the southern ocean. In such a case, it may not be appropriate to limit the variance to the climatological value.

Indeed, it may be desirable in certain situations to drive the system towards values other than the climatological mean and variance. The variance limitation can be viewed as a simple strategy to numerically stabilise the filtering process. As such it may be more beneficial to choose values differing from climatology which are given by numerical necessity rather than by dynamical and physical considerations. For example, in non-equilibrium situations where the dynamics is locally off the attractor the climatological values are not meaningful. Our approach still provides a means to control the dynamically natural overestimation to numerically stabilise the filtering procedure. However, this would imply careful tuning of the limiting values $\mathbf{A}_{\text{clim}}$ and $\mathbf{a}_{\text{clim}}$.

In this chapter we have studied systems where for sufficiently large observation intervals Δt_{obs} the variables acquire their true climatological mean and variance when the model is run. In particular we have not included model error. In the next chapter we examine the circumstances under which the variance limiting filter can help to control model error when the forecast model produces unrealistically large forecast covariances.

MODEL ERROR INDUCED COVARIANCE OVERESTIMATION: AN APPLICATION OF VARIANCE LIMITING KALMAN FILTERING

4.1 INTRODUCTION

Numerical simulations of geophysical fluids, such as in numerical weather prediction or climate dynamics tend to produce large errors at the grid resolution level, often linked to numerical instabilities. In the context of data assimilation this numerical “noise” can lead to an overestimation of forecast error covariances at small scales. This limits the reliability of the forecast over the forecast window and can produce unrealistically large forecast covariances. The standard approach to deal with numerical instabilities in large systems is to add various forms of artificial dissipation to the model (Durrán, 1999). However, this has the drawback of destroying conservation laws in the system which may be undesirable in long time simulations such as in climate modelling. Artificial viscosity may then lead to severe violations of important conservation laws such as mass conservation (Thuburn, 2008). Moreover, in the context of ensemble data assimilation artificial viscosity diminishes the spread of the forecast ensemble, causing detrimental error covariance underestimation (Houtekamer and Mitchell, 1998; Burgers et al., 1998; Anderson and Anderson, 1999) and the associated problems described in Chapter 2. For example, to control unwanted gravity wave activity severe divergence damping (see for example Skamarock and Klemp (1992)) is often introduced into the equations of motion to stabilise the numerical scheme, leading to an underestimation of the error covariances (Durrán, 1999).

The question we address in this chapter is: can one use numerical forecasting models which are not artificially damped, yet still control the resulting overestimation of forecast error covariances within the data assimilation procedure? This is particularly important in sparse observational networks where for the unobserved variables unrealistically large forecast covariances are not well controlled by observational data (Liu et al., 2008; Whitaker et al., 2009), as was discussed in the previous chapter. The main findings of this chapter will be that as model damping decreases, incorporating climatological information in the data assimilation procedure can improve the analysis skill over a standard ensemble transform Kalman filter (ETKF), however this comes at the expense of less improvement over a “poor man’s” analysis consisting of the observations and climatology only. We also find decreasing skill improvement for increasing observational noise.

In the previous chapter we proposed a variance limiting ensemble Kalman filter (VLKF) which controls covariance overestimation in sparse observational networks in a perfect model scenario. It was shown that for short observational intervals of up to 10 hours the VLKF produces superior skill when compared to the ETKF. Over long observation intervals the VLKF and ETKF ensembles both acquire the correct climatic variance, and so perform equally well. Here we will apply the VLKF in the case of an imperfect, underdamped forecast model, and for longer observation intervals. Unlike in the perfect model case where it was found that over large observation intervals the forecast error covariances did not exceed the climatic variance, when forecasting with underdamped forecast models the error covariances can often be larger than the climatic variance. This can be controlled by the VLKF by incorporating climatological information into the analysis step, leading to an improvement in the analysis skill. The positive effect of controlling unrealistic error covariances is especially present in the case of large observation intervals, where the forecast ensemble will have acquired large spread due to the underlying underdamped dynamics.

We remark that in the perfect model case, when $\mathbf{A}_{\text{clim}}$ and $\mathbf{a}_{\text{clim}}$ are given by the climatology, the forecast model in both the ETKF and VLKF will acquire comparable forecast error covariance $\mathbf{P}_f$ for large observational intervals Δt_{obs} . Hence both filters have approximately the same input parameters prior to the analysis step, and the two filters perform equally well as was shown in Chapter 3. This is not the case for imperfect models with an underdamped forecast model; when the variance limiting

constraint is switched on, the background covariance of the pseudo-observables is set to $\mathbf{A}_{\text{clim}} = \sigma_{\text{clim}}^2 \mathbf{I}$ for the VLKF, much smaller than that for the ETKF when using an underdamped model. Note that this implies that the forecast error covariance is also typically much smaller for the VLKF than the ETKF over short observation intervals.

We will study the effect of incorporating climatological information into data assimilation in the presence of model error using the standard Lorenz-96 model (Lorenz, 1996), where we assume that the truth is evolving to the standard set of parameters. To investigate whether underdamped models can be used as forecast models we then use forecast models with smaller values of the linear damping term in the Lorenz-96 system to generate ensembles with forecast spreads exceeding the true climatology. We will establish that the variance limiting weak constraint may be used as an adaptive alternative to artificial viscosity damping, counteracting model error.

In the next section we introduce the Lorenz-96 model with variable damping and describe its dynamical properties. In Section 4.3 we present results showing under which conditions the VLKF produces better skill than the standard ETKF. We conclude with a discussion and outlook in Section 4.4.

4.2 MODIFIED LORENZ-96 MODEL

Consider the following modified version of the Lorenz-96 model (c.f. (3.21))

$$\frac{dz_m}{dt} = z_{m-1}(z_{m+1} - z_{m-2}) - \gamma z_m + F \quad m = 1, \dots, D \quad (4.1)$$

with $\mathbf{z} = (z_1, \dots, z_D)^T$ and periodic $z_m = z_{m+D}$, where we have introduced a variable linear damping parameter γ . As in Chapter 3 we set the number of degrees of freedom as $D = 40$ and $F = 8$ for the forcing. For $\gamma = 1$, the classical value of the damping in the Lorenz-96 system, this corresponds to a time scaling of 5 days per model time unit, an e -folding time of 2.1 days and a distance between adjacent grid points of 750 km, roughly equalling the Rossby radius of deformation (Lorenz, 1996).

Figure 4.1 illustrates how the climatic mean μ_{clim} and variance σ_{clim}^2 depend on the value of the damping parameter γ . The climatic values are obtained from simulations over long time periods (1000 days). Note that in the Lorenz-96 system all variables z_m share the same statistical behaviour and so the climatic mean and variance are the same for all variables. Furthermore, we show the variances σ^2 when calculated from simulations running only over a finite length of time, such as analysis cycle length. As expected, the variance increases as the dynamics gets less damped for fixed forecast interval, and increase with the simulation length Δt_{obs} for fixed damping parameter γ (asymptoting to the climatic variance).

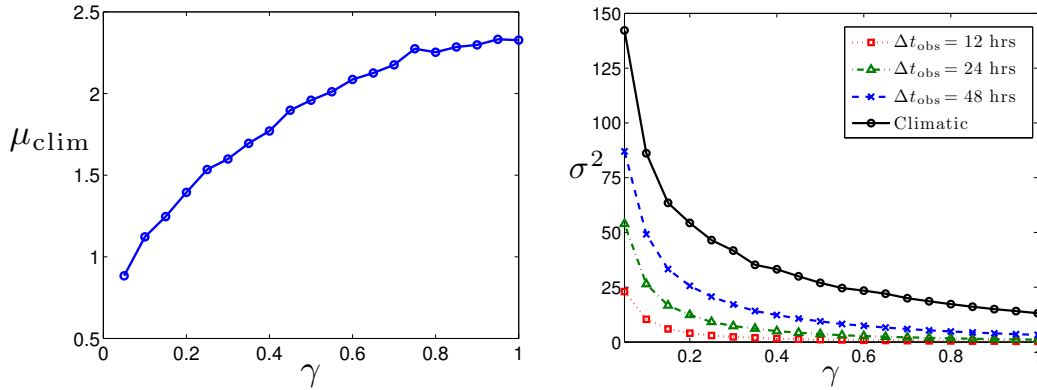


Figure 4.1.: Dependence of the climatic mean (left) and variance (right) on damping parameter γ . As well as the climatic variance (black, circles, solid) we show the average variance calculated over forecast intervals $\Delta t_{\text{obs}} = 12$ hours (red, squares, dotted), $\Delta t_{\text{obs}} = 24$ hours (green, triangles, dash-dotted) and $\Delta t_{\text{obs}} = 48$ hours (blue, crosses, dashed).

Two characteristic time scales of the Lorenz-96 system (4.1) will be important in the following, namely the e -folding time, measured here by the inverse of the maximal Lyapunov exponent λ_{max} and the advection time T_{adv} , which is a measure for the time taken for a front to propagate from one site m to the next $m + 1$. In Figures 4.2 and 4.3 we show how these characteristic time scales vary with γ . Firstly, we show the inverse of the maximal Lyapunov exponent λ_{max} which is a measure for the predictability time of the Lorenz-96 dynamics and correlates with the decorrelation time, as well as the number of positive Lyapunov exponents. For the original Lorenz-96 model with $\gamma = 1$ we reproduce the e -folding time of 50 hours, i.e. 2.1 days. In accordance with the observed increase in variance, the predictability time decreases with decreasing damping. Also, the number of positive Lyapunov exponents increases

with decreasing γ , indicating more unstable modes and therefore less predictability. As a second characteristic time scale we show the advective time T_{adv} as a function of γ in Figure 4.3. Since the nonlinear term in (4.1) can be interpreted as a finite difference discretisation of an advection term zz_x , information from one grid point m is mainly transported to its right neighbours in an asymmetric fashion (c.f. Figure 3.16). The characteristic time scale of this information transport will be important when discussing the influence of the observational network sparsity on the filtering skill. Figure 4.3 shows that this time scale decreases with increasing model error.

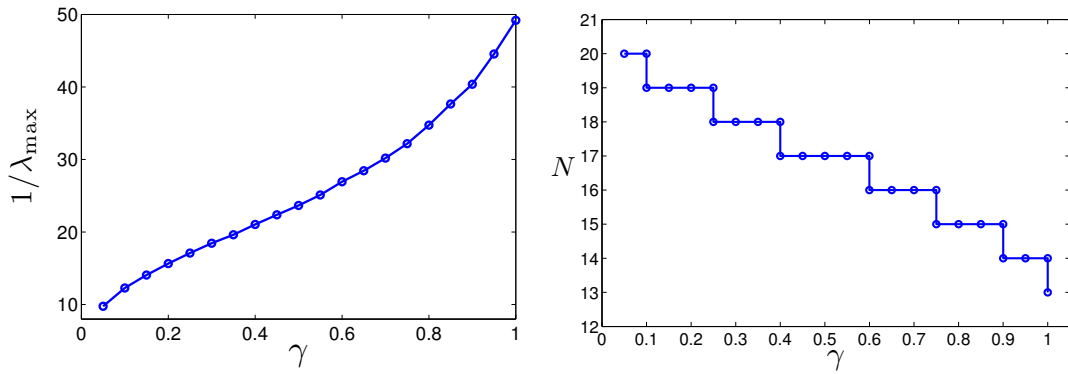


Figure 4.2.: Lyapunov exponents for the modified Lorenz-96 system (4.1) as a function of the damping parameter γ . Left: Inverse of the maximal Lyapunov exponent λ_{max} in hours. Right: Number of positive Lyapunov exponents N .

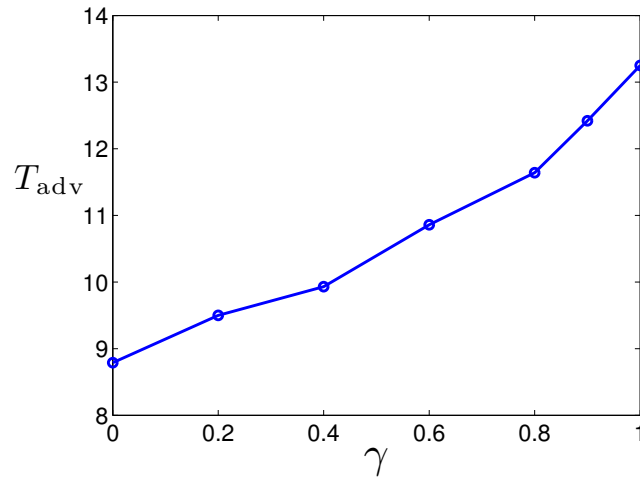


Figure 4.3.: Advection time T_{adv} in hours for the Lorenz-96 system (4.1) as a function of the damping parameter γ .

4.3 NUMERICAL RESULTS

We now investigate when and how the variance limiting Kalman filter can be used to control the increased covariances of an underdamped forecast model. In particular, we study the situation of data assimilation on a sparse observational grid.

We will address two issues. First, we will investigate under what conditions the VLKF as described in Section 3.2 produces better analysis skill than the ETKF. Second, we will investigate when the VLKF produces better skill than a worst-case analysis estimate which takes the observational data and the climatic mean for the observed and the unobserved variables, respectively (a “poor-man’s analysis”). This latter unfortunate situation can occur for example, for moderate observation intervals Δt_{obs} , as we will show. Besides the degree of underdamping γ , the skill will be seen to depend on the sparsity of the observational grid used as well as on the observation interval and the observational noise level.

We generate a truth $\mathbf{z}_t$ from a simulation of the Lorenz-96 system (4.1) with $\gamma = 1$. Observations $\mathbf{y}_o$ are then generated for the observables $\mathbf{H}\mathbf{z}$ with Gaussian error covariance $\mathbf{R}_o = (\eta\sigma_{\text{clim}})^2\mathbf{I}$, and are available after each observation interval Δt_{obs} at every N_{obs} site. The observations are then assimilated in an ensemble filter framework using the Lorenz-96 model (4.1) with damping parameter $\gamma < 1$ as forecast model, mimicking the situation of underdamped forecast models. We perform twin experiments where the same truth and same observations are used for data assimilation with both the ETKF and VLKF. Both filters are run with $k = 41$ ensemble members and the same forecast model (4.1) with $\gamma < 1$, making the forecasts underdamped with respect to the truth $\mathbf{z}_t$. As in Chapter 3, both filters are spun up for 100 cycles to ensure statistical stationarity of the ensembles, and we employ a constant global multiplicative covariance inflation factor of $\delta = 1.05$ in both filters. This value was found to be optimal for $N_{\text{obs}} = 4$ and $\Delta t_{\text{obs}} = 5$ hours, however we remark as in Section 3.4 that in a meteorological application it would be desirable to tune the inflation factor differently for different values of these parameters.

As in Section 3.4 we use an implicit mid-point rule (see for example [Leimkuhler and Reich \(2005\)](#)) to simulate the system (4.1) to a time $T = 60$ time units with a time step

$dt = 1/240$. The total integration time corresponds to 300 days, and the integration timestep dt corresponds to half an hour. The climatic mean and standard deviation over all variables with $\gamma = 1$ are estimated as temporal averages from a long trajectory as $\mu_{\text{clim}} = 2.34$ and $\sigma_{\text{clim}} = 3.63$ respectively, which determines our true climatology.

In Figures 4.4 and 4.5 we show sample analyses using both filters with model error $\gamma = 0.5$ for $\Delta t_{\text{obs}} = 24$ and $\Delta t_{\text{obs}} = 48$ hours, exemplifying moderate and large observation intervals, respectively. We show analyses for an arbitrary unobserved component and an arbitrary observed component of the Lorenz-96 model. Both filters track the truth for the observed components well for both moderate and large observation intervals, with the analysis tending more towards the observations for large Δt_{obs} . However, for the pseudo-observables with moderate observation interval $\Delta t_{\text{obs}} = 24$ hours, the ETKF tracks the truth better than the VLKF despite the inclusion of additional climatological information. For large observation intervals $\Delta t_{\text{obs}} = 48$ hours, neither filter tracks the truth for the pseudo-observables well. However, the additional input of the climatological information into the VLKF leads to an analysis with statistics which better resemble the true climatology, thereby improving the overall RMS errors and skill. It is also clearly seen that the main contribution to the error stems from the pseudo-observables, similar to our results in Chapter 3.

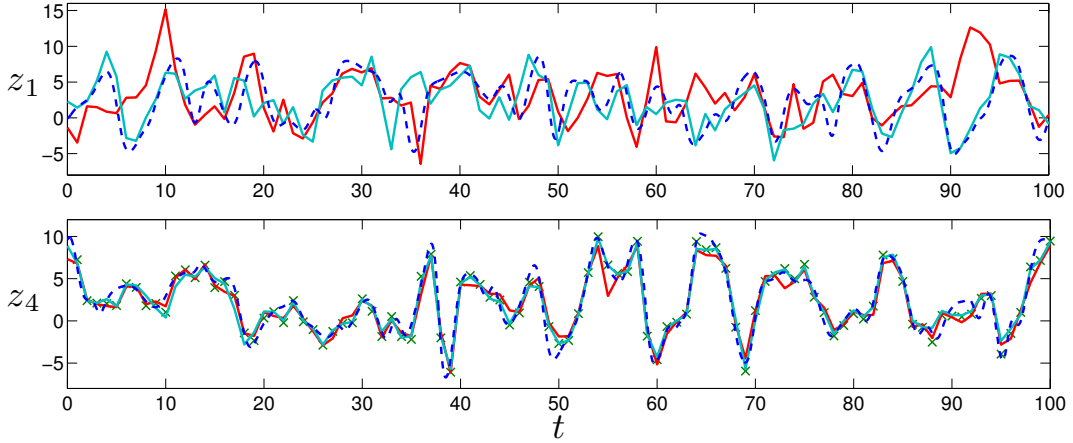


Figure 4.4.: Sample ETKF (blue-green, solid line) and VLKF (red, solid line) analyses over 100 days for observed z_4 (bottom panel) and unobserved z_1 (top panel) component for $\Delta t_{\text{obs}} = 24$ hours and model error $\gamma = 0.5$. The dashed line is the truth, the crosses are observations. Parameters used were $N_{\text{obs}} = 4$ and $\mathbf{R}_0 = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$.

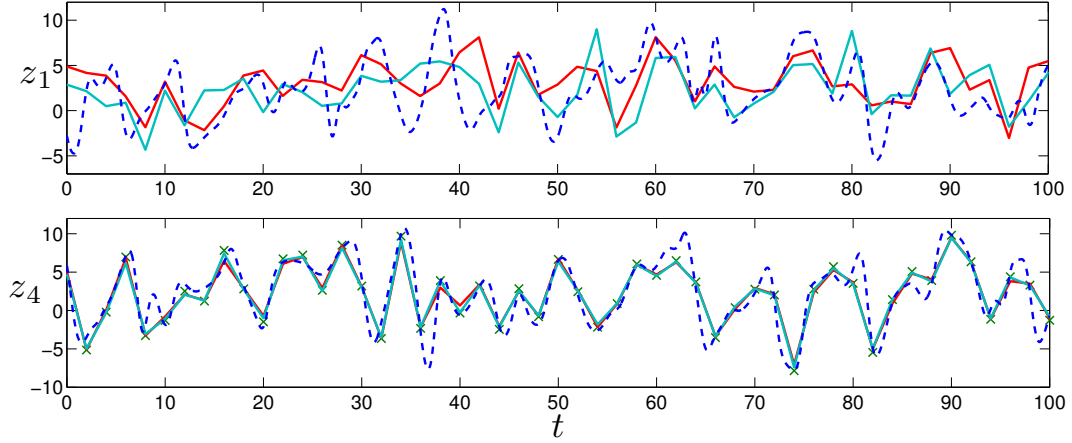


Figure 4.5.: Sample ETKF (blue-green, solid line) and VLKF (red, solid line) analyses over 100 days for observed z_4 (bottom panel) and unobserved z_1 (top panel) component for $\Delta t_{\text{obs}} = 48$ hours. The dashed line is the truth, the crosses are observations. Parameters as in Figure 4.4.

Figures 4.4 and 4.5 suggest that the notion of what constitutes a good performance of a filter depends on the length Δt_{obs} of the observation interval. When aiming for improved performance in the case of moderate observation intervals, one is interested in tracking the truth for both the observables and pseudo-observables. In the case of large observation intervals, one can only expect to track the truth for the observed variables and produce analyses which exhibit the same statistical behaviour as the truth for the pseudo-observables. We remark that for very large observational sparsity N_{obs} tracking cannot be demanded even for small observation intervals; for example with $\Delta t_{\text{obs}} = 6$ hours the analysis of an observational grid with $N_{\text{obs}} = 20$ does not produce tracking analyses for any variable z_m .

In Figure 4.6 we show the average maximal singular values s_H and s_h of the forecast and analysis error covariances projected onto the observation and pseudo-observation subspaces respectively. As expected the covariances increase with increasing model error, as well as with increasing observational sparsity. The variance constraint of the VLKF markedly controls the analysis covariance of the unobserved variables to assume values below the climatic variance. This causes the forecast covariances to be limited as well, albeit at values greater than the climatic variance σ_{clim}^2 . For large sparsity N_{obs} , the analysis covariance of the unobserved variables is set to the climatic variance

for the VLKF, whereas for the ETKF the analysis covariances grow larger than σ_{clim}^2 with increasing sparsity. Interestingly, the VLKF also limits the covariances for the observed variables. As we will see this may have negative effects and may lead to an underdispersive ensemble which spoils the VLKF analysis for the observed variables.

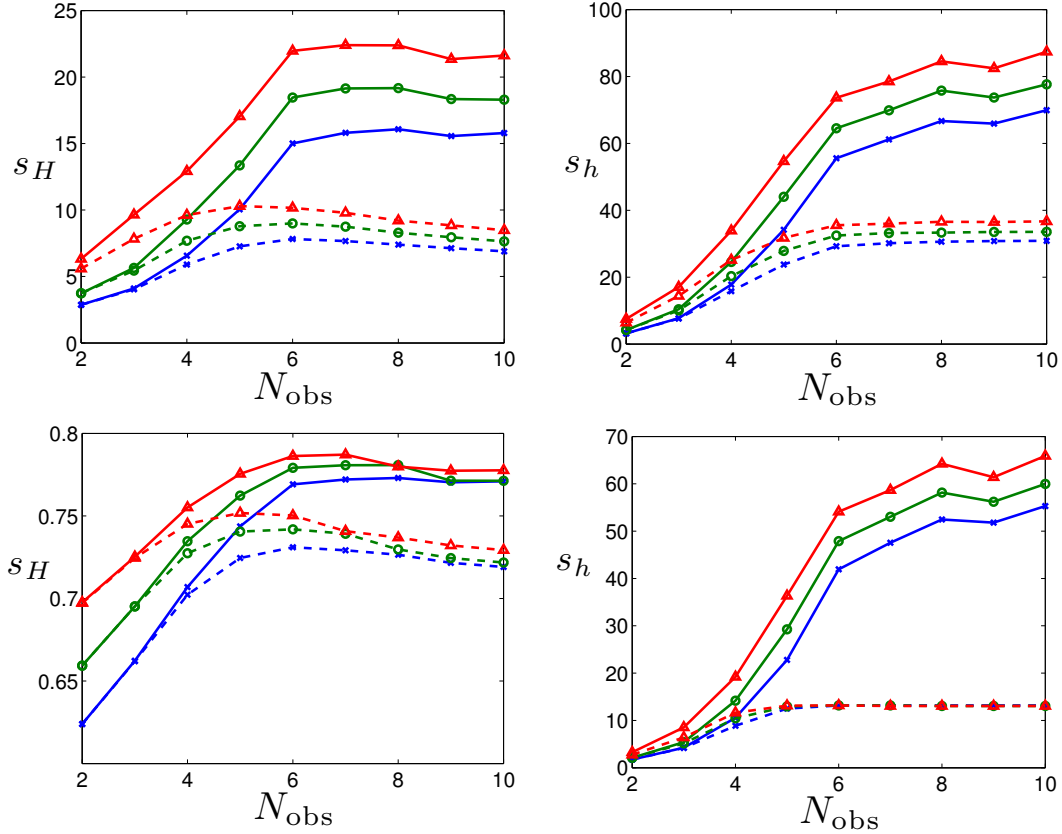


Figure 4.6.: Average maximal singular values s_H (left panels) and s_h (right panels) for $\mathbf{HP}_f \mathbf{H}^T$ and $\mathbf{hP}_f \mathbf{h}^T$ (top) and $\mathbf{HP}_a \mathbf{H}^T$ and $\mathbf{hP}_a \mathbf{h}^T$ (bottom), respectively, as a function of the sparsity N_{obs} . The solid lines depict values obtained from the ETKF and the dashed lines from the VLKF. We use $\gamma = 1$ (blue, crosses), $\gamma = 0.9$ (green, circles) and $\gamma = 0.8$ (red, triangles). We used $\mathbf{R}_0 = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$ and $\Delta t_{\text{obs}} = 48$ hours.

In Figure 4.7 we show how for long observation intervals $\Delta t_{\text{obs}} = 48$ hours the true climatic mean and variance are better preserved by the VLKF analysis than by the ETKF analysis. The true climatic mean μ_{clim} is reasonably well preserved by the VLKF analysis whereas the ETKF greatly underestimates it. The VLKF analysis still underestimates the true climatic variance σ_{clim}^2 , but not as severely as the ETKF does.

We will see that this leads to a marked increase in skill, especially for large model error $\gamma < 1$.

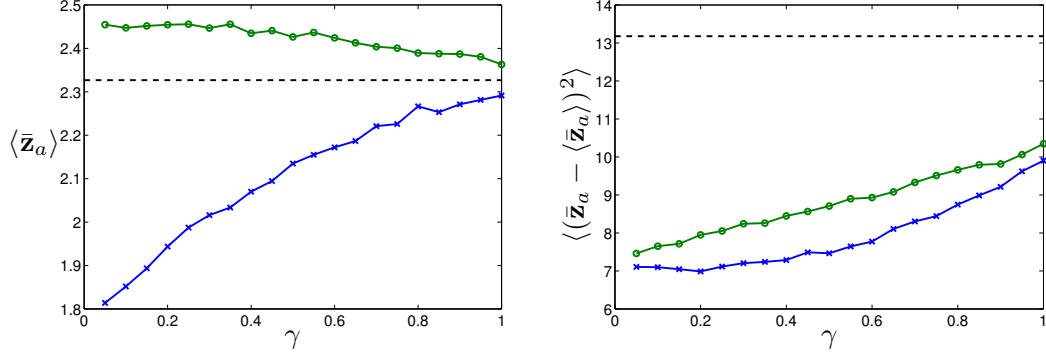


Figure 4.7.: Mean $\langle \bar{\mathbf{z}}_a \rangle$ (left) and variance $\langle (\bar{\mathbf{z}}_a - \langle \bar{\mathbf{z}}_a \rangle)^2 \rangle$ (right) of the analysis for ETKF (blue, crosses) and VLKF (green, circles), averaged over time and sites. The dashed black lines shows the climatic mean and variance μ_{clim} and σ_{clim}^2 of the truth with $\gamma = 1$. We used $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$, $N_{\text{obs}} = 4$, $\Delta t_{\text{obs}} = 48$ hours.

We now quantify the performance of the VLKF and the standard ETKF by the site-averaged RMS errors $\mathcal{E}$, $\mathcal{E}_o$ and $\mathcal{E}_{uo}$ as defined in (3.22). We also introduce a measure $\hat{\mathcal{E}}$ to quantify how well a filter performs against the poor-man's analysis of using the observations and the climatic mean for the observed and unobserved variables respectively, that is, taking $\mathbf{H}(\bar{\mathbf{z}}_a - \mathbf{z}_t) \sim \mathcal{N}(0, \mathbf{R}_o)$ and $\mathbf{h}(\bar{\mathbf{z}}_a - \mathbf{z}_t) \sim \mathcal{N}(0, \sigma_{\text{clim}}^2)$. Assuming that $\mathbf{H}(\bar{\mathbf{z}}_a - \mathbf{z}_t)$ and $\mathbf{h}(\bar{\mathbf{z}}_a - \mathbf{z}_t)$ are uncorrelated, we can calculate the RMS error $\hat{\mathcal{E}}$ in the poor-man's analysis as

$$\hat{\mathcal{E}}^2 = \frac{1}{L} \sum_{i=1}^L \left[\frac{1}{N_{\text{obs}}} \|\mathbf{H}(\bar{\mathbf{z}}_a(t_i) - \mathbf{z}_t(t_i))\|^2 + \left(1 - \frac{1}{N_{\text{obs}}}\right) \|\mathbf{h}(\bar{\mathbf{z}}_a(t_i) - \mathbf{z}_t(t_i))\|^2 \right]$$

where $L = \lfloor T/\Delta t_{\text{obs}} \rfloor$ is the number of analysis cycles as in (3.22), giving the RMS error

$$\begin{aligned} \hat{\mathcal{E}} &= \sqrt{\frac{1}{N_{\text{obs}}} \eta^2 \sigma_{\text{clim}}^2 + \left(1 - \frac{1}{N_{\text{obs}}}\right) \sigma_{\text{clim}}^2} \\ &= \sigma_{\text{clim}} \sqrt{1 + \frac{(\eta^2 - 1)}{N_{\text{obs}}}}, \end{aligned} \quad (4.2)$$

with $\mathbf{R}_o = (\eta\sigma_{\text{clim}})^2 \mathbf{I}$.

To quantify the proportional improvement in RMS errors for VLKF over ETKF we again use the analysis skill $\mathcal{S}$ as defined in equation (3.20). We also define the corresponding skills for proportional improvement of ETKF and VLKF over the poor man's analysis as

$$\hat{\mathcal{S}}^E = \frac{\hat{\mathcal{E}}}{\mathcal{E}^E}, \quad \hat{\mathcal{S}}^V = \frac{\hat{\mathcal{E}}}{\mathcal{E}^V},$$

respectively.

We first investigate the dependence of skill upon the observation interval Δt_{obs} . In Figure 4.8 the skills $\mathcal{S}$, $\hat{\mathcal{S}}^E$ and $\hat{\mathcal{S}}^V$ are shown as a function of the model error γ for different observation intervals Δt_{obs} . The figures reveal an intricate dependence of the skill on both the observation interval and the model error. We have checked that for $\Delta t_{\text{obs}} > 2.1$ days, the e -folding time for the Lorenz-96 model with $\gamma = 1$, the skill curves $\mathcal{S}(\gamma)$ collapse onto one curve, independent of the observation interval. This indicates that the forecast error covariance, which saturates for large observational times to the climatic covariance, is the main factor driving filter performance.

We will treat the near-perfect model case $\gamma \approx 1$ and the strong model error case with small $\gamma < 1$ separately. The perfect model case with $\gamma = 1$ was already considered in Chapter 3. We confirm here that VLKF is superior to the standard ETKF, its skill increasing with decreasing observation intervals Δt_{obs} . For smaller observation intervals skill improvements of more than 20% can be achieved, and the analysis of the pseudo-observables track the truth surprisingly well. This is consistent with our findings in Chapter 3. Note in particular that both filters perform better than the poor man's analysis with $\hat{\mathcal{S}}^E > 1$ and $\hat{\mathcal{S}}^V > 1$ for $\gamma = 1$. The variance limiting filter here successfully controls the overestimation of the analysis error covariance caused by the combination of finite ensemble sizes and sparse observations.

The case of model error with γ significantly smaller than 1 – our main objective in this chapter – shows an interesting dependence on both the observation interval and the model error. We first explain the different behaviour of the skill for moderate and large observation intervals and the dependence on the model error. For the discussion

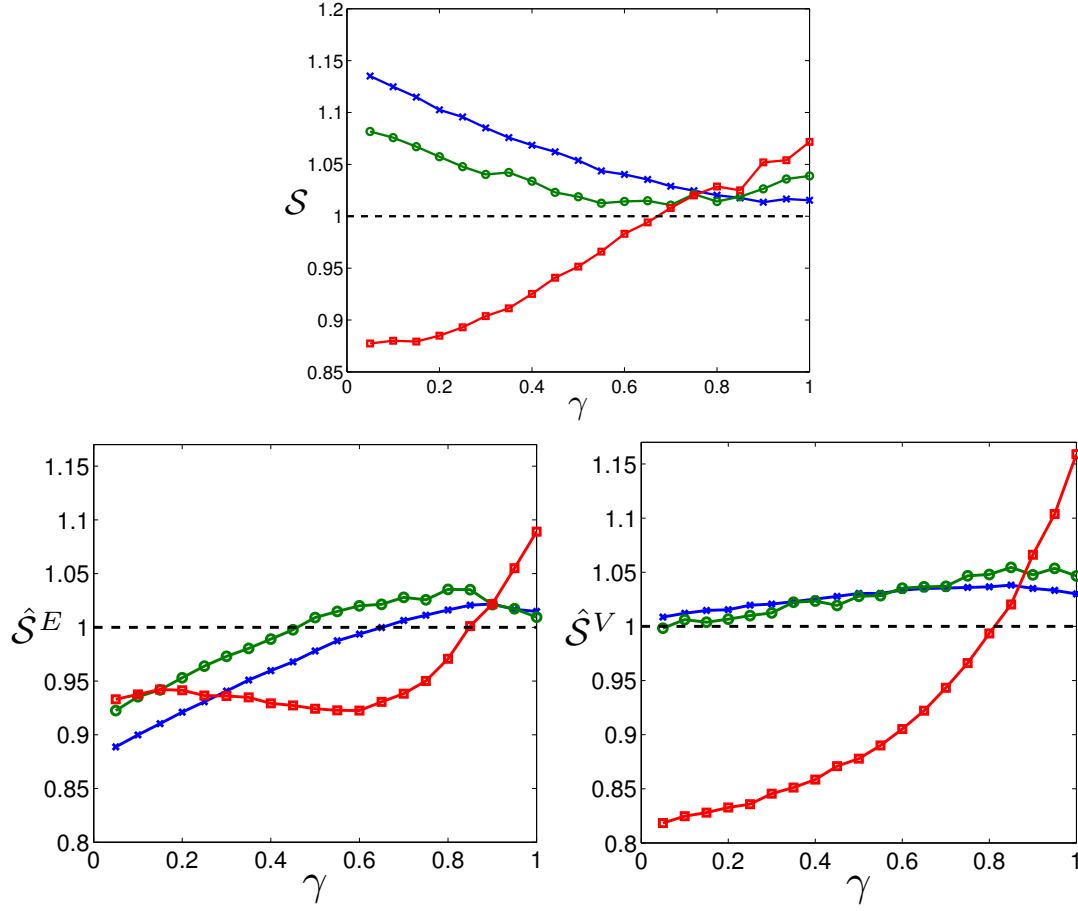


Figure 4.8.: Skill $\mathcal{S}$ as a function of damping γ for $\Delta t_{\text{obs}} = 24$ hours (red, squares), $\Delta t_{\text{obs}} = 36$ hours (green, circles) and $\Delta t_{\text{obs}} = 48$ hours (blue, crosses). We used $\mathbf{R}_0 = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$, $N_{\text{obs}} = 4$ and averaged all results over 200 realisations where one realisation was performed over 300 days.

below it is important to note that for sufficiently large model error the analysis error covariances in the directions of the pseudo-observables exceed the climatic variance σ_{clim}^2 ; hence the background error covariance matrices which are used to initialise an analysis cycle are larger than σ_{clim}^2 for the ETKF. On the other hand, the VLKF limits the analysis error covariance (projected onto the pseudo-observables) from above to σ_{clim}^2 , independent of the observation interval.

It is seen in Figure 4.8 that for moderate observation intervals Δt_{obs} (24 hours), the VLKF performs worse than the ETKF with skills $\mathcal{S} < 1$ decreasing with increasing model error (i.e. decreasing values of γ). On the contrary, for large observation

intervals with $\Delta t_{\text{obs}} > 36$ hours, the skill $\mathcal{S} > 1$ increases with increasing model error. Furthermore, for moderate observation intervals the VLKF performs increasingly worse than the poor man's analysis with increasing model error, but outperforms it for all values of γ for large observation intervals. However, the ETKF always performs worse than the poor man's analysis for all observation intervals and sufficiently large model error. We first explain the behaviour of the skill observed for moderate observation intervals and then for large observation intervals.

For moderate observation intervals the forecast ensemble may only acquire small spread, as already encountered in Figure 4.1. This small spread in conjunction with model error can lead to instances where the analysis mean performs large excursions away from the truth, thereby spoiling the analysis. This effect is stronger for the VLKF as it is initialised with a smaller ensemble spread associated with the constrained analysis error covariance as discussed above. Increasing the model error, i.e. decreasing γ , increases the probability of the ensemble mean diverging from the truth, thereby decreasing the skill. This explains the poor performance of the VLKF over ETKF with $\mathcal{S} < 1$ for moderate Δt_{obs} , and the poor performance of both the ETKF and VLKF over the analysis consisting of observations and climatology, with $\hat{\mathcal{S}}^E < 1$ and $\hat{\mathcal{S}}^V < 1$ as illustrated in Figure 4.8.

For large observation intervals the situation is different. Large observation intervals are characterised by the forecast dynamics having sufficiently decorrelated that the forecast ensemble will have equilibrated with (γ -dependent) covariance, having lost memory of the background initialisation. This has the effect that both filters will have acquired comparable forecast error covariance $\mathbf{P}_f$ over the forecast interval Δt_{obs} . The large spread acquired by the forecast ensemble during large observation intervals prohibits the analysed pseudo-observables tracking the truth well (cf. Figure 4.5). Skill improvements can then only be achieved by 1.) tracking the truth for the observed variables guided by the observations, and 2.) assuring the correct climatological behaviour for the pseudo-observables. This latter point is where the variance limiting method is beneficial, as illustrated in Figure 4.7. The more accurate statistical properties of the VLKF analysis results in improved skill with $\mathcal{S} > 1$ for all values of γ as shown in Figure 4.8.

However, Figure 4.8 also shows that the increased skill $\mathcal{S}$ for large observation intervals with increasing model error comes at the price of VLKF becoming less able to outperform the poor man's analysis with error $\hat{\mathcal{E}}$. As expected, $\hat{\mathcal{S}}^V$ asymptotes to 1 from above for decreasing values of γ , whereas $\hat{\mathcal{S}}^E$ becomes less than 1 for sufficiently large model error. We found that the VLKF performs better for all values of γ than the poor man's analysis for $\Delta t_{\text{obs}} > 36$ hours.

We conclude that the VLKF may be beneficial for ensemble based data assimilation with large underdamping model error and large observation intervals Δt_{obs} . For smaller observation intervals $\Delta t_{\text{obs}} < 18$ hours with sufficient model error (not shown) the standard ETKF performs better than the VLKF. However for small Δt_{obs} with a near-perfect model ($\gamma \approx 1$) it is still beneficial to use the VLKF, as shown in Chapter 3. For large observation intervals the overall skill increases with increasing model error γ due to the imposed control of the increased error covariances. However, there is a trade-off between increased skill improvement of the VLKF over ETKF, and of the skill improvement of the VLKF over the poor man's analysis.

We now look at the dependence of the skill upon the sparsity of the observational grid N_{obs} in the presence of model error. Numerical simulations show that the critical observation interval above which we observe skill improvement with $\mathcal{S} > 1$ for all values of γ decreases with increasing sparsity N_{obs} . Note that this impacts on the notion of what is considered to be moderate or large observation interval in the discussion above. This dependence of skill upon observational sparsity can be explained by noticing that sparser observations imply an increased forecast error covariance and ensemble spread for a given observation interval, as illustrated in Figure 4.6.

We have already seen that the main contribution to the overall RMS error stems from the pseudo-observables. Therefore one may be tempted to conclude that skill improvement consistently gets better for increasing N_{obs} . This is correct when only comparing the VLKF and ETKF. However, as in the case of increasing model error for fixed sparsity discussed above, the increased sparsity for fixed model error causes the VLKF to be less effective in outperforming our poor-man's analysis. Figure 4.9 illustrates how increased sparsity effects the overall RMS error $\mathcal{E}$. It is seen that the ETKF increasingly performs worse than the poor-man's analysis with increasing model error and increasing N_{obs} . In contrast, the RMS errors for the VLKF are below that

of the poor-man's analysis $\hat{\mathcal{E}}$ for all values of γ , and they asymptote to $\hat{\mathcal{E}}$ for large sparsity and large model error. This is in accordance with our observations in Figure 4.6 on the singular values of the error covariances of the VLKF. The trade-off between gain in skill of the VLKF over ETKF and less skill improvement with respect to a poor-man's analysis for increasing sparsity N_{obs} can be explained as follows. For sparse observations information from the observables cannot propagate to distant pseudo-observables within the observation interval Δt_{obs} (cf. Figure 4.3). This causes the forecast and analysis for those pseudo-observables to be essentially random variables with temporal mean μ_{clim} and variance σ_{clim}^2 for the VLKF, and $\mu(\gamma)$ and $\sigma^2(\gamma)$ for the ETKF. Hence increased sparsity leads to $\hat{\mathcal{E}}^V$ asymptoting from below to $\hat{\mathcal{E}}$ for all values of γ , with increasingly worse performance for decreasing damping γ . In contrast, $\hat{\mathcal{E}}^E > \hat{\mathcal{E}}$ for sufficiently large sparsity, for all values of γ .

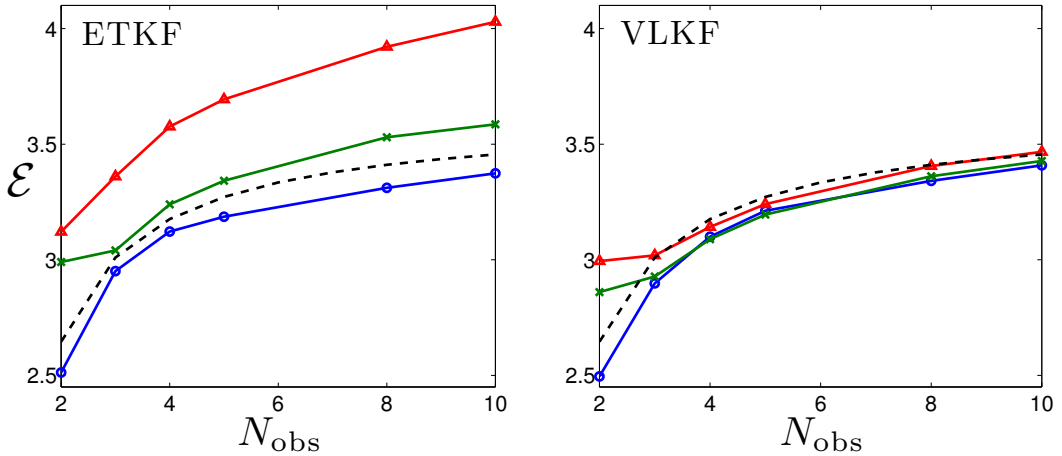


Figure 4.9.: Overall RMS error $\mathcal{E}$ as a function of the sparsity of the observational grid N_{obs} for different values of γ . We show the perfect model case $\gamma = 1$ (circles, blue), $\gamma = 0.5$ (crosses, green) and $\gamma = 0.05$ (triangles, red). The dashed line denotes $\hat{\mathcal{E}}$. The left panel shows results for the ETKF, the right panel for the VLKF. We used $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$ and $\Delta t_{\text{obs}} = 48$ hours.

It is illuminating to see how the RMS errors are distributed over the observables and the pseudo-observables. In Figure 4.10 we depict the RMS errors $\mathcal{E}_o$ for the observables and $\mathcal{E}_{uo}$ for the pseudo-observables, as defined in (3.22). We restrict our discussion here to large observation intervals $\Delta t_{\text{obs}} = 48$ hours. Consistent with our observation that the overall skill improvement is dominated by the superior analysis of the pseudo-observables, $\mathcal{E}_{uo}$ exhibits the same behaviour and trends as the overall RMS error $\mathcal{E}$ (cf.

Figures 4.8 and 4.9). The error for the observed variables shows different behaviour. It is seen that the standard ETKF produces better or equal RMS errors than the VLKF for all values of the model error γ . The inferior performance of the VLKF over ETKF for the observed variables is an instance of underestimation of the error covariances – the familiar situation in ensemble data assimilation described in previous chapters. As seen in Figure 4.6, where the average maximal singular value s_H of $\mathbf{H}\mathbf{P}_a\mathbf{H}^T$ for the ETKF and VLKF is shown, the variance constraint of the VLKF for the pseudo-observations also decreases the covariance in the observed subspace. The observed poor performance of the VLKF for the observed variables suggests that the limited covariance in the observed subspace is insufficient to provide enough ensemble spread for a reliable analysis.

For small model error, both filters produce worse analyses than an analysis consisting of observations only. This is again linked to the insufficient spread of the forecast ensembles. We have confirmed that for larger ensemble sizes $\mathcal{E}_o < \sqrt{\mathbf{R}_o}$ for all γ with both the ETKF and VLKF. Underestimation of error covariances due to finite size effects in ensemble filters can be controlled by multiplicative covariance inflation, as discussed in Chapters 2 and 3. In Figure 4.11 we show how the RMS error of the observed variables $\mathcal{E}_o$ decreases below the RMS error of the observed variables for $\gamma = 1$ and increasing inflation factor δ . We note that for very large values of δ in the perfect model situation the ETKF and VLKF produce analyses of equal RMS error. Increasing the forecast covariance inflation has the same effect as increasing the model error (i.e. decreasing the forecast model damping γ) in producing larger forecast ensemble spread.

Furthermore, it is seen in Figure 4.10 that for sufficiently large model error, the RMS errors of the observables $\mathcal{E}_o$ for VLKF and ETKF converge to the same γ -independent value $\tilde{\mathcal{E}}_o < \sqrt{\mathbf{R}_o}$. This can be explained by the greater inflation for less damped forecast models; larger model damping increases the forecast error covariance $\mathbf{P}_f$, and so the analysis tends to trust the observations more. As such, $\mathcal{E}_o \rightarrow \sqrt{\mathbf{R}_o}$ as $\gamma \rightarrow 0$ for both filters, albeit slightly reduced by the Kalman filtering. Due to the variance constraint the RMS error of the observed variables $\mathcal{E}_o$ asymptotes to $\tilde{\mathcal{E}}_o$ at larger model error for a given observational sparsity.

The value of the model damping γ below which the forecast variance is large enough for the filter to trust the observations depends on the sparsity N_{obs} . The larger the sparsity N_{obs} , the larger the value of γ for which the RMS error $\mathcal{E}_o$ of the two filters

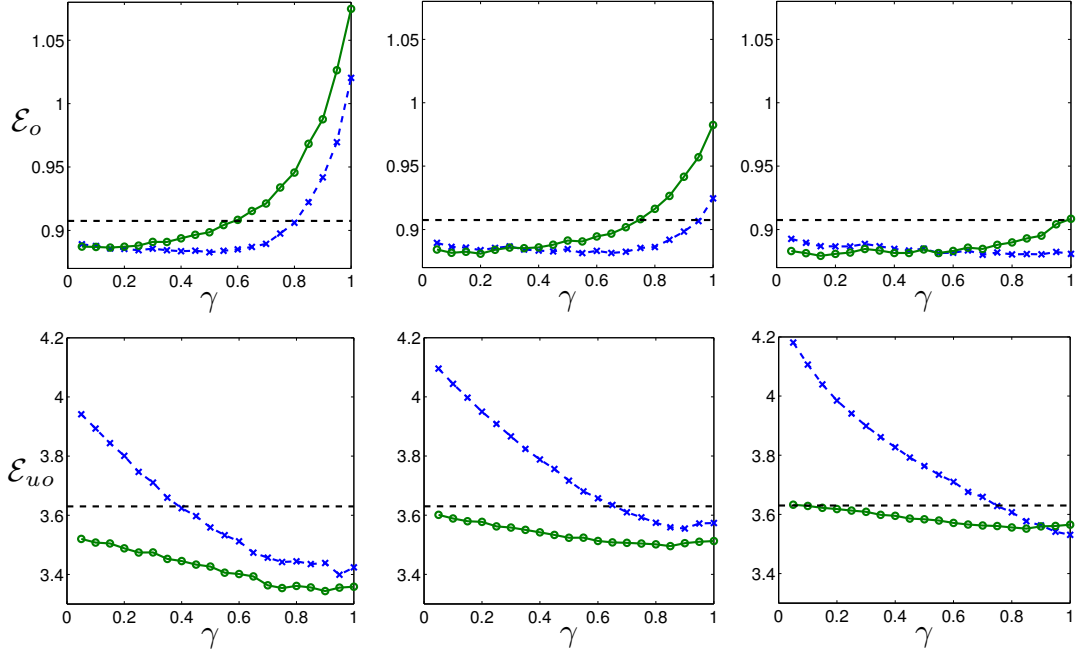


Figure 4.10.: RMS errors of observed variables $\mathcal{E}_o$ (top) and of unobserved variables $\mathcal{E}_{uo}$ (bottom) for the ETKF (blue, dashed, crosses) and VLKF (green, dashed, circles) for $N_{\text{obs}} = 3$ (left), $N_{\text{obs}} = 4$ (centre) and $N_{\text{obs}} = 8$ (right). Top and bottom horizontal dashed lines show $\sqrt{\mathbf{R}_o}$ and σ_{clim} , respectively. We used $\mathbf{R}_o = (0.25\sigma_{\text{clim}})^2 \mathbf{I}$ and $\Delta t_{\text{obs}} = 48$ hours.

collapses onto the constant value $\tilde{\mathcal{E}}_o$. For $N_{\text{obs}} = 8$, $\mathcal{E}_o \approx \tilde{\mathcal{E}}_o$ for all values of γ for the ETKF and $\mathcal{E}_o$ assumes this value at roughly $\gamma < 0.7$ for the VLKF. For $N_{\text{obs}} = 3$, $\mathcal{E}_o \approx \tilde{\mathcal{E}}_o$ for $\gamma < 0.7$ for the ETKF and for $\gamma < 0.25$ for the VLKF. This is linked again to the increased forecast error covariances caused by the sparsity, as illustrated in Figure 4.6. The top left panel shows that for a given observation interval and model error γ , the forecast error covariance increases with sparsity N_{obs} , causing the observations to be weighted more by the analysis.

Finally, in Figure 4.12 we show how the overall skill, as well as the climatic skills $\hat{\mathcal{S}}^E$ and $\hat{\mathcal{S}}^V$, vary with the observational noise level η , where $\mathbf{R}_o = (\eta\sigma_{\text{clim}})^2 \mathbf{I}$. The VLKF outperforms both the ETKF and the observations/climatology for large enough noise $\eta \geq 0.25$. Large observational noise gives preference to the pseudo-observables, whose unrealistically large error covariances as seen in Figure 4.6 are limited by the VLKF. This is particularly true for large model error. It is seen that for sufficiently large

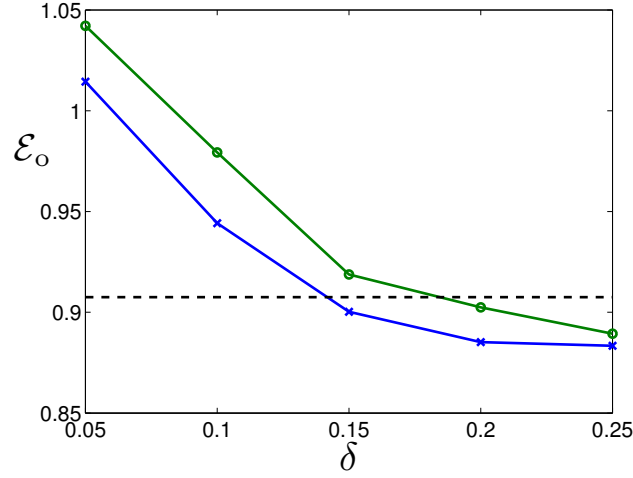


Figure 4.11.: RMS errors $\mathcal{E}_o$ for ETKF (blue, dashed, crosses) and VLKF (green, dashed, circles) for increasing forecast covariance inflation δ . The dashed line shows $\sqrt{\mathbf{R}_o}$. We used $\gamma = 1$, $\Delta t_{\text{obs}} = 48$ hours, $N_{\text{obs}} = 3$.

observational noise $\eta \geq 0.25$, the VLKF performs better than a poor-man's analysis for all values of the model error γ .

4.4 DISCUSSION

Our numerical results suggest that including climatic information on the mean and variance of the unobserved variables in an ensemble filter is beneficial in the case when the forecast error covariances are overestimated. In particular, this occurs when sparse observations are taken at large observation intervals and the forecast model is sufficiently underdamped. However, we have also highlighted situations where the model error causes error covariances to be underestimated, and hence the additional variance constraint provided by the VLKF spoils the analysis (albeit only for the observables and not, as has to be stressed, for the overall RMS error). This is the case for moderate observation intervals with large model error, and also for large observation intervals in the case of near-perfect models, but only if one is interested in the analysis of just the observables. This latter situation is not particularly limiting, as RMS errors for the

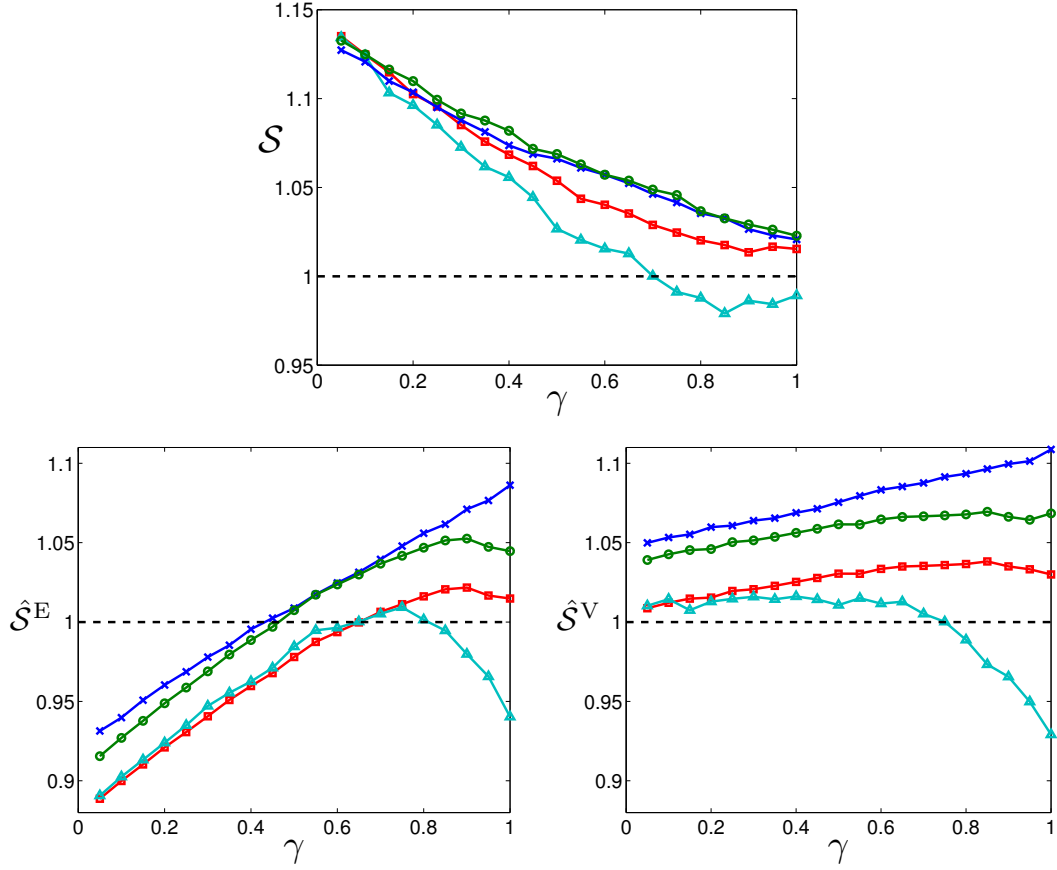


Figure 4.12.: Overall skill $\mathcal{S}$ (top) and climatic skills $\mathcal{S}^E$ (bottom left) and $\mathcal{S}^V$ (bottom right) as a function of γ for different levels of observational noise η , where $\mathbf{R}_o = (\eta\sigma_{\text{clim}})^2\mathbf{I}$. Here $\eta = 1$ (blue, crosses), $\eta = 0.5$ (green, circles), $\eta = 0.25$ (red, squares) and $\eta = 0.15$ (cyan, triangles). We used $\Delta t_{\text{obs}} = 48$ hours and $N_{\text{obs}} = 4$.

observables are relatively small compared to those for pseudo-observables, and the finite size effects can be eliminated here by basic multiplicative covariance inflation.

We argue that model error associated with an underdamped forecast model leads to an increased forecast error covariance and increased associated ensemble spread by the same mechanism by which covariance inflation is beneficial. The variance limiting filter controls this increased spread and keeps it within climatologically reasonable bounds, thereby leading to a lower RMS error and improving the overall analysis skill.

In the previous and current chapters we incorporated climatological information into the data assimilation procedure through the analysis step. In the next chapter we will take a different perspective, instead incorporating climatological information through the forecast step of the ensemble Kalman filter algorithm. We will consider another approach to the problem of covariance underestimation, this time through the very different process of stochastic parametrisation of the numerical forecast model.

USING HOMOGENISED STOCHASTIC PARAMETRISATIONS FOR DATA ASSIMILATION IN MULTISCALE SYSTEMS

5.1 INTRODUCTION

In this chapter we address the problem of stochastic parametrisations of subgrid-scale processes in multiscale systems. We ask whether and why stochastic parametrised climate models can be beneficial in the data assimilation procedure. We will focus here on the particular technique of *homogenisation*, because this method is underpinned by rigorous theorems (Khasminsky, 1966; Kurtz, 1973; Papanicolaou, 1976) in the limit of infinite time scale separation. Hence, we eliminate any issues related to possible unphysical or inaccurate parametrisations of the forecast model.

The main result of this chapter is that reduced stochastic models can indeed improve the analysis skill when used in place of the perfect full deterministic forecast model in data assimilation. We find in particular that reduced stochastic models can better detect transitions between metastable regimes in a multiscale system. The reason why stochastic climate models are capable of producing superior skill in an ensemble setting is due to the finite size of the ensemble; it is found that ensembles produced by the perfect deterministic forecast model lack sufficient spread even for moderate ensemble sizes. On the other hand, stochastic forecast models provide a natural way to guarantee large enough ensemble spread to detect transitions between regimes. The conclusion is that stochastic parametrisations present an attractive framework for data assimilation despite their sensitivity to uncertainties in the reduced model parameters.

A secondary aim of this chapter is to investigate how the predictability of a multiscale system is determined by the properties of the system, specifically when in the presence of fast and slow regimes. In an ensemble data assimilation context this is relevant as the analysis ensemble should represent the directions of growing errors. Sensitivity of error growth rates in systems with regimes suggests caution is required when assimilating data on such systems. We will present the surprising result that error growth rates can be a highly nonlinear function of the initial error amplitude in some multiscale systems. This will be seen to manifest as nonmonotonic behaviour in the spectrum of finite size Lyapunov exponents, and will be shown to be due to the presence of fast and slow regimes in the system.

How to effectively parameterise subgrid-scale processes is an important question in the atmospheric sciences, and has been an active field of research in recent times. Since the pioneering papers by [Leith \(1975\)](#) and [Hasselmann \(1976\)](#), the stochastic parametrisation of processes which cannot be spatially and/or temporally resolved have gained popularity across disciplines in the atmospheric, oceanographic and climate sciences ([Palmer, 2001](#)). Applications range from the study of atmospheric low-frequency variability ([Franzke et al., 2005](#)), deep convection and cloud modelling in global circulation models ([Lin and Neelin, 2002](#); [Plant and Craig, 2008](#)), decadal climate changes such as the El Niño Southern Oscillation (ENSO) ([Kleeman, 2008](#)) to paleoclimatic modelling ([Ditlevsen, 1999](#); [Kwasniok and Lohmann, 2009](#)). For further applications we refer the reader to the books edited by [Imkeller and von Storch \(2001\)](#) and by [Palmer and Williams \(2010\)](#) which contain excellent overviews of current trends in stochastic climate modelling.

There exists a plethora of different methods to construct stochastic subgrid-scale parametrisations, including ad hoc methods such as randomisation of existing deterministic parametrisation schemes (e.g. [Buizza et al. \(1999\)](#)) and energetic backscattering (e.g. [Shutts \(2005\)](#)), data-driven techniques such as Markov chains (e.g. [Crommelin and Vanden-Eijnden \(2008\)](#)), and systematic approaches using stochastic homogenisation (e.g. [Majda et al. \(1999, 2003\)](#)). The specific functional forms and parameter values of the respective parametrisation scheme can be heuristically postulated on the grounds of the particular physics involved (e.g. [Lin and Neelin \(2000\)](#)), or estimated using time series analysis (e.g. [Wilks \(2005\)](#)). In the case of multiscale dynamics, however, the parameters can be systematically derived using averaging and homogenisation

techniques (e.g. Majda et al. (1999, 2001, 2008)). Our work will be concerned with the latter approach. The general theory of stochastic averaging and homogenisation for multiscale dynamical systems goes back to the seminal papers of Khasminsky (1966), Kurtz (1973) and Papanicolaou (1976). These mathematically rigorous ideas have recently been applied in the atmospheric context in a series of papers starting with Majda et al. (1999).

We will introduce a simple deterministic multiscale toy model in which a slow degree of freedom with multiple metastable states, resembling slow weather or climate regimes (cf. Ditlevsen (1999); Kwasniok and Lohmann (2009)), is driven by a fast chaotic subsystem which also involves regimes. Although simple, this model involves core phenomena found in realistic applications such as slow and fast regimes as caricatures of climate and weather regimes respectively, and rare transitions between them. The model is amenable to the theory of homogenisation (see the reviews by Givon et al. (2004) and Pavliotis and Stuart (2008)), and we will derive a reduced stochastic climate model describing the slow dynamics of the full higher-dimensional model. Whereas for non-systematic stochastic parametrisations the results are sensitive to details of the assumed processes, we will verify here that the homogenised reduced model faithfully represents the slow dynamics of the full system even for moderate time scale separation.

To understand how forecasting with stochastic reduced climate models can be beneficial in data assimilation, we will first investigate characteristic time scales of a multiscale system with slow and fast regimes. This will be necessary to understand when and why stochastic reduced models perform better in the forecast step than the full deterministic model. An important question is how to measure the radius of predictability in multiscale systems; there are a range of time scales which characterise predictability on various spatial and temporal scales. One well-known measure is the maximal Lyapunov exponent $\lambda_{\max}$. A generic situation is that the fast degrees are strongly chaotic, causing the Lyapunov exponent of the whole system to be large, indicating poor predictability. The large scale slow behaviour, however, can still be forecast with reasonable accuracy for much longer times than $\mathcal{O}(1/\lambda_{\max})$. This is usually due to the small-scale instabilities growing faster but becoming nonlinearly saturated at a much smaller level than the large scale instabilities (Lorenz, 1969). One particular aspect of predictability we focus on in this chapter is the case when the small scale fast process involves regimes. This is highly relevant for atmospheric and climate variability, where fast metastable

processes may determine the predictability of dominant slower processes. For example, metastable synoptic weather events such as Rossby-wave breaking and high-latitude blocking episodes with life times of 5-10 days are important for the North Atlantic Oscillation, the dominant pattern of atmospheric variability over the Atlantic, and may give rise to its low-frequency variability on interannual and longer time scales (Greatbatch, 2000; Woollings et al., 2008). Similarly, the ENSO variability on time scales of seasons to years can be produced or maintained by faster subseasonal long-lived transient westerly wind bursts and the Madden-Julian Oscillation with life times of 30-90 days (Frauen and Dommenges, 2010; Roulston and Neelin, 2000; Zavala-Garay et al., 2003). On the mesoscale, it was shown (Nguyen et al., 2011) that during Hurricane *Katrina* two to six mesovortices formed around the inner edge of the barotropically unstable eyewall, caused by convection. These fast structures with life times of a few hours lead to mixing of heat and momentum between the eye and the eyewall, thereby removing the instability and temporarily pausing the intensification of the storm.

The second important problem we investigate is the sensitivity of reduced stochastic models to uncertainties in the drift and diffusion terms. We verify here that the homogenised reduced model with appropriately determined drift and diffusion terms faithfully represents the slow dynamics of the full model even for moderate time scale separation. Despite this, we find that the statistics of the climate model are sensitive to uncertainties in the drift and diffusion terms. It will be shown however that this sensitivity to drift and diffusion can actually be beneficial in the data assimilation process.

The third aim of this chapter is to address the following questions: under what circumstances can reduced stochastic climate models improve the skill of an ensemble based data assimilation scheme when used as forecast models? Furthermore, if they do, why so? Harlim and Majda (2008, 2010) studied ensemble filtering on the Lorenz-96 model in the fully turbulent regime using a stochastic forecast model which was constructed by radically replacing all nonlinear terms by linear stochastic processes whose statistics are estimated by fitting the model to the climatological variance, the decorrelation time and the observation time correlation. They found skill improvement for a stochastic climate forecast model which was constructed by radically replacing all nonlinear terms by linear stochastic processes. In those works statistics of these processes were estimated by fitting the model to the climatological variance, correlation time and

observation time correlation. Here we study the performance of systematically derived climate models, however our results will also shed light on skill improvements of other ensemble based stochastic filtering methods. An important result is that inaccurately reproducing the statistics of the slow dynamics does not preclude stochastic reduced models from being beneficial in data assimilation. We identify a range of observation intervals in which the reduced stochastic climate model actually outperforms the full deterministic model in terms of analysis skill. The range of observation intervals for which skill improvement is observed will be related to the characteristic time scales of the system. In particular we find that the observation interval has to be larger than the typical time taken for the slow state to switch regimes, and smaller than the decay rate of the autocorrelation function for the slow variable. Hence, stochastic climate models will be beneficial when studying weather or climate systems on time scales which resolve regime switches. The main message of this chapter will be that the superior performance of reduced stochastic models in data assimilation is linked not to their accurate approximation of the statistical slow behaviour but rather to a controlled increase of the ensemble spread associated with their inherent dynamic stochasticity, with an effect similar to inflating covariances or an increasing ensemble size.

The remainder of this chapter is organised as follows: In the next section we introduce the model and derive the reduced stochastic climate model for the slow dynamics using homogenisation techniques. Estimation of the drift and diffusion terms in the climate model is performed in Section 5.3. In Section 5.4 we deal with the issue of predictability through examining some of the relevant characteristic time scales of the dynamics. In Section 5.5 we study the sensitivity of these time scales to uncertainties in the time scale separation parameter as well as in the drift and diffusion terms of the climate model. We present numerical results showing the skill improvement in using the reduced climate model over the full deterministic model in an ensemble Kalman filter data assimilation procedure in Section 5.6. We conclude with a discussion in Section 5.7.

5.2 THE MODELS

We study multiscale systems of the form

$$\frac{dx}{dt} = x - x^3 + \frac{1}{\varepsilon}f(y) \quad (5.1)$$

$$\frac{dy}{dt} = \frac{1}{\varepsilon^2}g(y) , \quad (5.2)$$

in which a slow degree of freedom $x \in \mathbb{R}$ describes an overdamped degree of freedom in a double-well potential $V(x) = x^4/4 - x^2/2$ which is driven by a fast chaotic process $y \in \mathbb{R}^3$. Unless stated otherwise we set $\varepsilon^2 = 0.01$.

The primary model we investigate in this chapter has the fast chaotic subsystem given by the famous Lorenz-63 system (Lorenz, 1963)

$$\frac{dx}{dt} = x - x^3 + \frac{\kappa}{\varepsilon}y_2 \quad (5.3)$$

$$\frac{dy_1}{dt} = \frac{10}{\varepsilon^2}(y_2 - y_1) \quad (5.4)$$

$$\frac{dy_2}{dt} = \frac{1}{\varepsilon^2}(28y_1 - y_2 - y_1y_3) \quad (5.5)$$

$$\frac{dy_3}{dt} = \frac{1}{\varepsilon^2}(y_1y_2 - \frac{8}{3}y_3) . \quad (5.6)$$

The fast dynamics contains regimes consisting of the two respective lobes of the “butterfly” attractor. We note that the Lorenz-63 system is metastable but the two metastable states do not correspond to the different lobes of the attractor (Froyland and Padberg, 2009). We will therefore refer to regimes instead of metastable states for this model, where a regime means one lobe of the Lorenz attractor. This system was introduced in (Givon et al., 2004), and as in that work we use here $\kappa = 4/90$. In Figure 5.1 we show a trajectory of the slow variable x obtained from a simulation of the full deterministic system (5.3)-(5.6). The trajectory of the slow variable appears to randomly switch between two metastable states around the minima of the potential $V(x)$ at $x^* = \pm 1$.

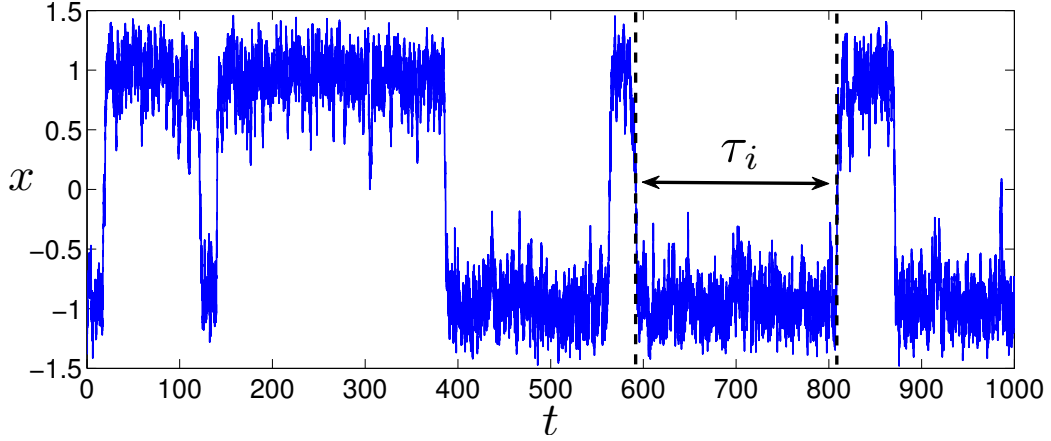


Figure 5.1: Sample trajectory of the slow variable x of the deterministic model (5.3)-(5.6). The dynamics appears to randomly switch between two metastable states with randomly distributed sojourn times τ_i .

In order to study the influence of fast regimes on the predictability of the slow mode x , we also introduce a model containing a fast chaotic subsystem with no metastable states. We consider the multiscale model (5.1)-(5.2) with a fast Rössler subsystem

$$\frac{dx}{dt} = x - x^3 + \frac{\kappa}{\varepsilon}(y_2 - \bar{y}_2) \quad (5.7)$$

$$\frac{dy_1}{dt} = \frac{1}{\varepsilon^2}(-y_2 - y_3) \quad (5.8)$$

$$\frac{dy_2}{dt} = \frac{1}{\varepsilon^2}(y_1 + 0.432y_2) \quad (5.9)$$

$$\frac{dy_3}{dt} = \frac{1}{\varepsilon^2}(2 + y_3(y_1 - 4)) . \quad (5.10)$$

The coupling $f(y)$ is chosen so that the forcing has mean zero; we estimate the mean of the driving Rössler variable as $\bar{y}_2 = -0.939$ from a long trajectory. However, unlike the Lorenz model the Rössler model is asymmetrical, and the probability density functions for each variable are sharply peaked unimodal with broad tails. This makes the mean $\bar{y}_2$ difficult to estimate accurately, which will have impacts in estimating time scales as seen in Section 5.4. We choose the coupling parameter $\kappa = 0.525$ so that the coupling term $f(y)$ produces fluctuations in the slow variable x of approximately the same size as those for the model with fast Lorenz subsystem (5.3)-(5.6). We have checked that the probability density function for the slow dynamics of (5.7)-(5.10) is comparable to that of (5.3)-(5.6). Unlike the Lorenz system (5.4)-(5.6), the fast chaotic Rössler system has only one unstable fixed point and does not support regimes. We will derive a reduced

stochastic climate model which describes the effective slow dynamics X for both models (5.3)-(5.6) and (5.7)-(5.10) in the next section.

Derivation of the climate model

The ergodic and mixing properties of the fast subsystem allows for an application of stochastic model reduction techniques, by which the fast chaotic degrees of freedom are parametrised by a one-dimensional stochastic process. Ergodicity and mixing have been proven for the Lorenz-63 system (by Tucker (1999) and Luzzatto et al. (2005) respectively), and numerical simulations suggest that this is the case for the Rössler system as well. The replacement of the fast dynamics by a stochastic process can be heuristically justified provided the fast processes decorrelate rapidly enough; then the slow variables experience the sum of uncorrelated fast dynamics during one slow time unit. According to the (weak) Central Limit Theorem this converges to approximate Gaussian noise in the limit when the time scale separation becomes infinite. We will formalise this and apply stochastic singular perturbation theory (homogenisation) (Khasminsky, 1966; Kurtz, 1973; Papanicolaou, 1976) to deduce the following reduced stochastic model for the effective slow dynamics X

$$\frac{dX}{dt} = X(1 - X^2) + \sigma \frac{dW}{dt} \quad (5.11)$$

with 1-dimensional Wiener process W , where σ is given by the total integral of the autocorrelation function of the coupling function $f(y)$ with

$$\begin{aligned} \frac{\sigma^2}{2} &= \int_0^\infty \left[\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(y(s))f(y(t+s))ds \right] dt \\ &= \kappa^2 \int_0^\infty \left[\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T y_2(s)y_2(t+s)ds \right] dt \end{aligned} \quad (5.12)$$

for our choice of functions $f(y)$ in (5.3)-(5.6) and (5.7)-(5.10). The proof is lengthy but straightforward, and is given in modern expositions of the theory of stochastic homogenisation, for example in (Givon et al., 2004; Pavliotis and Stuart, 2008). The existence of the diffusion limit for purely deterministic systems of the form (5.1)-(5.2) was proven by Melbourne and Stuart (2011). In contrast, the pioneering works

of (Khasminsky, 1966; Kurtz, 1973; Papanicolaou, 1976) consider classes of multiscale stochastic systems.

We will analyse the system (5.1)-(5.2) in the framework of the backward Kolmogorov equation for the conditional expectation value of some sufficiently smooth observable $\phi(x, y)$ defined as

$$v(x_0, y_0, t) = \mathbb{E} [\phi(x(t), y(t)) \mid x(0) = x_0, y(0) = y_0] .$$

with $y = (y_1, y_2, y_3)$ for our choice of the fast subsystems. Here the expectation value is taken with respect to the ergodic measure induced by the fast dynamics. We drop the 0-subscripts for ease of exposition from here on. We study the following Cauchy problem, the backward Kolmogorov equation, for $t \in [0, \infty)$

$$\begin{aligned} \frac{\partial}{\partial t} v(x, y, t) &= \mathcal{L} v(x, y, t) \\ v(x, y, 0) &= \phi(x, y) , \end{aligned} \tag{5.13}$$

with the generator

$$\mathcal{L} = \frac{1}{\varepsilon^2} \mathcal{L}_0 + \frac{1}{\varepsilon} \mathcal{L}_1 + \mathcal{L}_2 ,$$

where

$$\mathcal{L}_0 \equiv g(y) \cdot \nabla_y \tag{5.14}$$

$$\mathcal{L}_1 \equiv f(y) \frac{\partial}{\partial x} \tag{5.15}$$

$$\mathcal{L}_2 \equiv x(1 - x^2) \frac{\partial}{\partial x} , \tag{5.16}$$

and $g(y)$ denotes the scaled vectorfield of the fast subsystem $g(y)$ (here either the Lorenz or Rössler system).

Pioneered by Kurtz (1973) and Papanicolaou (1976) a perturbation expansion can be made according to

$$v(x, y, t) = v_0 + \varepsilon v_1 + \varepsilon^2 v_2 + \dots . \tag{5.17}$$

Substituting the series (5.17) into the backward Kolmogorov equation (5.13) we obtain at lowest order, $\mathcal{O}(1/\varepsilon^2)$,

$$\mathcal{L}_0 v_0 = 0 . \quad (5.18)$$

Assuming ergodicity of the fast subsystem, the expectation value does not depend on initial conditions, and we obtain

$$v_0 = v_0(x, t)$$

as the only solution of (5.18). Ergodicity also implies the existence of a unique invariant density induced by the fast chaotic dynamics given by the unique solution of the associated Fokker Planck-equation

$$\mathcal{L}_0^* \rho = 0 ,$$

where $\mathcal{L}_0^*$ is the formal adjoint¹ in the L_2 sense of the generator $\mathcal{L}_0$. We label this solution as $\rho_\infty(y)$. We assume here the existence of absolutely continuous measures so we are guaranteed the existence of a probability density.

At the next order, $\mathcal{O}(1/\varepsilon)$, we obtain

$$\mathcal{L}_0 v_1 = -\mathcal{L}_1 v_0 . \quad (5.19)$$

To assure boundedness of v_1 (and thereby of the asymptotic expansion (5.17)) a solvability condition prescribed by the Fredholm alternative must be satisfied. Equation (5.19) is only solvable provided the right-hand-side is in the space orthogonal to the (one-dimensional) null space of the adjoint $\mathcal{L}_0^*$, i.e. if

$$\langle \mathcal{L}_1 v_0 \rangle_{\rho_\infty} = -\langle f(y) \rangle_{\rho_\infty} \frac{\partial}{\partial x} v_0(x, t) = 0 ,$$

where we have introduced the average of an observable $h(x, y)$ over the fast ergodic density as $\langle h \rangle_{\rho_\infty} := \int h(x, y) \rho_\infty(y) dy$. Note that the vanishing of the average of the fast perturbation in the slow equation with respect to the invariant measure induced

¹ Not to be confused with the adjoint of meteorological literature which is the transposed tangent linear model, here the adjoint means the operator which satisfies $\mathcal{L}_0^* \rho = -\nabla \cdot (g\rho)$. For further details see for example (Pavliotis and Stuart, 2008).

by the fast system implies that classical averaging would only produce trivial reduced dynamics $\dot{x} = \mathcal{O}(\varepsilon)$. Since $\langle f(y) \rangle_{\rho_\infty} = 0$, there exists a solution of (5.19) which we can formally write as

$$v_1(x, y, t) = -\mathcal{L}_0^{-1} \mathcal{L}_1 v_0 + R(x) , \quad (5.20)$$

where $R(x)$ lies in the kernel of $\mathcal{L}_0$.

At the next order, $\mathcal{O}(1)$, we obtain

$$\mathcal{L}_0 v_2 = \frac{\partial}{\partial t} v_0 - \mathcal{L}_1 v_1 - \mathcal{L}_2 v_0 , \quad (5.21)$$

which yields the desired evolution equation for v_0 as the associated solvability condition

$$\frac{\partial}{\partial t} v_0 = \langle \mathcal{L}_2 v_0 \rangle_{\rho_\infty} + \langle \mathcal{L}_1 v_1(x, t) \rangle_{\rho_\infty} . \quad (5.22)$$

The reduced slow backward Kolmogorov equation is then

$$\frac{\partial}{\partial t} v_0 = x(1 - x^2) \frac{\partial}{\partial x} v_0 - \langle f(y) \mathcal{L}_0^{-1} f(y) \rangle_{\rho_\infty} \frac{\partial^2}{\partial x^2} v_0 . \quad (5.23)$$

This can be simplified further. For every function h with $\langle h \rangle_{\rho_\infty} = 0$ we define

$$H(x, y) = - \int_0^\infty e^{\mathcal{L}_0 t} h \, dt ,$$

which, upon using the fact that $\mathcal{L}_0$ corresponds to an ergodic process, implies that

$$\begin{aligned} \mathcal{L}_0 H &= - \int_0^\infty \mathcal{L}_0 e^{\mathcal{L}_0 t} h \, dt \\ &= - \int_0^\infty \partial_t \left(e^{\mathcal{L}_0 t} h \right) \, dt \\ &= - \lim_{t \rightarrow \infty} e^{\mathcal{L}_0 t} h + h \\ &= \langle h \rangle_{\rho_\infty} + h \\ &= h. \end{aligned}$$

For further details on the calculation see (Pavliotis and Stuart, 2008). Hence we can evaluate

$$\begin{aligned}\langle f(y) \mathcal{L}_0^{-1} f(y) \rangle_{\rho_\infty} &= - \int_0^\infty \langle f(y) e^{\mathcal{L}_0 t} f(y) \rangle_{\rho_\infty} dt \\ &= - \int_0^\infty \left\{ \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(y(s)) f(y(t+s)) ds \right\} dt ,\end{aligned}$$

where the last identity follows from employing Birkhoff's ergodic theorem. The reduced backward Kolmogorov equation (5.23) can then be written as

$$\frac{\partial}{\partial t} v_0 = x(1-x^2) \frac{\partial}{\partial x} v_0 + \frac{\sigma^2}{2} \frac{\partial^2}{\partial x^2} v_0 , \quad (5.24)$$

with σ^2 given by (5.12). Note that $R(x)$ does not contribute to the dynamics. Without loss of generality we can therefore choose $R(x) = 0$ in order to ensure that $\langle v \rangle_{\rho_\infty} = \langle v_0 \rangle_{\rho_\infty} + \mathcal{O}(\varepsilon^2)$. The slow reduced Langevin equation associated with the reduced backward Kolmogorov equation (5.24) is then given by our stochastic climate model (5.11).

The unique invariant density of the gradient system (5.11) is readily determined as the unique stationary solution $\hat{\rho}(x)$ of the associated Fokker-Planck equation

$$\frac{\partial}{\partial t} \rho(x) = \frac{\partial}{\partial x} \left(\frac{dV}{dx} \rho \right) + \frac{\sigma^2}{2} \frac{\partial^2}{\partial x^2} \rho . \quad (5.25)$$

We find

$$\hat{\rho}(x) = \frac{1}{Z} e^{-\frac{2}{\sigma^2} V(x)} \quad (5.26)$$

with

$$Z = \int_{-\infty}^{\infty} e^{-\frac{2}{\sigma^2} V(x)} dx .$$

Note that the unique invariant density of the full deterministic dynamics (5.1)-(5.2) is now approximated by $\rho(x, y) = \hat{\rho}(x) \rho_\infty(y) + \mathcal{O}(\varepsilon)$.

5.3 PARAMETER ESTIMATION

The value of the diffusion coefficient σ given by (5.12) cannot be determined analytically for the reduced climate model as the unique invariant densities $\rho_\infty(y)$ of the fast Lorenz and Rössler subsystems (5.4)-(5.6) and (5.8)-(5.10) are not explicitly known. We therefore need to evaluate the diffusion coefficient σ numerically from simulations of the full deterministic model (5.1)-(5.2) under the assumption that its slow dynamics is well approximated by the stochastic climate model (5.11). We do so by coarse-graining the full deterministic system and estimating conditional averages (Siegert et al., 1998; Gardiner, 2003; Stemler et al., 2007). This method has been used by the meteorological community to study time series in diverse contexts such as synoptic variability of midlatitude sea surface winds, Icelandic meteorological data, planetary waves and paleoclimatic ice-core data (Sura and Barsugli, 2002; Sura, 2003; Egger and Jónsson, 2002; Berner, 2005; Ditlevsen, 1999). This approach also allows us to determine the drift term in the climate model, which we write for convenience here as

$$dx = d(x)dt + \sigma dW, \quad (5.27)$$

with drift term $d(x) = x(1 - x^2)$. Numerically integrating the full deterministic system (5.1)-(5.2) with a small time step dt , we create a long time series $x(t_i)$, $i = 0, 1, \dots, n$ with $t_i = i dt$. To estimate the parameters of the Langevin system (5.27) we subsample the time series $x(t_i)$ at a coarse sampling time $h \gg dt$. We choose h such that during h the finely sampled trajectory $x(t_i)$ performs roughly 3-4 fast oscillations.

We estimate the diffusion coefficient σ (and the drift term $d(x)$) by partitioning the state space into bins X_i of bin size ΔX . We coarse grain by mapping x into X for $x \in (X, X + \Delta X)$. To estimate the diffusion coefficient we define the average conditioned on the bins $[X, X + \Delta X]$

$$S(X) = \frac{1}{h} \langle (x^{n+1} - x^n)^2 \rangle \Big|_{x^n \in (X, X + \Delta X)}, \quad (5.28)$$

where $x^n = x(nh)$. The angular brackets denote a conditional ensemble average over microscopic realisations $x(t_i)$ which fall into a coarse macroscopic bin $[X, X + \Delta X]$.

In the limit when the deterministic dynamics can be approximated by the stochastic climate model (5.27) and for small values of the sampling time h , we can evaluate

$$S(X) = \sigma^2 + X^2(1 - X^2)^2 h ,$$

where we have used $\langle dW \rangle = 0$. In the limit $h \rightarrow 0$ we estimate $S(X) = \sigma^2$. Similarly the drift $d(x)$ can be estimated by

$$D(X) = \frac{1}{h} \langle (x^{n+1} - x^n) \rangle \Big|_{x^n \in (X, X+\Delta X)} , \quad (5.29)$$

with $D(X) \approx d(x)$ in the limit $h \rightarrow 0$.

There are several pitfalls one may encounter when using this method to estimate the drift and diffusion coefficients of stochastic differential equations related to the choice of the sampling time h (see for example Pavliotis and Stuart (2007)). In this case, if h is chosen smaller than the typical time scales on which the fast subsystem decorrelates, the diffusion coefficient does not exist and $S(X)$ will be close to zero in the limit $h \rightarrow 0$. On the contrary, if h is chosen too large the estimator for the diffusion coefficient will be swamped by the deterministic part of the dynamics and we will obtain $S(X) \approx X^2(1 - X^2)^2 h$. Similarly there are problems with estimating the drift $d(X)$ for sampling times h taken too large. For such sampling times, the coarse-grained dynamics will randomly switch from bin to bin according to the invariant density $\hat{\rho}(x)$ given by (5.26). Since in our case $\hat{\rho}(x)$ is symmetric, we can write the continuous approximation of the direct estimator for the drift as

$$D(X) = \frac{1}{h} \int (Y - X) \hat{\rho}(Y) dY = -\frac{X}{h} .$$

This may lead to erroneous classifications of stochastic processes as Ornstein-Uhlenbeck processes. We remark that if the time series is not sufficiently long, the estimated drift coefficient in our case of a bimodal probability density function will be erroneously estimated as $D(X) = -(X - X_0)/h$ due to the presence of the two metastable states around $X_0 = \pm 1$.

This procedure can be readily extended to study multi-dimensional stochastic processes where the bins are hypercubes; however the interpretation of the results in higher

dimensions and the deducing functional dependencies may be difficult. We remark that one may also use Kalman filters to estimate the parameters (Cornick et al., 2009).

In Figure 5.2 we show results from the parameter estimation procedure for the Lorenz-driven model (5.3)-(5.6). We choose a coarse-grain bin size of $\Delta X = 0.05$ and sample the slow variable x at constant sampling intervals $h = 0.005$ corresponding to approximately 4 fast oscillations. We used a trajectory of the full system (5.3)-(5.6) with $\varepsilon^2 = 0.0005$ and evolved until $T = 3 \times 10^4$ time units. We confirm the validity of climate model (5.11) by estimating the drift coefficient clearly as

$$D(X) = X(1 - X^2)$$

in Figure 5.2(a) and estimating a constant diffusion coefficient $S(X) = \sigma^2 = 0.113$ in Figure 5.2(b). We note that this is slightly smaller than the value $\sigma^2 = 0.126 \pm 0.003$ which was found by Givon et al. (2004) using a different method.

In Figure 5.2(c) we show how the diffusion coefficient $S(X)$ depends on the sampling time h as discussed above. It is clearly seen that the diffusion coefficient σ cannot be unambiguously determined (at least not using this method). We will investigate in Section 5.5 how sensitive the statistics of the climate model (5.11) are to this degree of uncertainty in estimation of the diffusion coefficient.

5.4 TIME SCALES

An important parameter in data assimilation is the observation interval Δt_{obs} . We will see in Section 5.6 that, like our results for the Lorenz-96 model in Chapters 3 and 4, analysis skill improvement is heavily dependent on Δt_{obs} . We therefore now analyse a number of different predictability time scales for the Lorenz-driven system (5.3)-(5.6). Except for the inverse finite size Lyapunov exponent (FSLE) in Section 5.4.4, we do not calculate time scales for the Rössler-driven model (5.7)-(5.10) because of the sensitivity of these time scales to small errors in estimation of the mean $\bar{y}_2$. This sensitivity arises because small errors in $\bar{y}_2$ lead to a slight tendency of the slow variable x to favour one steady state x^* over the other. We find that over long integration times as used here

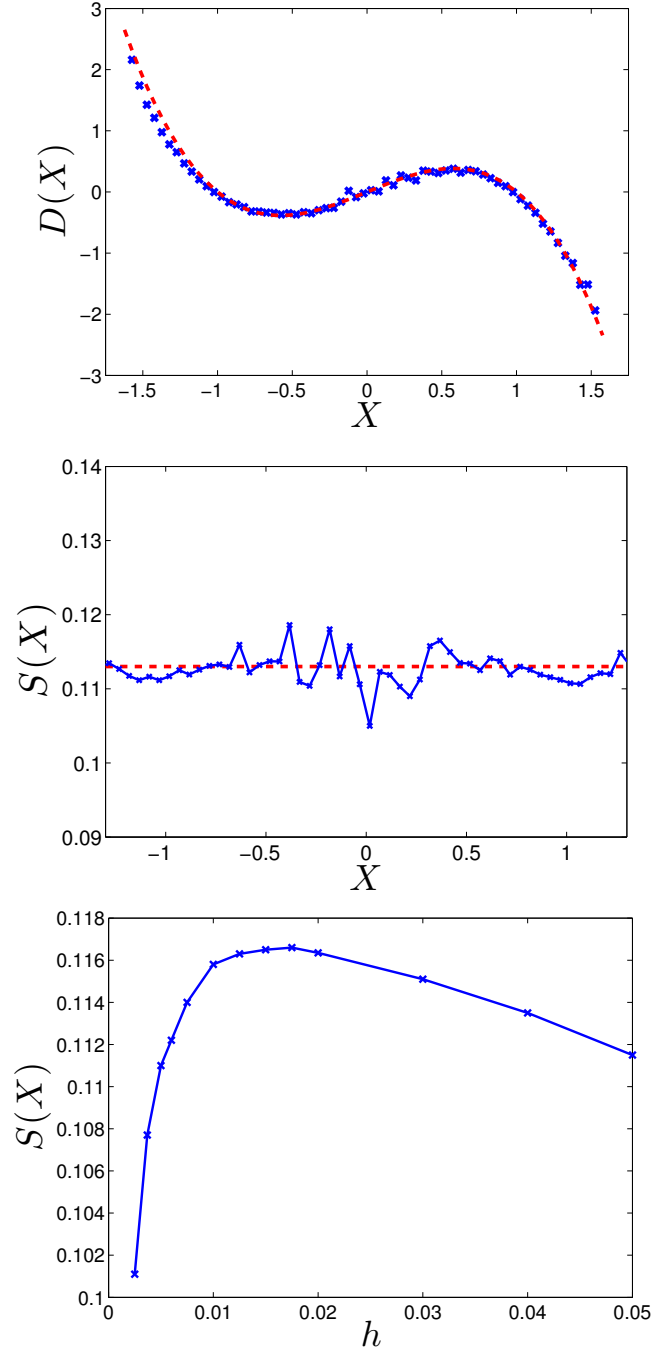


Figure 5.2.: Parameter estimation for the (a) drift term $D(X)$, and (b) diffusion coefficient $S(X)$ from a long trajectory of the full deterministic Lorenz system (5.3)-(5.6) with $\epsilon^2 = 0.0005$ and sample interval $h = 0.005$. The dashed line depicts the theoretically expected drift from the climate model (5.11) in (a), and the constant diffusion coefficient $\sigma^2 = 0.113$ in (b). (c): Diffusion coefficient $S(X)$ averaged over all X , as a function of the sampling time h .

for estimating time scales, the resulting differences in residence times τ_i between the slow metastable states at $x^* = 1$ and $x^* = -1$ adversely affect the calculation of some characteristic time scales. However, in Section 5.4.4 we will still calculate the FSLE spectrum for the Rössler-driven model in order to illustrate an important point about predictability in systems with fast and slow regimes.

5.4.1 Autocorrelation time

The first predictability time scale we estimate is the decorrelation time τ_{corr} , which is linked to the decay rate of the autocorrelation function of the slow variable

$$C(\tau) = \frac{1}{C_0} \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x(s)x(\tau + s) ds , \quad (5.30)$$

where $C_0 = C(0)$. The decorrelation time is estimated as the e -folding time of $C(\tau)$. The autocorrelation function $C(\tau)$ can be estimated from a long trajectory of x obtained from a simulation of the full multiscale model (5.1)-(5.2). For small values of τ it decays approximately exponentially, the exponent of which gives the inverse of the decorrelation time τ_{corr} . For the full deterministic Lorenz model (5.3)-(5.6) we measure a decay rate of 0.00481 corresponding to an e -folding time of $\tau_{\text{corr}} \approx 208$ time units. For the climate model (5.11) with $\sigma^2 = 0.113$ we measure a decay rate of 0.00451, giving the decorrelation time $\tau_{\text{corr}} \approx 222$ time units. We show the autocorrelation functions for the full deterministic model and the associated climate model in Figure 5.3.

5.4.2 Sojourn and exit times

As a second time scale we consider the mean sojourn time

$$\bar{\tau} = \frac{1}{M} \sum_{i=1}^M \tau_i , \quad (5.31)$$

for (5.3)-(5.6), where the sojourn times τ_i measure the amount of time spent in one of the slow metastable states around $x^* = \pm 1$ as depicted in Figure 5.1. Firstly, we

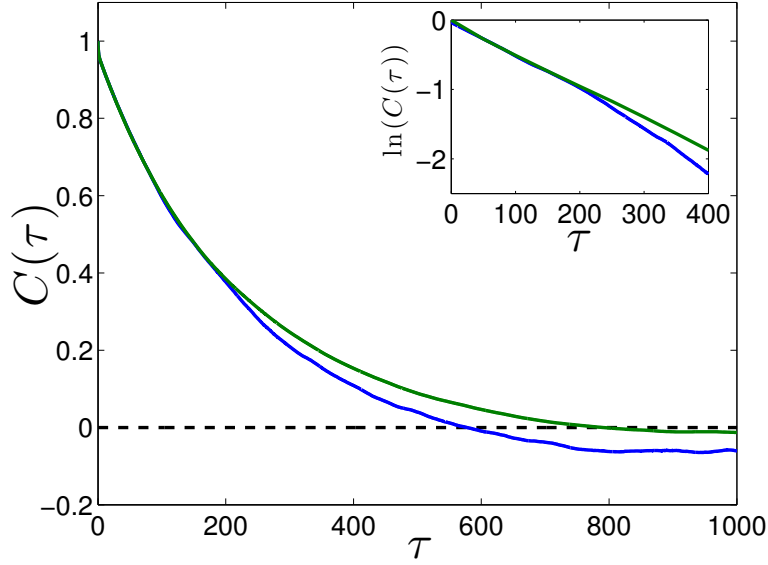


Figure 5.3.: Autocorrelation function $C(\tau)$ as a function of the time lag τ for the slow variable of the multiscale model with Lorenz subsystem (5.3)-(5.6) (blue) and reduced climate model (5.11) (green).

estimate the mean sojourn time directly from a long trajectory of the slow variable x using (5.31) as $\bar{\tau} \approx 218.2$ time units.

Secondly, the succession of rapid transitions and relatively long residence times in the potential wells suggests that successive jumps between metastable states can be treated as independent random events, and that the sojourn times are a Poisson process with

$$P(\tau_i) = \frac{1}{\bar{\tau}} \exp\left(-\frac{\tau_i}{\bar{\tau}}\right) \quad (5.32)$$

and cumulative probability distribution function

$$P_c(\tau_i) = 1 - \exp\left(-\frac{\tau_i}{\bar{\tau}}\right). \quad (5.33)$$

To avoid sampling errors due to binning we use (5.33) instead of (5.32) to numerically estimate the mean sojourn time $\bar{\tau}$. In Figure 5.4 we show the empirical ranked histogram $P_c(\tau_i)$ for the sojourn times measured from a long trajectory $x(t)$ of (5.3)-(5.6), allowing us to determine the mean sojourn time $\bar{\tau} \approx 213.8$ time units. Calculating the mean sojourn time by either the average of individual sojourn times (5.31) or via the Poisson process approximation (5.33) yields results differing by only 2%, indicating that for $\varepsilon^2 =$

0.01 the fast subsystems have almost fully decorrelated and that the rare transitions between the metastable states are approximately a Poisson process. In the following we will use the average of the two obtained values for the mean sojourn time and set $\bar{\tau} = 216.0$.

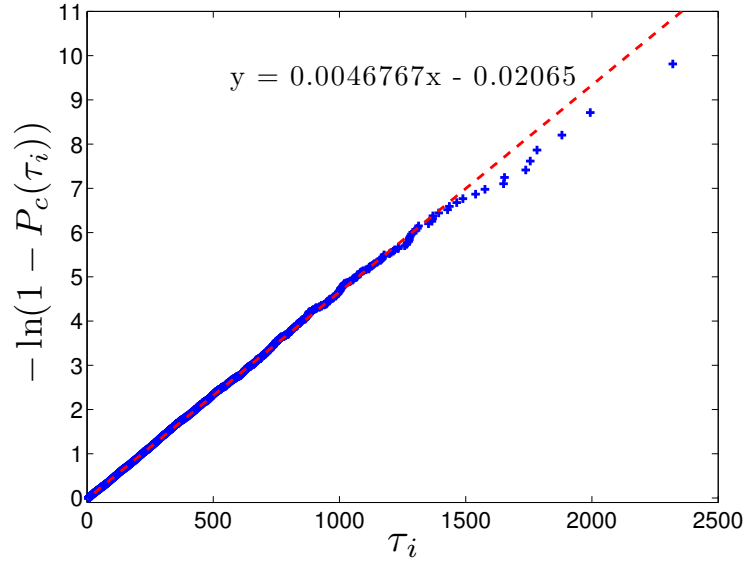


Figure 5.4.: Log-plot of the normalised histogram of sojourn times τ_i of the slow variable x in each metastable state around $x^* = \pm 1$ for the full deterministic Lorenz system (5.3)-(5.6). The dashed line shows a least-square fit. Using (5.33) the mean sojourn time can be estimated via the inverse of the slope of the least-square fit as $\bar{\tau} \approx 213.8$.

These numerically obtained results for the full deterministic system (5.3)-(5.6) compare well with the value calculated analytically for the climate model (5.11) using Kramer's theory (Kramers, 1940) to calculate the first exit time

$$\tau_e = \frac{1}{2} \bar{\tau} .$$

We define the first exit time τ_e to be the average time it takes from the potential well at $x = \pm 1$ to reach the saddle of the potential $V(x)$ at $x = 0$. To calculate the exit time τ_e we solve the following Cauchy problem (see for example Zwanzig (2001))

$$\mathcal{L}_{\text{clim}} \tau_e = -1 , \tag{5.34}$$

where

$$\mathcal{L}_{\text{clim}} = -V'\partial_x + \frac{\sigma^2}{2}\partial_{xx}$$

is the generator of the climate model (5.11), with boundary conditions $\partial_x \tau_e(x = \pm 1) = 0$ and $\tau_e(0) = 0$. Note that the first exit times for the symmetric potential V out of the potential wells at $x = -1$ or $x = 1$ are identical. We can write the first exit problem as

$$\beta^{-1} e^{\beta V(x)} \partial_x \left(e^{-\beta V(x)} \frac{\partial \tau_e}{\partial x} \right) = -1 ,$$

with $\beta = 2/\sigma^2$. Upon using the boundary conditions, we solve (5.34) to obtain

$$\tau_e = \beta \int_{-1}^0 dy e^{\beta V(y)} \int_{-1}^y e^{-\beta V(z)} dz , \quad (5.35)$$

which can be numerically evaluated to yield $\tau_e = 117.8$ for $\sigma^2 = 0.113$. The exact analytical value calculated from the reduced climate model compares well with the numerically estimated values of the first exit times of the full deterministic model $\tau_e = 108.0$, confirming the ability of the homogenised model to faithfully represent the statistics of the full parent model.

We can also approximate τ_e using the saddle point method (see for example Zwanzig (2001)) by first replacing the lower limit of integration for the inner integral in (5.35) by $-\infty$ so that

$$\begin{aligned} \int_{-1}^y e^{-\beta V(z)} dz &= \int_{-\infty}^y e^{-\beta V(z)} dz \\ &\approx \int_{-\infty}^{\infty} e^{-\beta V(z)} dz \\ &\approx \int_{-\infty}^{\infty} \exp(-V(-1)) \exp\left(-\frac{V''(-1)}{2}(x+1)^2\right) dz \\ &= \exp\left(\frac{\beta}{4}\right) \sqrt{\frac{\pi}{\beta}}. \end{aligned}$$

The second line is justified by assuming that $\beta \gg 1$ so that the integral is dominated by the potential value $V(-1) = -1/4$, and the third line is a Taylor series expansion around $x = -1$.

Similarly, for the outer integral in (5.35) we obtain

$$\begin{aligned} \int_{-1}^0 e^{\beta V(y)} dy &\approx \int_{-\infty}^0 e^{\beta V(y)} dy \\ &\approx \int_{-\infty}^0 \exp(\beta V(0)) \exp\left(\beta \frac{V''(0)}{2} x^2\right) dy \\ &= \sqrt{\frac{\pi}{2\beta}} \end{aligned}$$

where the approximation on the first line assumes that the integral is dominated by the potential at the saddle point $V(0)$, and the second line is again a Taylor series for $V(x)$. We thus obtain $\tau_e = (\pi/\sqrt{2}) \exp(\beta/4) = 185.5$ for $\sigma^2 = 0.113$. We remark that this poor approximation to the numerically obtained value $\tau_e = 117.5$ is because the potential values $V(\pm 1)$ do not sufficiently dominate the above integrals. As such the saddle point method is not preferable for this problem.

5.4.3 Transit time

The third time scale we study is the mean transit time τ_t . This is the average over the shortest transit time $\tau_{t,i}$ between the two slow metastable states around $x^* = \pm 1$. We define this time to be the mean of

$$\tau_{t,i} = \min\{(t_i - t_j) | x(t_i) = \pm 1 \wedge x(t_j) = \mp 1\}. \quad (5.36)$$

For a long trajectory containing 5000 transitions between the slow metastable states we numerically estimate $\tau_t = 5.90$ for the full deterministic Lorenz model (5.3)-(5.6).

Similarly to the mean sojourn time $\bar{\tau}$, the transit time τ_t can be analytically calculated for the climate model (5.11) (see for example Gardiner (2003)) as

$$\tau_t = 4 \frac{\Pi_{+1}(0) - \Pi_{-1}(0)}{\sigma^2 \hat{\rho}_{-1}(0)}, \quad (5.37)$$

where

$$\begin{aligned}\Pi_{+1}(x) &= \hat{\rho}_{+1}(x) \int_x^1 dx' \hat{\rho}^{-1}(x') \int_{-1}^{x'} \hat{\rho}_{-1}(z) \hat{\rho}(z) dz \\ \Pi_{-1}(x) &= \hat{\rho}_{-1}(x) \int_{-1}^x dx' \hat{\rho}^{-1}(x') \int_{-1}^{x'} \hat{\rho}_{-1}(z) \hat{\rho}(z) dz ,\end{aligned}$$

with the splitting probabilities

$$\begin{aligned}\rho_{-1}(x) &= \frac{\int_x^1 dz \hat{\rho}^{-1}(z)}{\int_{-1}^1 dz \hat{\rho}^{-1}(z)} \\ \rho_{+1}(x) &= \frac{\int_{-1}^x dz \hat{\rho}^{-1}(z)}{\int_{-1}^1 dz \hat{\rho}^{-1}(z)} .\end{aligned}$$

For the climate model with $\sigma^2 = 0.113$ we evaluate this numerically to be $\tau_t = 5.66$ which again compares well with the numerically obtained value of the transit time for the full deterministic Lorenz system (5.3)-(5.6).

5.4.4 Finite Size Lyapunov Exponents

The final time scale we consider is the inverse of the finite size Lyapunov exponent (FSLE). Although the calculation of this time scale will not be essential for our investigation into the usability of stochastic reduced models in data assimilation, we will present our results here which expose new and surprising results on FSLE spectra for systems with multiple regimes.

The FSLE was introduced to extend the idea of measuring the divergence of nearby trajectories to resolve predictability measures for processes developing on various scales (Aurell et al., 1997; Boffetta et al., 1998a,b). Similar ideas have independently led to the development of *breeding vectors* to produce optimal perturbations for ensemble forecasting (Toth and Kalnay, 1993, 1997). Their average growth rate is closely related to the FSLE (Boffetta et al., 2002; Peña and Kalnay, 2004). FSLEs have been successfully used in recent years to study mixing and transport problems in diverse geophysical

applications (D'Ovidio et al., 2004; Tew Kai et al., 2009; Károlyi et al., 2010; Hernández-Carrasco et al., 2011).

In order to study the problem of predictability in systems with fast and slow regimes, we numerically determine the FSLE spectrum for multiscale systems of the form (5.1)-(5.2) using only the data of the slow x -variable for three cases when the fast subsystem is: A.) the chaotic Rössler system (5.7)-(5.10) with no regimes, B.) the chaotic Lorenz-63 system (5.3)-(5.6) with two regimes and C.) the reduced stochastic system (5.11). We find for these low-dimensional toy multiscale models that, depending on the nature of the fast chaotic system the FSLE exhibits widely different behaviour which can be explained using a simple heuristic argument.

The FSLE $\lambda(\delta)$ introduced in (Aurell et al., 1997; Boffetta et al., 1998a,b) measures the growth rate of a perturbation of finite size δ . It is defined as

$$\lambda(\delta) = \left\langle \frac{1}{T_r(\delta)} \right\rangle_t \ln r = \frac{1}{\langle T_r(\delta) \rangle_{\text{ens}}} \ln r, \quad (5.38)$$

where $T_r(\delta)$ is the time taken for a perturbation of size δ to grow by an amplification factor r , which we take to be $r = 1.1$ throughout. The first average $\langle \cdot \rangle_t$ is taken over the invariant measure of the dynamics which is approximated (assuming ergodicity) by the ensemble average $\langle \cdot \rangle_{\text{ens}}$ over many realisations (we use 5000 realisations throughout). Two trajectories are created starting from an initial separation δ_0 , and the separation δ is measured as the trajectories diverge over time. The perturbations δ are assumed to be already aligned with the most unstable direction, which we guarantee by initialising the estimation with an initial perturbation size of $\delta_0 = 10^{-9}$. The small scale FSLEs with $\delta \rightarrow 0$ correspond to the maximal Lyapunov exponent λ_{max} .

Figure 5.5 shows the FSLE as a function of the perturbation size δ for the Rössler-driven model (5.7)-(5.10). It shows the typical behaviour seen for multiscale systems where for small values of the perturbation size δ a plateau is seen corresponding to the maximal Lyapunov exponent $\lambda_{\text{max}} \approx 10.1$ (Aurell et al., 1997; Boffetta et al., 1998a,b). For larger perturbations δ the FSLE monotonically decreases to reach a minimum with the large scale FSLE $\lambda_{\text{LS}} \approx 0.00227$ at δ slightly larger than 1, suggesting a large-scale predictability time scale of around 440 time units. In particular the associated predictability time is much larger than $1/\lambda_{\text{max}} = 0.0990$. For values $\delta > 1.25$ the

FSLE increases, indicating that trajectories diverge quicker for larger separations. The large scale FSLE measures the predictability associated with the transitions of the slow variable from one slow metastable states near $x^* = \pm 1$ to the other, whereas the small scale FSLE $\lambda_{\max}$ describes fluctuations within each of the slow metastable states.

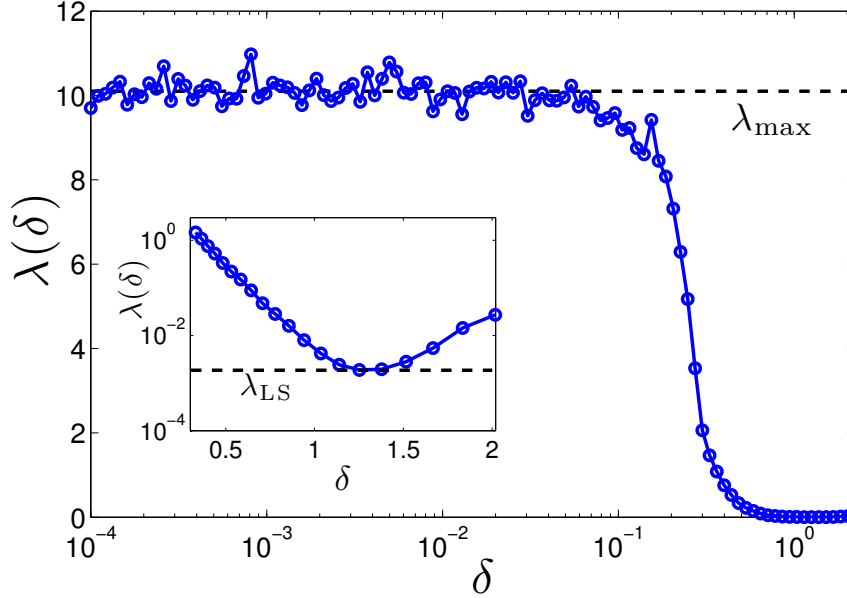


Figure 5.5.: FSLE λ as a function of perturbation size δ for the Rössler-driven system (5.7)-(5.10) with no fast regimes. The inset shows a zoom for large scale disturbances.

In Figure 5.6 we show the FSLE as a function of perturbation size δ for the model with fast Lorenz subsystem (5.3)-(5.6). As in Figure 5.5, for small values of the perturbation size δ we find a plateau corresponding to the maximal Lyapunov exponent which we estimate to be $\lambda_{\max} \approx 97.4$ (for $\varepsilon^2 = 0.01$), and for larger perturbations δ the large-scale FSLE reaches approximately $\lambda_{\text{LS}} \approx 0.00428$ as shown in the inset, suggesting a large-scale predictability time scale of around 234 time units. We note that this is close to the decorrelation time for this system, estimated as $\tau_{\text{corr}} \approx 208$ time units. Like for the Rössler-driven system, the FSLE reaches a minimum at δ slightly larger than 1. Surprisingly, the FSLE spectrum contains several peaks, notably near $\delta = 0.0278$ and $\delta = 0.0790$. This is in stark contrast to the behaviour seen in Figure 5.5 and reported in (Aurell et al., 1997; Boffetta et al., 1998a,b). Note that the FSLE at the first peak is much larger than the maximal Lyapunov exponent $\lambda_{\max}$, suggesting a far greater loss of predictability at those scales.

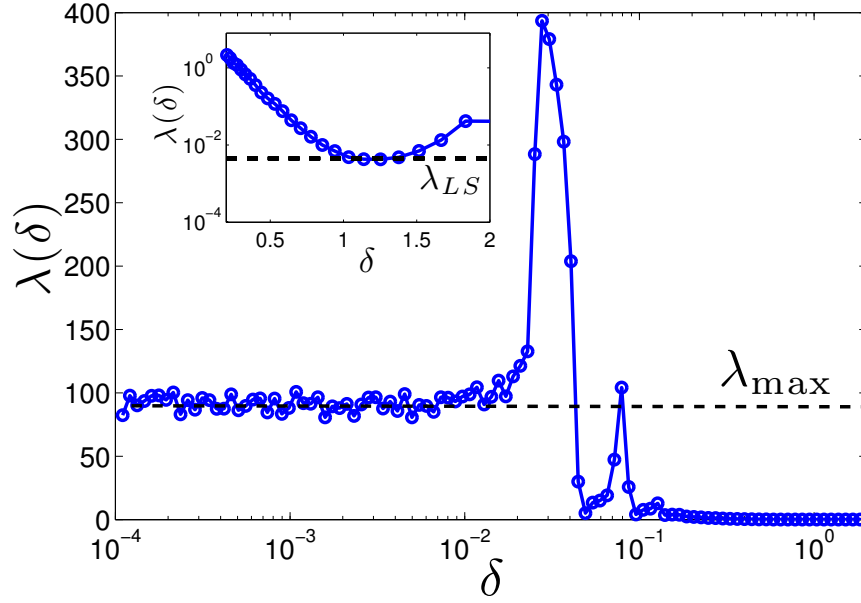


Figure 5.6.: FSLE λ as a function of perturbation size δ for the Lorenz-driven system (5.3-5.6). The inset shows a zoom for large scale disturbances.

We now present simple heuristic analytical arguments explaining the nonmonotonic behaviour of the FSLE $\lambda(\delta)$ observed at both small and large scales. Firstly we explain the peaks observed in the small scale part of the FSLE spectrum, and then the troughs observed in the large scale part of the spectrum. We find that the peaks are related to the presence of multiple regimes in the fast subsystem, and that the troughs are due to the presence of slow metastable states. We then corroborate these findings with numerical experiments.

Firstly, we explain the small scale peaks observed for the FSLE spectrum of the Lorenz-driven system (5.3)-(5.6), in terms of the existence of regimes in the fast process and the switching of the fast dynamics between the two lobes of the butterfly attractor. Figure 5.7 depicts the slow x and the fast y_2 variables of two typical trajectories which are used to calculate the FSLE. The slow variable x evolves in a step-like fashion with step size Δx . Separations between nearby trajectories of $\delta = m\Delta x$ can only occur when the y_2 components of each trajectory are on opposite lobes of the Lorenz attractor. The time scale of one (fast) revolution around a lobe of the Lorenz attractor is $T_f = 0.00691$. Integrating the slow dynamics (5.3) over one fast period T_f and

assuming that no transitions between the metastable states at $x^* = \pm 1$ occur, so that $\int_0^{T_f} x(t)(1 - x^2(t))dt = 0$, we can approximate the step size Δx by

$$\Delta x = \frac{4}{90\epsilon} T_f \langle y_2 \rangle_t, \quad (5.39)$$

where $\langle y_2 \rangle_t = \frac{1}{T_f} \int_0^{T_f} y_2 dt \approx \pm 10$ is the average value of y_2 in each of the fast regimes. This yields a step size of $\Delta x \approx \pm 0.0307$ which corresponds reasonably well with the observed location of the first FSLE peak in Figure 5.6 at $\delta = 0.0278$. The location of the second peak at $\delta = 0.0790$ is roughly approximated by $\delta = 2\Delta x = 0.0614$ according to the above argument.

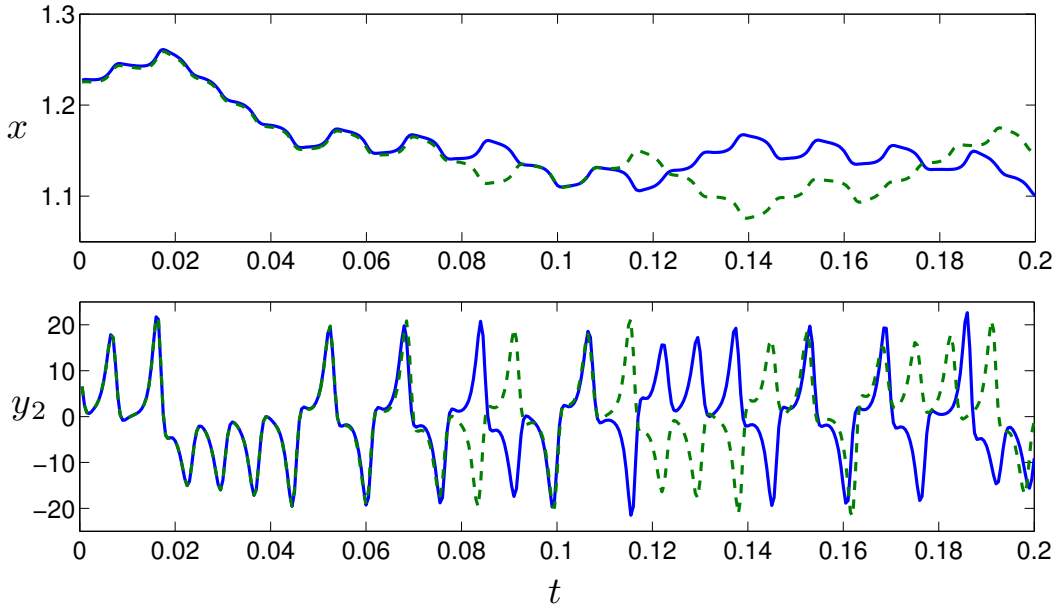


Figure 5.7.: Two trajectories obtained from integrating (5.3)-(5.6) from nearby initial conditions with separation $\delta_0 = 10^{-3}$, showing short time dynamics of the slow x variable, and how increments correlate with switches between regimes of the y_2 -component of the chaotic Lorenz subsystem.

The corresponding FSLEs $\lambda(\delta)$ of the peaks can be estimated as follows. We assume that the separation of slow x trajectories over short times $t = \mathcal{O}(T_f)$ is approximately linear, and that the initial separation δ of trajectories prior to taking a “step” in opposite directions is small. We define the times t_m and t_{rm} at which trajectories are separated by $m\Delta x$ and $rm\Delta x$ respectively. Assuming trajectories move apart at a constant rate of $2\Delta x/T_f = m\Delta x/t_m = rm\Delta x/t_{rm}$ (where $2\Delta x$ is the separation of trajectories after

one step taken in opposite directions, see Figure 5.8), the time $T_r(m\Delta x)$ taken for a perturbation of size $m\Delta x$ to grow to size $rm\Delta x$ is

$$\begin{aligned} T_r(m\Delta x) &= t_{rm} - t_m \\ &= rt_m - t_m \\ &= mT_f(r - 1)/2. \end{aligned}$$

We can therefore approximate the FSLE for the m th separation $m\Delta x$ using (5.38) as

$$\lambda(m\Delta x) = \frac{2 \ln(r)}{m(r - 1)T_f}.$$

For $r = 1.1$ we find $\lambda(\Delta x) = 276$ and $\lambda(2\Delta x) = 138$, which, given the crude approximations we made, provide a rough estimate of the numerically observed peaks $\lambda(0.0278) = 394$ and $\lambda(0.0790) = 104$ in Figure 5.6.

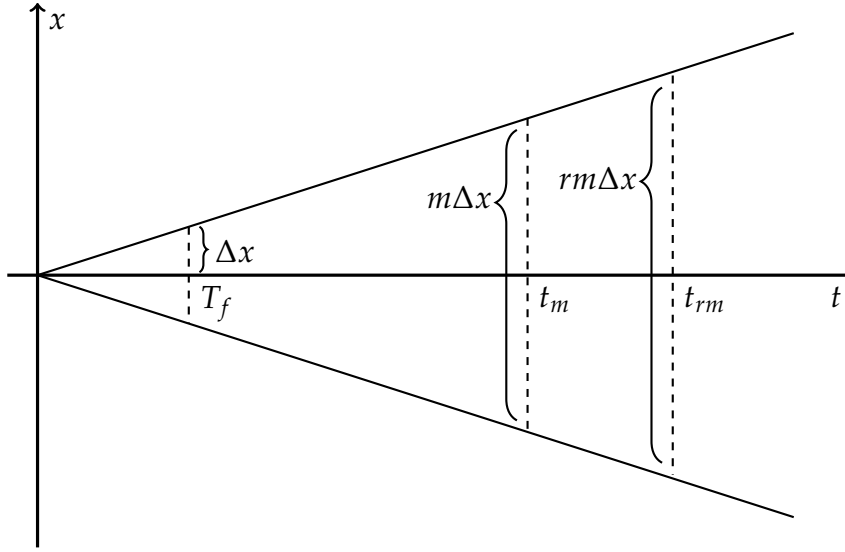


Figure 5.8.: Schematic diagram showing growth of a small perturbation to sizes Δx , $m\Delta x$ and $rm\Delta x$. Using similar triangles we can obtain the relation $2\Delta x/T_f = m\Delta x/t_m = rm\Delta x/t_{rm}$.

Secondly, the functional dependency of the FSLE $\lambda(\delta)$ exhibiting a minimum at large scales can be understood by considering the bimodal probability density of the slow variables. Each of the two maxima in the pdf (5.26) has a support of roughly 1 (see for example Figure 5.12). Perturbations larger than this size are therefore likely to be on opposite sides of the saddle in the potential $V(x)$ at $x = 0$. Therefore, perturbations of

around this critical size will separate quickly, as we have seen that the transit time τ_t between states across the saddle is short compared with the exit time τ_e .

We now corroborate the above explanations for the peaks and troughs in the FSLE spectrum in terms of fast and slow regimes respectively with numerical experiments. To show the robustness of our argument explaining the location of the peaks we vary the coupling parameter κ in (5.3) and track the resulting locations of the peaks. To show that slow metastable states are responsible for producing distinct minima in the FSLE spectrum we examine how the location of the large scale FSLE depends on the location of the slow metastable state b , where the drift term

$$d(x) = ax(b^2 - x^2)$$

determines the slow metastable states $x^* = \pm b$. To assure invariance of the pdf of x for varying b we also scale the coupling coefficient κ with b , so that $\kappa \rightarrow \frac{4}{90}b^2\kappa$ in (5.3).

In Figure 5.9 we show how the location of the peaks and minima depend on the coupling and drift terms respectively. In the left panel we show how the location of the first two peaks varies with the coupling strength κ in (5.3). The figure shows that our argument above is robust to variations in the coupling strength between fast and slow subsystems; our estimate for the peak location Δx in (5.39) remains very good for both the first and second peaks, even when increasing the coupling by approximately 10 times, confirming that the presence of peaks in the FSLE spectrum is indeed due to the presence of regimes in the fast subsystem. The right panel of Figure 5.9 shows how the location of the large scale FSLE depends on b . The linear behaviour confirms that the location of the minimum of the FSLE spectrum is a signature of slow metastable states in the system.

The reduced stochastic model (5.11) does not involve any metastable states associated with the fast subsystem, hence there are no peaks in its FSLE spectrum, as seen in Figure 5.10. We show the FSLE spectrum for $\sigma^2 = 0.113$, for which the full dynamics of the Lorenz-63 driven system (5.3)-(5.6) is best approximated as in Section 5.3. We find that the FSLE for small δ of the stochastic system do not reproduce the values of their parent systems, and no plateau is observed in the FSLE spectrum for small δ . This is as expected, as the maximal Lyapunov exponent $\lambda_{\max}$ is not defined for the stochastic system. However, at large scales δ which are depicted in the inset, we

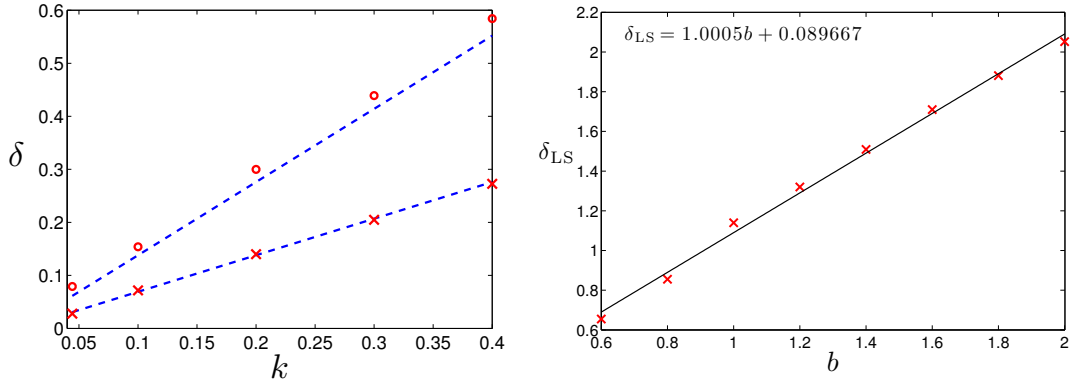


Figure 5.9.: Left: location of the FSLE peaks δ as a function of the coupling parameter κ in (5.3)-(5.6). The crosses show the locations of the first peak, the circles the location of the second. The dashed lines are the locations predicted by (5.39). Right: location of large scale minimum δ_{LS} (crosses) as a function of the slow metastable state b . The solid black line is a least-squares fit.

find $\lambda_{LS} = 0.00485$ for $\sigma^2 = 0.113$, giving a large scale predictability time of 206.4 time units, close to that obtained for the full parent model (5.3)-(5.6). We note that the long time slow dynamics of the Rössler system (5.7)-(5.10) is not well approximated by the reduced stochastic system (5.11) because small errors in the estimation of the mean $\bar{y}_2$ dominate the behaviour at long times. This violates the solvability condition $\langle f(y) \rangle_{\rho_\infty} = \langle y_2 \rangle_{\rho_\infty} = 0$ which was necessary for the derivation of the reduced stochastic model.

The anomalous behaviour of FSLE peaks was not observed by Boffetta et al. (1998b), who studied a system of nonlinearly coupled fast and slow Lorenz systems. This is entirely due to the nature of the coupling used; we check below that for linear coupling again large peaks in the FSLE spectrum are observed. Linearly coupled Lorenz-63 systems were studied for example in (Peña and Kalnay, 2004) to model ENSO events.

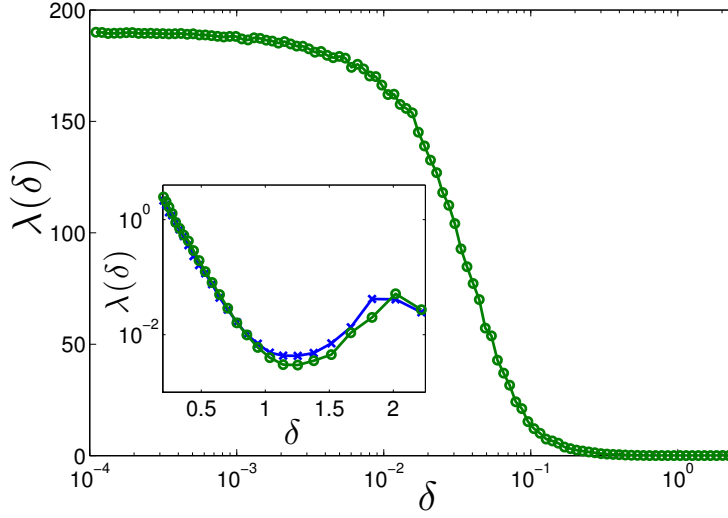


Figure 5.10.: FSLE λ as a function of perturbation size δ for the reduced stochastic model (5.11) with $\sigma^2 = 0.113$. The inset shows larger-scale FSLEs for the reduced stochastic model (green, circles) and the full deterministic model (blue, crosses), showing good agreement in the location of the minimum and the large-scale FSLE.

We consider a similar system to the two coupled Lorenz models of (Boffetta et al., 1998b)

$$\frac{dx_s}{dt} = \sigma(y_s - x_s) \quad (5.40)$$

$$\frac{dy_s}{dt} = r_s x_s - y_s - x_s z_s - \mathcal{K}_s \quad (5.41)$$

$$\frac{dz_s}{dt} = x_s y_s - b z_s \quad (5.42)$$

$$\frac{dx_f}{dt} = \frac{\sigma}{\varepsilon}(y_f - x_f) \quad (5.43)$$

$$\frac{dy_f}{dt} = \frac{1}{\varepsilon}(r_f x_f - y_f - x_f z_f) + \mathcal{K}_f \quad (5.44)$$

$$\frac{dz_f}{dt} = \frac{1}{\varepsilon}(x_f y_f - b z_f), \quad (5.45)$$

where we use $\mathcal{K}_s = k_s y_f$ and $\mathcal{K}_f = k_f y_s$. As in Boffetta et al. (1998b) we choose the canonical parameters $\sigma = 10$, $r_s = 28$, $b = 8/3$ for the slow system, and $\varepsilon = 0.1$, $r_f = 45$ for the fast system. Setting $\mathcal{K}_s = k_s y_f x_f$ and $\mathcal{K}_f = k_f y_s x_f$ in equations (5.41) and (5.44) recovers the original system of Boffetta et al.

In Figure 5.11 we show how the FSLE depends on δ for Boffetta *et al.*'s original model and our modified system with linear coupling. In both models, for small values of δ we see a plateau corresponding to the maximal Lyapunov exponent of the fast subsystem, $\lambda_{\max} = 12.17$. There is a transition to another regime for perturbations larger than $\delta > 0.01$ and for both models the large scale FSLE plateaus at $\lambda_{\text{LS}} \approx 0.95$. However, for perturbations $\delta \approx 0.01$ there is a small region where for the system with linear coupling trajectories separate extremely quickly and we observe FSLEs up to 4 times larger than $\lambda_{\max}$. This is not observed for the nonlinearly coupled system because such a choice of $\mathcal{K}_s$ does not induce rapid variations in the slow dynamics of the form seen in Figure 5.7. The nonlinearly coupled example in Boffetta *et al.* (1998b) shows that the existence of fast regimes is not sufficient for the occurrence of peaks in the FSLE spectrum.

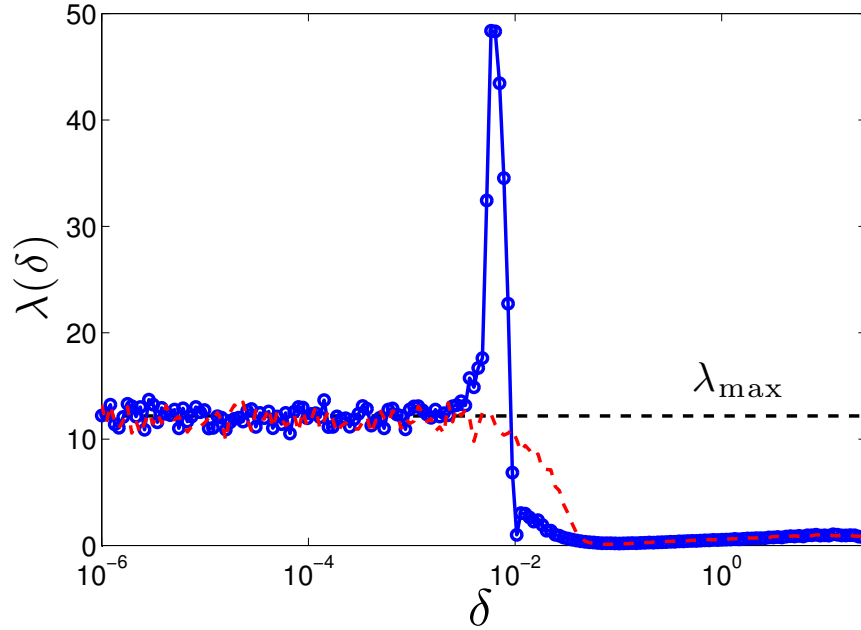


Figure 5.11.: FSLE λ as a function of perturbation size δ for the coupled Lorenz system (5.40)-(5.45) with linear coupling (blue, circles) and the nonlinear coupling of Boffetta *et al.* (1998b) (red, dashed). We used $r = 1.1$ and averaged over 5000 realisations. The trajectories were started from an initial perturbation size of $\delta_s(0) = 10^{-12}$ for the slow dynamics and $\delta_f(0) = 10^{-8}$ for the fast dynamics.

Finally, we remark that in Boffetta *et al.* (1998b) a minimum in the FSLE spectrum was found for a coupled map system which has a unimodal probability density function but the dynamics of which consists of laminar phases interrupted by intermittent large

amplitude bursts. This is in accordance with the reasoning of this section of a critical perturbation size above which the dynamics dramatically increases sensitivity.

Connecting back now to our main objective in this section of identifying the characteristic time scales of the multiscale model (5.1)-(5.2), we summarise the values of the four time scales introduced in this section for the full Lorenz-driven deterministic model and the associated climate model in Table 5.1. These results are for $\sigma^2 = 0.113$, the best estimate of the driving noise for the reduced stochastic model. The percentage error estimates in the table show that all time scales are well reproduced by the stochastic reduced model for this value of the diffusion coefficient σ^2 . How well the time scales are reproduced for different values of σ^2 is the subject of the next section.

Table 5.1.: Characteristic time scales of the full Lorenz-driven model (5.3)-(5.6) with $\varepsilon^2 = 0.01$ and the climate model (5.11) with $\sigma^2 = 0.113$. We indicate the error with respect to the full model in percentage.

	full model	climate model ($\sigma^2 = 0.113$)
Large-scale predictability time $\frac{1}{\lambda_{LS}}$	233.7	206.4 (13.2%)
Decorrelation time τ_{corr}	208.0	221.7 (6.6%)
Exit time τ_e	106.9	117.8 (10.2%)
Transit time τ_t	5.9	5.7 (4.1%)
Small-scale predictability time $\frac{1}{\lambda_{\text{max}}}$	0.0103	n.a.

5.5 VALIDITY OF THE CLIMATE MODEL

In this section we numerically investigate to what degree the reduced stochastic model (5.11) faithfully reproduces the slow dynamics of the full deterministic model with fast Lorenz subsystem (5.3)-(5.6). It is clear that such a correspondence can only be of a statistical nature. The close correspondence of the time scales of the full deterministic system and the climate model for $\sigma^2 = 0.113$ and $\varepsilon^2 = 0.01$ already indicates the validity of the stochastic climate model in reproducing statistical aspects of the slow dynamics. The rigorous theory by Kurtz (1973) and Papanicolaou (1976) assures weak convergence of the climate model in the limit of $\varepsilon \rightarrow 0$. Here we discuss the effect of

finite time scale separation ε . Furthermore we study the sensitivity of the climate model to uncertainties in the diffusion coefficient σ^2 and drift term as we have encountered in Section 5.3.

5.5.1 Probability density function

We first examine how the probability density function $\hat{\rho}(x)$ for the slow variables varies for different values of the time scale separation parameter ε . In Figure 5.12 we show empirical densities for the slow variable x obtained from a long simulation of the full deterministic system (5.3)-(5.6) for different values of ε . The obtained estimates of the empirical invariant densities exhibit a bimodal structure reflecting the two slow metastable states near $x^* = \pm 1$. The probability density functions are reasonably insensitive to changes in ε^2 , with the maxima of the densities at $x^* = \pm 1$ differing for $\varepsilon^2 = 0.0005$ and $\varepsilon^2 = 0.01$ by less than 1%, and by approximately 5% for $\varepsilon^2 = 0.01$ and $\varepsilon^2 = 0.1$. This demonstrates numerically the weak convergence of solutions of the homogenised stochastic climate model to solutions of the full deterministic slow dynamics.

Next we show in Figure 5.13 how sensitive the empirical density of the climate model (5.11) is to changes in the diffusion coefficient σ^2 . Unlike for changes in the time scale parameter ε , the climate model is very sensitive to changes in the diffusion coefficient. The invariant density of the full deterministic model is best estimated by the climate model with $\sigma^2 = 0.113$; larger values of σ^2 mean that transitions between states occur more frequently and hence broaden the peaks, while smaller values of σ^2 lead to less frequent transitions and so thinner and taller peaks.

Finally we examine how the statistics of the climate model (5.11) depends on uncertainties in the estimation of the drift term $d(X) = aX(b^2 - X^2)$. The parameter b controls the location of the metastable states near $x^* = \pm b$ and their separation. The height of the potential barrier $\Delta V(x) = ab^4/4$ is controlled by both a and b . In Figure 5.14 we show the invariant density (5.26) for the climate model with $\sigma^2 = 0.113$ and several combinations of the drift term parameters a and b . The invariant density is more sensitive to changes in b than in a (not shown) since b simultaneously affects the

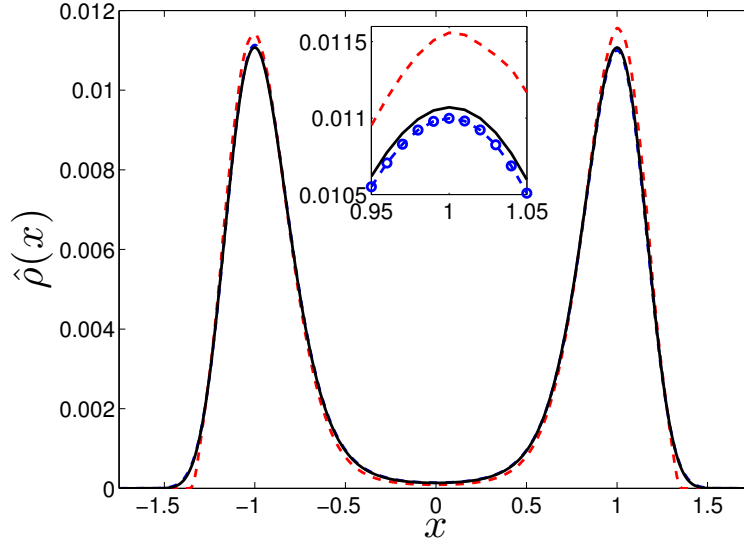


Figure 5.12.: Normalised empirical densities for the slow variable x of the full deterministic model (5.3)-(5.6) with $\epsilon^2 = 0.1$ (dashed line), $\epsilon^2 = 0.01$ (solid line) and $\epsilon^2 = 0.0005$ (circles). The empirical densities were obtained from long trajectories integrated to time $T = 10^9$ time units.

distance of the minima of the potential as well as the height of the potential barrier. The location of the maxima of the probability density function changes by approximately 22 % and the actual values of the maxima change by approximately 27 % over the range of b values shown. Varying a produces similar changes in $\hat{\rho}(x)$, but as expected the locations of the maxima of the probability density function are not shifted.

5.5.2 Time scales

Differences in the probability density function imply different statistics of the dynamics. In particular, we will now examine how the characteristic time scales of the full model (5.3)-(5.6) change with the time scale separation ϵ , and how they change for the climate model (5.11) with varying diffusion coefficients σ^2 .

Consistent with our earlier observation that the probability density function does not vary greatly with the time scale separation parameter ϵ (provided it is sufficiently small), we confirm here that the time scales are insensitive to a decrease of ϵ for $\epsilon^2 < 0.01$. In

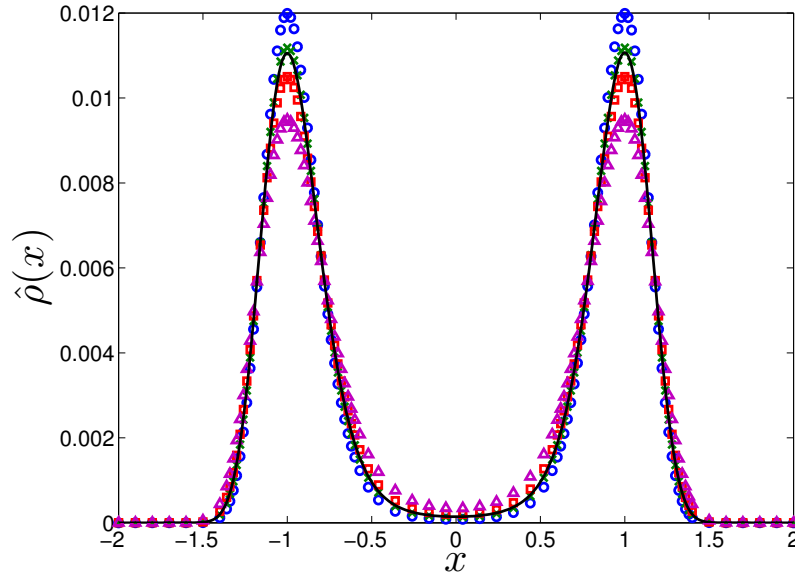


Figure 5.13.: Invariant density (5.26) for the climate model (5.11), with $\sigma^2 = 0.1$ (circles), $\sigma^2 = 0.113$ (crosses), $\sigma^2 = 0.126$ (squares) and $\sigma^2 = 0.15$ (triangles), and empirical density for the slow variable x of the full deterministic model (5.3)-(5.6) with $\varepsilon^2 = 0.01$ (solid line).

particular the transit time τ_t exhibits very little variation with ε . We summarise the relevant time scales for different values of ε^2 in Table 5.2.

Table 5.2.: Characteristic time scales of the full model (5.3)-(5.6) for different values of ε^2 .

	$\varepsilon^2 = 0.0005$	$\varepsilon^2 = 0.01$	$\varepsilon^2 = 0.1$
$\frac{1}{\lambda_{LS}}$	333.4	233.7	546.2
τ_{corr}	201.6	208.0	331.4
τ_e	97.6	107.1	390.3
τ_t	5.9	5.9	6.0

With the exception of the transit time τ_t , we find that the characteristic time scales of Section 5.4 are very sensitive to uncertainties in the estimation of the diffusion coefficient σ^2 . For example, in Figure 5.15 the autocorrelation function $C(\tau)$ is shown as estimated from a long trajectory of x obtained from a simulation of the full deterministic model (5.3)-(5.6), and from simulations of the climate model (5.11) with different values of the diffusion coefficient σ^2 . Using $\sigma^2 = 0.113$ in (5.11) produces the best fit to the shape

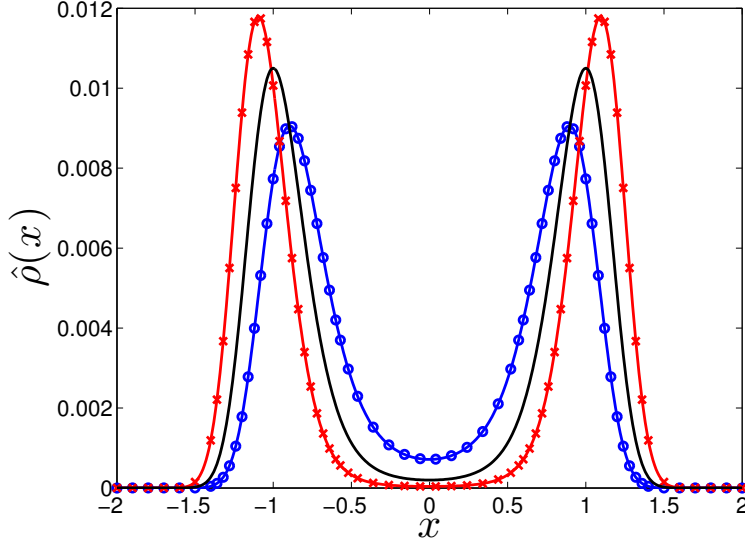


Figure 5.14.: Invariant density (5.26) for the climate model (5.11), with $\sigma^2 = 0.113$ and $(a, b) = (1, 1)$ (solid line), $(a, b^2) = (1, 0.8)$ (circles) and $(a, b^2) = (1, 1.2)$ (crosses).

of $C(\tau)$ for the full deterministic model (5.3)-(5.6). Sensitivity of the autocorrelation function to changes in drift and diffusion can also be found in (Franzke et al., 2005; Franzke and Majda, 2006) on homogenisation of barotropic fluids, who found that stochastic reduced models do not necessarily reproduce the autocorrelation function of the full deterministic system well. We note however that Franzke et al. (2005) applied a method which is only valid in the limit of large time scale separation to a model with no time scale separation; considering this their results reproduce the autocorrelation function surprisingly well.

Table 5.3 lists the values of the characteristic time scales for the climate model (5.11) for various values of the diffusion coefficient σ^2 . We show the percentage difference of each value to that for the full deterministic model (5.3)-(5.6) with $\varepsilon^2 = 0.01$ in brackets. All time scales vary by approximately a factor of 5 over the range of values indicated, except for the transit time which varies by just 13% and approximates the transit time of the full deterministic model better as σ^2 decreases. The transit time is determined by the rate of divergence associated with the unstable fixpoint at $x = 0$ of the deterministic drift term, making it relatively insensitive to changes in the diffusion coefficient. The transit times of the full deterministic model are slightly underestimated, since the definition (5.36) does not take into account the skewness of the probability

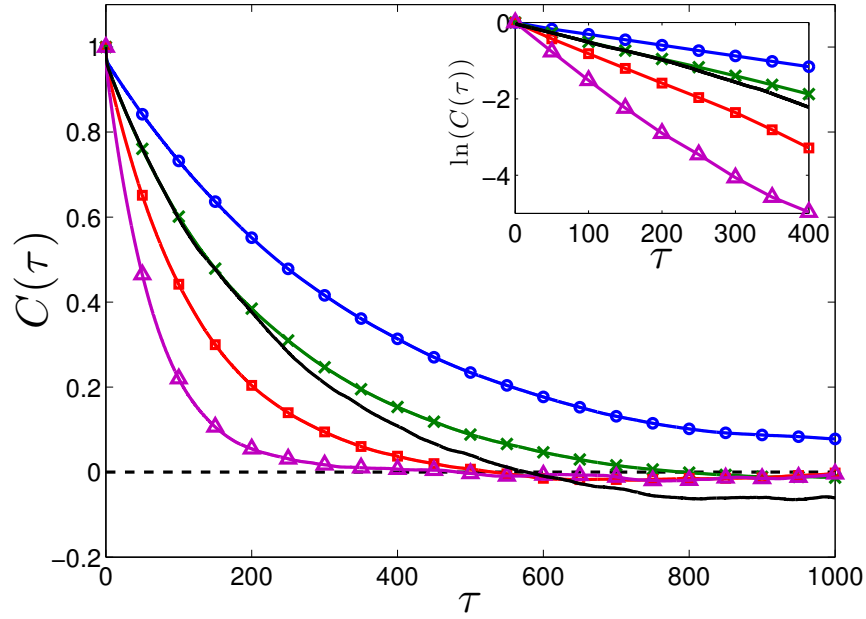


Figure 5.15.: Autocorrelation function $C(\tau)$ as a function of the time lag τ for the full deterministic model (5.3)-(5.6) (black solid line) and for the stochastic climate model (5.11) with $\sigma^2 = 0.1$ (blue, circles), $\sigma^2 = 0.113$ (green, crosses), $\sigma^2 = 0.126$ (red, squares) and $\sigma^2 = 0.15$ (cyan, triangles).

density function (see Figure 5.12). This skewness means that the trajectory spends more time within the range $x \in [-1, 1]$ than outside this range. The table indicates that all time scales (except the transit time τ_t) are best approximated by the climate model with $\sigma^2 = 0.113$, consistent with the better approximation of the probability density function for that value of σ^2 . However, we will see in Section 5.6 that in the context of ensemble data assimilation it is not necessarily ideal to use this value for σ^2 ; in fact it is preferable to use models with larger diffusion to create a more reliable ensemble with superior analysis skill.

Table 5.3.: Characteristic time scales of the climate model (5.11) for different values of the diffusion coefficient σ^2 . In brackets we indicate the error when compared to values obtained from the full deterministic model (5.3)-(5.6) with $\varepsilon^2 = 0.01$.

	$\sigma^2 = 0.1$	$\sigma^2 = 0.113$	$\sigma^2 = 0.126$	$\sigma^2 = 0.15$
τ_{corr}	353.9 (70.1%)	221.7 (6.6%)	129.0 (61.2%)	70.5 (195%)
τ_e	205.7 (90.5%)	117.8 (9.1%)	75.6 (42.6%)	40.8 (164.7%)
τ_t	5.86 (0.07%)	5.66 (4.2%)	5.48 (7.7%)	5.17 (14.1%)

As for the diffusion coefficient, differences in the approximation of the empirical density imply sensitivity of the statistics to changes in the drift term parameters. In Table 5.4 we show how the characteristic time scales change for varying drift term parameters. Note that now the transit time τ_t varies, because varying a modifies the rate of divergence of the unstable fixed point. As for the invariant density the time scales are consistently much more sensitive to changes in b than in a . All time scales except for the transit time τ_t are higher for larger a or b , because this increases the height of the potential barrier $\Delta V(x)$. As expected, the transit time decreases for larger a and b , due to the increase in steepness of the potential barrier.

Table 5.4.: Characteristic time scales of the climate model (5.11) with $\sigma^2 = 0.126$ for different values of the drift coefficients a and b .

	$a = 0.8$ $b = 1$	$a = 1.2$ $b = 1$	$a = 1$ $b^2 = 0.8$	$a = 1$ $b^2 = 1.2$
$\frac{1}{\lambda_{LS}}$	115.6	416.7	93.3	1079.9
τ_{corr}	77.3	256.3	41.1	645.7
τ_e	43.7	136.1	31.1	212.9
τ_t	6.4	4.8	7.2	4.5

In the next Section we perform data assimilation for the toy model introduced in Section 5.2. We examine when and why the climate model (5.11) can be used as a forecast model to improve the analysis skill. In particular, we will relate the range of validity of the climate model to the relation between the observation interval and the characteristic time scales introduced in Section 5.4.

5.6 CLIMATE MODEL FOR FORECASTING IN DATA ASSIMILATION

We now investigate how the ETKF performs when either the full 4D deterministic system (5.3)-(5.6) or the reduced stochastic 1D climate model (5.11) is used for forecasting when the truth evolves according to the full deterministic model. We are concerned here with the following questions: can the dimension reduced climate model produce a

better analysis than the full deterministic model? If so, under what circumstances does the climate forecast model perform better and why?

We perform twin experiments using the ETKF as described in Chapter 2 to compare how well the full model and reduced stochastic models perform when used in the forecast step. We generate a truth $\mathbf{z}_t = (x_t, y_{1t}, y_{2t}, y_{3t})$ by integrating the full model (5.3)-(5.6) with $\varepsilon^2 = 0.01$ using a fourth order Runge-Kutta scheme with timestep $dt = \varepsilon^2/20$. Observations of the slow variables only are then generated at equidistant times separated by the observation interval Δt_{obs} according to $x_{\text{obs}} = x_t + \eta$ where $\eta \sim \mathcal{N}(0, \sqrt{R_o})$ with $R_o = 0.5\sigma^2$ and $\sigma^2 = 0.126$. Note that this value is slightly larger than the optimal value $\sigma^2 = 0.113$ which was found in Section 5.3 to best approximate the full deterministic model. The choice of σ^2 on our results will be discussed later in this section.

To integrate the forecast model we use a fourth order Runge-Kutta scheme with the same timestep dt for the full deterministic model, and an Euler-Maruyama scheme for the reduced stochastic climate model using the same timestep. One advantage of using a reduced stochastic model in the forecast step is the ability to use a much larger timestep in the model integration than required to stably integrate the stiff deterministic multiscale system. However, such computational aspects will not be our primary focus here. Unless stated otherwise we use $\sigma^2 = 0.126$ for the climate forecast model, again slightly larger than the optimal value $\sigma^2 = 0.113$ found in Section 5.3. We include a spin-up time of 100 analysis cycles in all experiments. We perform twin experiments, where both forecast models are started from the same initial conditions, and are then subsequently assimilated using the same truth and same observations over 5000 time units. In order to avoid rare occasions of filter divergence due to an underestimation of the forecast covariances we employ a multiplicative inflation factor of $\delta = 1.02$ (Anderson and Anderson, 1999). We note that this level of inflation has negligible impact on the RMS errors we will calculate below.

Figure 5.16 shows two sample analysis time series created by assimilating the same truth and observations at regularly spaced intervals of $\Delta t_{\text{obs}} = 50$ where the climate model and the full model are used as the respective forecast models. Both forecast models track the truth well. The climate model, however, is able to more accurately

track the transitions between the slow metastable states around $x^* = \pm 1$, for example in the interval $3200 < t < 4000$.

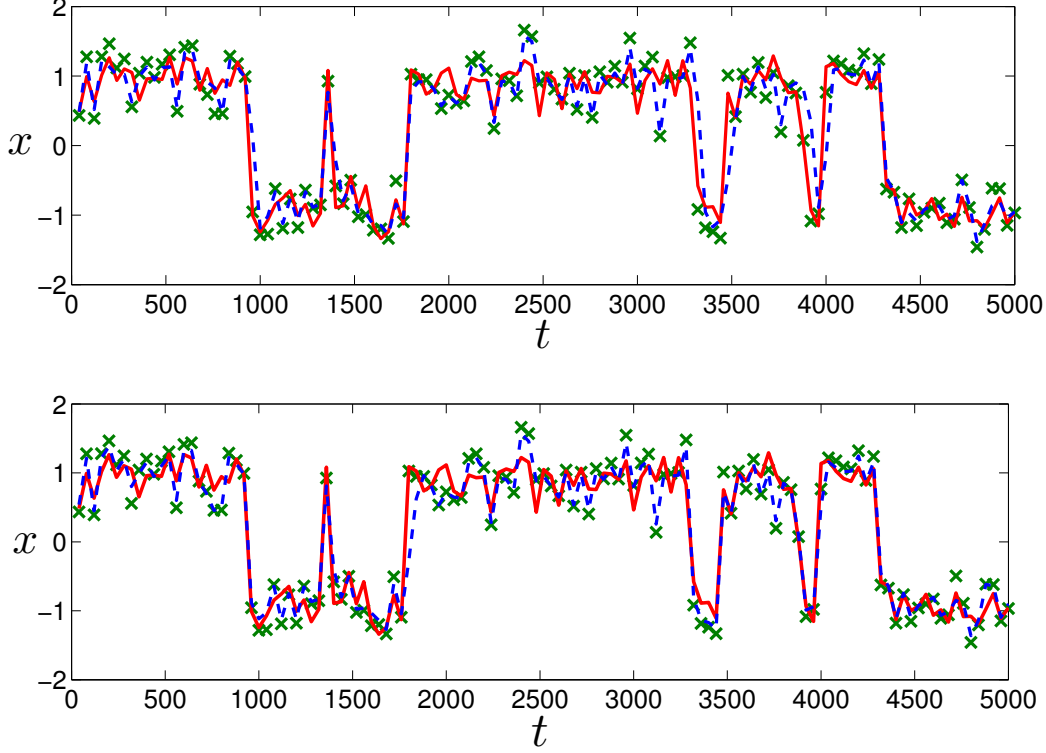


Figure 5.16.: Sample truth (solid line) and ETKF analyses (dashed line) for the observed x -component using the full deterministic model (5.3)-(5.6) (top panel) and the reduced stochastic climate model (5.11) (bottom panel) as the forecast model. The crosses are observations with error variance $R_o = 0.5\sigma^2$ ($\sigma^2 = 0.126$). Here we have used an observation interval $\Delta t_{\text{obs}} = 50$ and used $k = 15$ ensemble members.

As in Chapters 3 and 4, we quantify the improvement made using the reduced climate model by measuring the RMS error $\mathcal{E}$ between the observed truth x_t and the analysis mean x_a ,

$$\mathcal{E} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_a(t_i) - x_t(t_i))^2},$$

where N is the total number of analysis cycles. Note that unlike in previous chapters, $\mathcal{E}$ is calculated here over the slow observed variables x only. In Figure 5.17 we show $\mathcal{E}$ as averaged over 1000 realisations, where we have used both the full deterministic

model and the reduced stochastic climate model as forecast models, as a function of the observation interval Δt_{obs} . Both filters exhibit the same accuracy for small observation intervals $\Delta t_{\text{obs}} < 10$, when the system is frequently observed, and for very large observational times $\Delta t_{\text{obs}} > 500$, when the forecast is much larger than all the characteristic time scales we discussed in Section 5.4. For very large observation intervals (not shown), the ensemble of both forecast models will have explored the state space of the slow variable with variance $\sigma_{\text{clim}}^2 \approx 1$. The resulting analysis error variance can be estimated using the addition rule of covariances (2.12) as $P_a^{-1} = \sigma_{\text{clim}}^{-2} + R_o^{-1}$ as $P_a = 0.244^2$ for $R_o = 0.5\sigma^2$ and $\sigma^2 = 0.126$, which corresponds well with the measured asymptote of the RMS error $\mathcal{E}$ in Figure 5.17. Interestingly, for moderate observation intervals Δt_{obs} the climate model performs considerably better than the full deterministic model. Moreover, the analysis using the full deterministic forecast model exhibits analysis errors worse than the observational error of $\sqrt{R_o} = 0.251$. In contrast, when the climate model is used in the forecast step, the analysis error is always below the observational error, meaning that performing data assimilation is desirable over just trusting the observations (albeit with an improvement in RMS error of only around 5% for large Δt_{obs}).

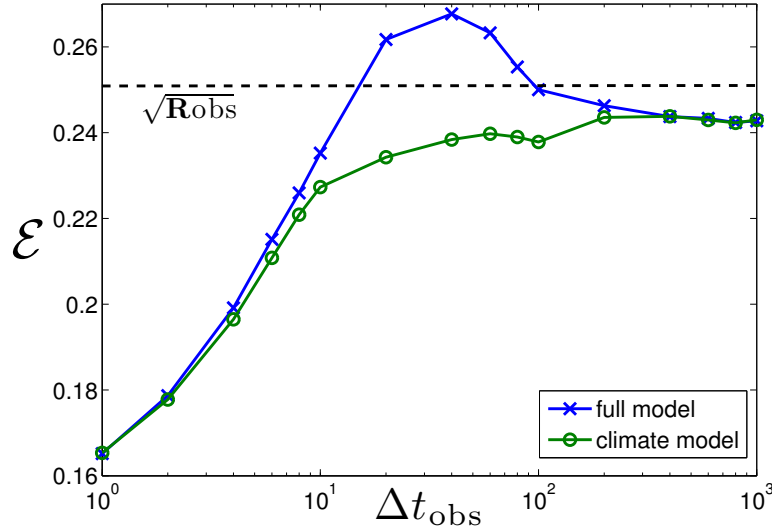


Figure 5.17.: RMS errors $\mathcal{E}$ of the analysis using the full deterministic model (5.3)-(5.6) (crosses) and the climate model (5.11) with $\sigma^2 = 0.126$ (circles) as forecast models, as a function of the observation interval Δt_{obs} . The horizontal dashed line indicates the observational error, $\sqrt{R_o} = \sigma/\sqrt{2} = 0.251$ ($\sigma^2 = 0.126$). The results are averaged over 1000 realisations.

Analogously to previous chapters, we introduce the proportional improvement in the RMS error $\mathcal{E}$ or skill

$$\mathcal{S} = \frac{\mathcal{E}_{\text{full}}}{\mathcal{E}_{\text{clim}}},$$

where $\mathcal{E}_{\text{full}}$ and $\mathcal{E}_{\text{clim}}$ are the RMS errors when the full deterministic or reduced stochastic model is used in the forecast step respectively. Values of $\mathcal{S} \geq 1$ imply that it is beneficial to use the reduced stochastic climate model. Note that when $\mathcal{S} = 1$ it is still beneficial to use the climate model because of the reduced computational expense. Figure 5.18 shows $\mathcal{S}$ as a function of the observation interval Δt_{obs} , averaged again over 1000 realisations, with the time scales τ_t , τ_e and τ_{corr} superimposed. Skill improvements occur only for observation intervals larger than the mean transit time τ_t , with the maximum skill improvement over all forecasts occurring for Δt_{obs} slightly smaller than the mean exit time τ_e . If forecasts are longer than τ_{corr} , the full and climate models are essentially indistinguishable with $\mathcal{S} = 1$.

Figure 5.18 also shows how the skill $\mathcal{S}$ is distributed over the whole set of the analyses including metastable states and transitions between them (circles), as well as over only those analyses which remain within a metastable state near either $x^* = -1$ or $x^* = 1$ (crosses), and those which fall in the transitions between the metastable states (squares). Note that for large observation intervals $\Delta t_{\text{obs}} > \tau_{\text{corr}}$ half of all forecasts switch between regimes, so the notion of analyses which remain within one metastable regime becomes obsolete. As such, we do not plot skill as calculated over the metastable regimes for $\tau > \tau_{\text{corr}}$ in this case. Figure 5.18 illustrates that the major contribution of the skill improvement stems from the climate model better detecting the transitions between slow metastable states. While there is actually a minor degradation in skill of around 3% for forecasts with $\tau_t < \Delta t_{\text{obs}} < \tau_{\text{corr}}$ made for analyses which remain within metastable states, skill improvements of over 25 % are made for those analyses which fall into the transitions between the metastable states. This is because analyses which detect transitions between metastable states incorrectly contribute large RMS errors ($\mathcal{O}(1)$), compared to the small errors ($\mathcal{O}(\sqrt{R_0})$) made by incorrect analyses within the metastable states. These large errors have a major impact upon the total skill $\mathcal{S}$. We conclude that if one is primarily interested in detecting transitions between metastable states then it can be desirable to use an overdifusive reduced forecast model in place of the full deterministic forecast model.

There is also skill improvement as calculated over the transition periods for small values of the observation interval Δt_{obs} . However, there is no corresponding overall skill improvement since for small observation intervals the number of instances where the truth remains within a metastable regime is overwhelmingly larger, thereby swamping the skill improvement obtained over the rare transitions.

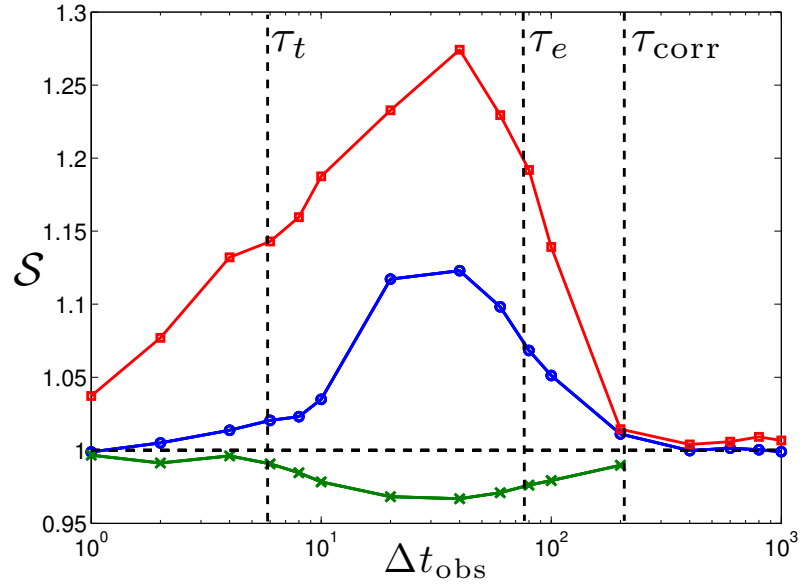


Figure 5.18.: Proportional skill improvement $\mathcal{S}$ of the climate model (5.11) over the full deterministic model (5.3)-(5.6) as a function of the observation interval Δt_{obs} . We show $\mathcal{S}$ as calculated over all analyses (circles), the metastable regimes only (crosses), and the transitions only (squares). Dashed vertical lines indicate the characteristic time scales of the slow dynamics: the decorrelation time τ_{corr} , first exit time τ_e and transit time τ_t . Parameters are as in Figure 5.17.

We note that the skill $\mathcal{S}$ is insensitive to the noise level of the observations, provided R_o is sufficiently small so that observations do not misrepresent the actual metastable state or regime in which the truth resides. The skill $\mathcal{S}$ is roughly the same as in Figure 5.18 for $R_o = 0.1\sigma^2$ and $R_o = 0.25\sigma^2$ where $\sigma^2 = 0.126$, and we observe no skill improvement $\mathcal{S} \approx 1$ when we take very poor observations with $R_o = 0.5 \approx 4\sigma^2$, half the variance of the full slow system.

5.6.1 *Sensitivity of analysis skill to uncertainties in drift and diffusion*

As for the time scales in Section 5.5, we examine how the analysis skill $\mathcal{S}$ depends on the diffusion coefficient σ^2 and the drift term. Figure 5.19 shows the skill $\mathcal{S}$ as a function of the observation interval Δt_{obs} for several values of σ^2 . For very small diffusion and for very large diffusion (not shown) the climate model actually performs worse than the full deterministic model for all observation intervals Δt_{obs} . There exists a range of values of σ^2 for which the climate model outperforms the full model with roughly similar skill values $\mathcal{S}(\Delta t_{\text{obs}}) > 1$ for $\tau_t < \Delta t_{\text{obs}} < \tau_{\text{corr}}$.

The dependence of analysis skill upon the diffusion coefficient can be understood as follows: for small diffusion ($\sigma^2 = 0.1$) the climate model is less diffusive than the truth and the deterministic forecast model, and its associated forecast is likely to remain in a metastable state. This produces small ensemble spread and therefore causes more instances of the forecast not detecting a transition between regimes. On the other extreme, for large diffusion $\sigma^2 \gg 0.15$, the climate model exhibits regime switches too frequently, thereby producing a large forecast error which contaminates the analysis. For moderate values of the diffusion coefficient, the increase in spread of the forecast ensemble helps the detection of regime switches. So long as the climate model is sufficiently, but not overwhelmingly more diffusive than the full model, the exact value taken for σ^2 is not very important; we note that there is little difference in the skill curves for $\sigma^2 = 0.126$ and $\sigma^2 = 0.15$.

We also investigate how uncertainties in the drift term parameters a and b may affect the skill $\mathcal{S}$ in an ETKF data assimilation procedure. Figure 5.20 shows skill curves for fixed value of the diffusion coefficient $\sigma^2 = 0.126$, for different combinations of a and b . As with the characteristic time scales, the skill $\mathcal{S}$ is more sensitive to changes in the parameter a than in b . The sensitivity to changes in the drift term is of a similar nature to that for diffusion. Whereas decreasing a or b by 20 % produces little change in skill, increasing a or b by 20% makes the climate model again less skilful than the full model. This is readily explained by noticing that increasing b increases the difference between the two metastable states and also the height of the potential barrier. Therefore, for fixed diffusion σ^2 , this inhibits transitions between the metastable states, thereby decreasing the spread of the ensemble. Similarly, decreasing a (while keeping b fixed) increases the

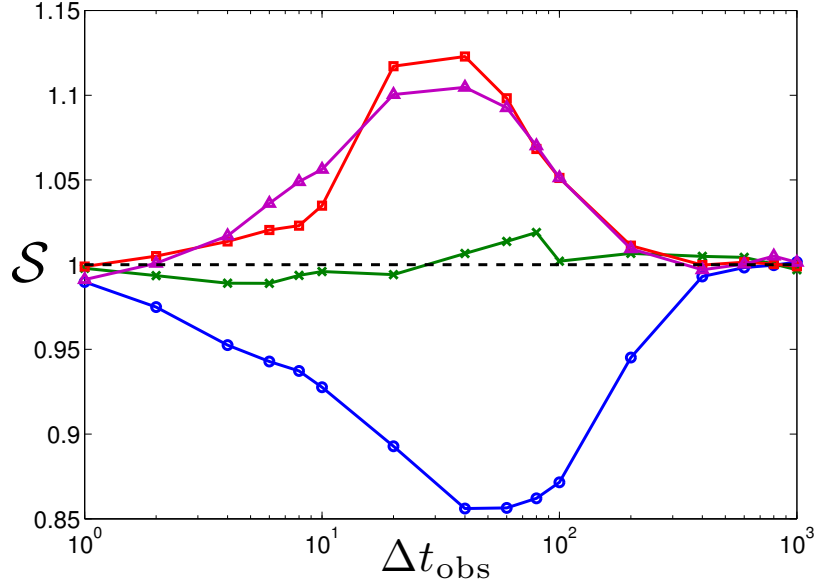


Figure 5.19.: Proportional skill improvement $\mathcal{S}$ as a function of the observation interval Δt_{obs} with $\sigma^2 = 0.1$ (circles), $\sigma^2 = 0.113$ (crosses), $\sigma^2 = 0.126$ (squares) and $\sigma^2 = 0.15$ (triangles). All other parameters are as in Figure 5.17.

potential barrier, inhibiting transitions and thereby decreasing the ensemble spread. As for the diffusion coefficient σ^2 , the values of the parameters of the drift term which are closest to the actual values $a = b = 1$ for which the climate model best approximates the slow dynamics of the full deterministic model is not the optimal value in terms of skill improvement. Values which increase the ensemble spread are more favourable.

5.6.2 Effect of ensemble size and covariance inflation

The results from the previous subsection suggest that the observed skill improvement is linked to the increased forecast ensemble spread of the climate model, rather than to the accuracy of the stochastic parametrisation. This is exemplified by the fact that for $\sigma^2 = 0.113$ for which the climate model best approximates the statistics of the full deterministic model, there is no skill improvement $\mathcal{S} \approx 1$, and skill improvement $\mathcal{S} > 1$ occurs for $\sigma^2 > 0.113$ where transitions between the metastable states are better captured.

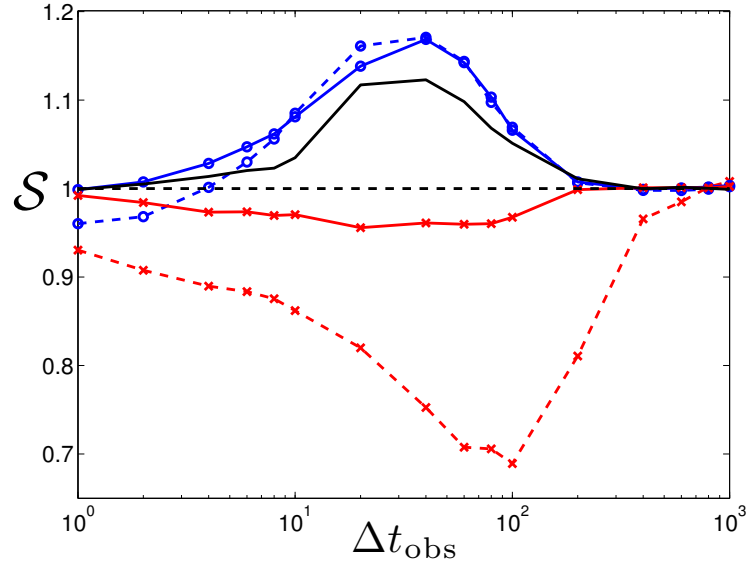


Figure 5.20.: Proportional skill improvement $\mathcal{S}$ as a function of the observation interval Δt_{obs} for $\sigma^2 = 0.126$, with $(a, b) = (1, 1)$ (black solid line), $(a, b) = (0.8, 1)$ (solid, circles), $(a, b) = (1.2, 1)$ (solid, crosses), $(a, b^2) = (1, 0.8)$ (dashed, circles) and $(a, b^2) = (1, 1.2)$ (dashed, crosses). All other parameters are as in Figure 5.17.

The lack of sufficient ensemble spread when using the full deterministic system as a forecast model is caused by the finite size of the ensemble (Ehrendorfer, 2007). As discussed in previous chapters, a common approach to account for insufficient spread is covariance inflation (Anderson and Anderson, 1999), whereby the forecast variance $\mathbf{P}_f$ is artificially inflated through multiplication by a factor $\delta^2 > 1$. We now investigate how the ensemble size k and the inflation factor δ affect the skill. In particular we show that the RMS error $\mathcal{E}$ using the full deterministic model can be reduced by either increasing the ensemble size k or by using larger inflation factors δ .

Figure 5.21 shows $\mathcal{E}$ as a function of the ensemble size k for $\Delta t_{\text{obs}} = 40$ and $\sigma^2 = 0.126$ (which is close to the observation interval yielding maximal skill improvement $\mathcal{S}$; see Figure 5.18), and averaged over 200 realisations. For large ensembles with $k \geq 35$ the overdiffusive climate model and full deterministic model perform equally well as forecast models in the data assimilation scheme. We have checked that when using a large ensemble ($k = 50$) the climate and full models perform equally well for the range of observation intervals used in Figure 5.17. For smaller ensemble sizes k the overdiffusive climate model outperforms the full deterministic model. This illustrates that the skill improvement is linked to the insufficient ensemble spread of the deterministic model, which is compensated in the climate model by the slightly increased diffusion coefficient.

Similarly, for fixed ensemble size k , the ensemble spread can be artificially increased by inflating the forecast covariance with a constant factor δ . Figure 5.22 shows $\mathcal{E}$ as a function of the inflation factor δ , for $1 \leq \delta \leq 3$. It is seen that for very large values of $\delta > 2$ the RMS error decreases when the full deterministic model is used as a forecast model, whereas it is relatively insensitive to changes in δ when using the overdiffusive climate model (again here with $\sigma^2 = 0.126$). For even larger values of $\delta > 3$ the RMS errors for the two forecast models eventually become equal (not shown).

We comment here also on one of the computational benefits of using the stochastic reduced model, namely the ability to use a larger model integration timestep for a stiff problem. Instances where the numerical integrator breaks down due to stiffness are dramatically reduced by using the climate model with sufficiently large values of the diffusion coefficient $\sigma^2 \in (0.126, 0.15)$. For large values of δ , we found that forecasts made with the full deterministic model break down approximately 78.0 % of the time,

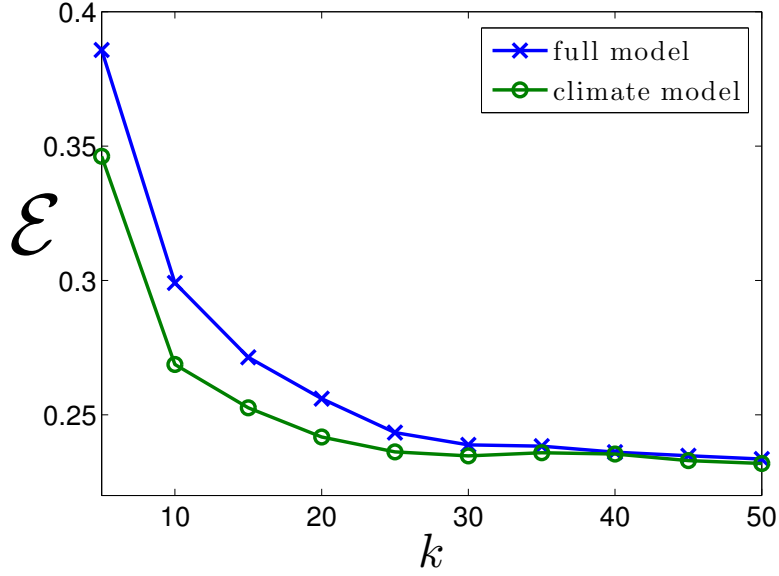


Figure 5.21.: RMS error $\mathcal{E}$ as a function of ensemble size k , using the full deterministic model (5.3)-(5.6) (blue, crosses) and the climate model (5.11) with $\sigma^2 = 0.126$ (green, circles) as forecast models. We used $\Delta t_{\text{obs}} = 40$; all other parameters as in Figure 5.17.

compared to only 1.7% of the time when using the climate model with $\sigma^2 = 0.126$. This breakdown is due to the chosen timestep dt being too large to integrate the full deterministic model stably. This demonstrates the computational benefit of using the stochastic reduced model in allowing one to use a larger model integration timestep. Instances of breakdown in the full model could be reduced by making the timestep dt smaller.

5.6.3 Reliability and ranked probability histograms

Another way of understanding why the ensemble variance of the climate model produces superior skill to that of the full deterministic model is through ranked probability histograms (also known as Talagrand diagrams) (Anderson, 1996; Hamill and Colucci, 1997; Talagrand et al., 1997). To create probability density histograms we sort the members of the forecast ensemble $\mathbf{X}_f = [x_{f,1}, x_{f,2}, \dots, x_{f,k}]$ from lowest to highest value and create bins $(-\infty, x_{f,1}]$, $(x_{f,1}, x_{f,2}]$, $\dots$, $(x_{f,k}, \infty)$ at each forecast step. We then

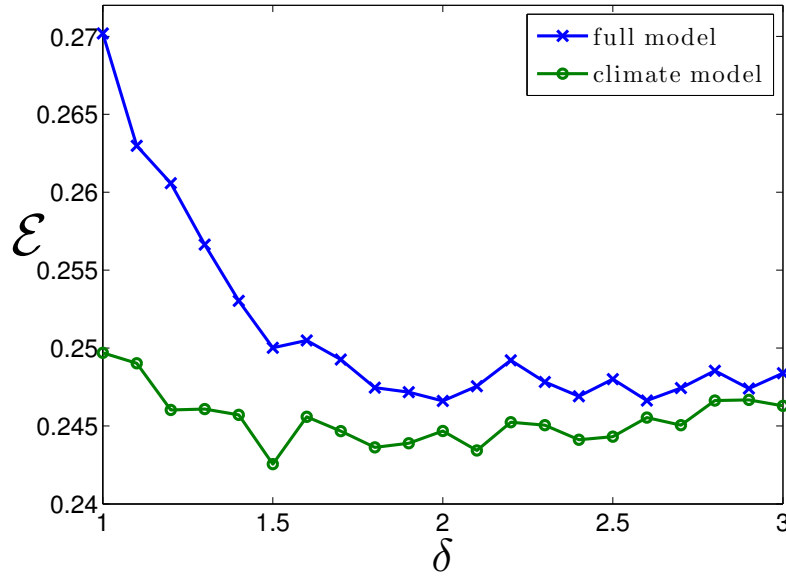


Figure 5.22.: RMS error $\mathcal{E}$ as a function of inflation factor δ , using the full deterministic model (5.3)-(5.6) (blue, crosses) and the climate model (5.11) with $\sigma^2 = 0.126$ (green, circles) as forecast models. We used $\Delta t_{\text{obs}} = 40$; all other parameters are as in Figure 5.17.

increment whichever bin the actual truth falls into after each forecast to produce a histogram of probabilities P_i of the truth being in bin i . A convex probability density histogram indicates an ensemble which underestimates the variance of the truth, while a concave diagram indicates an ensemble which overestimates the variance. A flat probability density histogram is seen as indicating a reliable ensemble for which each ensemble member has equal probability of being nearest to the truth. We direct the interested reader to the book by Wilks (2006) for a detailed discussion of ranked probability histograms.

Hamill (2001) suggested that a single ranked probability histogram is not sufficient for dynamical systems with several dynamical regimes such as our climate model, but that in fact one must look at the variability of the ensemble in the different dynamical regions individually. We therefore construct ranked probability histograms (for both forecast models) over the metastable regime with characteristic time $\bar{\tau}$ only and over the complementary regime of transitions with characteristic time τ_t only, as we did in Figure 5.18 for the distribution of the skill over those regimes.

In Figure 5.23, we plot probability density histograms of the forecast with $\Delta t_{\text{obs}} = 40$ for the full deterministic model (5.3)-(5.6) and the climate model (5.11) with various values of the diffusion coefficient σ^2 , (a) over all forecasts, (b) over the forecasts which fall near the metastable states $x^* = \pm 1$ and (c) over the forecasts which fall in the transitions between metastable states. We use a long cycle containing 250,000 forecasts. Although the full model produces the most reliable ensemble when averaged over all forecasts, it overestimates the variance in the metastable regions and underestimates the variance across the transitions. For the climate model, the more underdiffusive model with $\sigma^2 = 0.1$ produces the most reliable ensemble in the metastable regions while the overdiffusive models with $\sigma^2 = 0.126$ and $\sigma^2 = 0.15$ both produce more uniform ranked probability histograms than the full model does in the transition regions. The reduced stochastic climate model with optimal diffusion coefficient $\sigma^2 = 0.113$ behaves similarly to the full deterministic model as expected. It is thus more desirable to favour the more diffusive climate models because a comparatively large improvement in RMS error can be made in the transition regions compared to in the metastable regions. This agrees with our earlier observation that the skill improvement is entirely due to the climate model being more accurate in the transition regions where large errors can be accrued by misclassifying the regime (see Figure 5.18).

5.7 DISCUSSION

We have investigated a homogenised stochastic climate model of a bistable multiscale system with a chaotic fast subsystem, and its usage in the forecast step of an ensemble Kalman filter. Homogenised climate models replace in a controlled fashion the accumulative effect of the fast chaotic dynamics on the slow dynamics by white noise. The reduced stochastic climate model was numerically shown to faithfully reproduce the statistics of the slow dynamics of the full deterministic model even in the case of finite time scale separation where the theorems underpinning homogenisation fail. We estimated the analytical expressions for the drift and diffusion terms of the reduced stochastic model using coarse-graining and quadratic variations, and found that the diffusion coefficient is very sensitive to the applied undersampling time. The statistics of the climate model, i.e. the probability density function and averaged characteristic time scales, were found to be very sensitive to the diffusion and drift terms used.

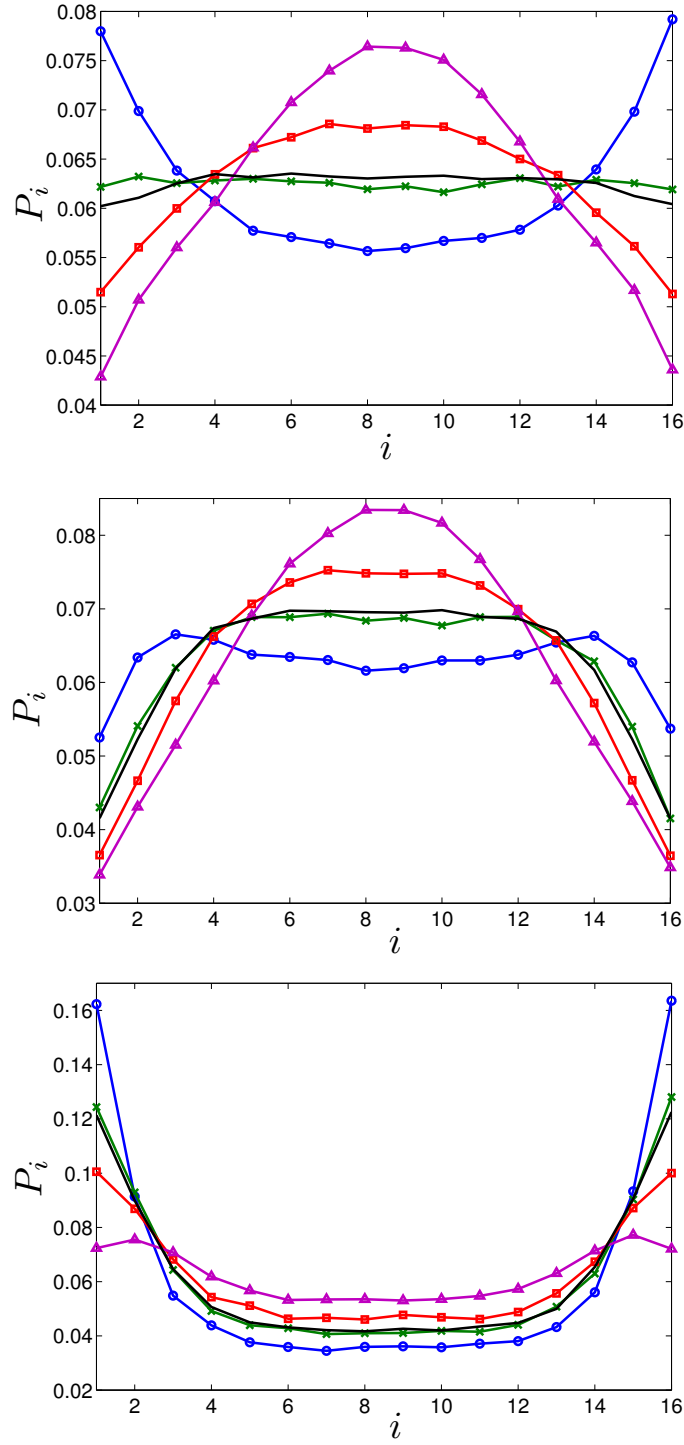


Figure 5.23.: Ranked probability histograms for the full model (black solid line) and climate models with $\sigma^2 = 0.1$ (circles), $\sigma^2 = 0.113$ (crosses), $\sigma^2 = 0.126$ (squares) and $\sigma^2 = 0.15$ (triangles) over all forecasts (top), as well as only those forecasts which fall near the metastable states $x^* = \pm 1$ (middle) and those which fall in the transitions between metastable states (bottom). We used $\Delta t_{\text{obs}} = 40$.

Despite this sensitivity and the implied uncertainties associated with the “correct” stochastic climate model, we found that stochastic climate models can be beneficial as forecast models for data assimilation in an ensemble filter setting. We have shown numerically that using such a climate model has the advantage of 1.) being computationally more efficient to run due to the reduced dimensionality and avoidance of stiff dynamics, and 2.) producing a superior analysis mean for a range of observation intervals Δt_{obs} . These skill improvements occur for observation intervals larger than the time taken to switch between slow metastable states τ_t but less than the decorrelation time τ_{corr} .

The observed skill improvement is due to the associated larger ensemble variance of the stochastic climate model, which allows the ensemble to explore the full state space of the slow variable better than the full deterministic model does. Forecasts made by the climate model will generally contain more ensemble members which cross the potential barrier between the slow metastable states (weather or climate regimes) than full model forecasts will. Consequently, there is a greater chance of one or more climate model ensemble members crossing the potential barrier during the forecast to the opposite metastable state than in the full deterministic model ensemble. If an observation appears near the opposite metastable state to where the forecast began, the Kalman filter analysis equation (2.9) trusts the observation more than the forecast, correctly producing an analysis which changes between the regimes. Thus the climate model with large enough diffusion better detects transitions between slow metastable states and so has lower RMS error near these transitions in which large errors can be accrued.

We remark that multimodal probability density functions are not necessary for the existence of metastable regimes, as was pointed out in (Wirth, 2001; Majda et al., 2006). In cyclostationary systems, i.e. systems which involve time-periodic coefficients, or systems in which “regimes” occur intermittently without a well-defined periodicity, the probability density function may be unimodal whereas if restricted to time windows focussing on the regimes the “hidden” regimes can be identified. This has important implications for interpreting atmospheric data (Majda et al., 2006) where one needs to reconcile the apparent paradox of multimodal probability density functions of planetary waves obtained from (relatively short) observational data (Kimoto and Ghil, 1993; Cheng and Wallace, 1993; Corti et al., 1999) and unimodal but non-Gaussian probability density

functions obtained from long time simulations of atmospheric general circulation models (Branstator and Berner, 2005; Franzke and Majda, 2006; Berner and Branstator, 2007). For a critical account on the analysis of short climatic data see (Hsu and Zwiers, 2001; Stephenson et al., 2004).

It is pertinent to mention that the results found here are not dependent on the existence of metastable states and multi-modal probability density functions, but only require dynamics with rapid large amplitude excursions as for example in intermittent systems. In (Harlim and Majda, 2008, 2010) a system without multiple equilibria was investigated and superior filter performance was found for stochastic parametrisations (in this case by radically replacing all nonlinear terms by linear Ornstein-Uhlenbeck processes in Fourier space). We believe that the increased analysis skill they observed can be explained by the increased ensemble spread, counteracting finite sampling errors of the underlying ensemble Kalman filter approach as we did. The advantage when homogenisation can be used is that the slow dynamics and its mean are still well resolved and reproduced and therefore the ensemble spread increasing stochasticity in a more controlled fashion. The use of homogenised stochastic climate models with adjusted larger diffusivity therefore amounts to incorporating deliberate but controlled model error into the data assimilation scheme.

We find that the same skill for the reduced forecast model can be obtained using the full deterministic model either by using a larger ensemble, or by increasing the covariance inflation factor. The first is undesirable for large models because of the computational difficulty of simulating large ensembles and their covariances, while the second is undesirable because it introduces unphysical inflation (here a multiplicative factor of $\delta > 2$ was needed) and requires expensive empirical tuning of the inflation factor.

Using stochastic reduced models for multiscale systems provides significant computational advantages. Firstly, the dimension reduction allows for a gain in computational speed. Furthermore, by replacing a stiff ordinary differential equation by a non-stiff stochastic differential equation means that coarser integration time steps can be used which otherwise would lead to a break down when simulating the stiff multiscale system.

Homogenised stochastic climate models therefore provide a cheap way of simulating large ensembles, and a more natural way of incorporating forecast covariance inflation into the data assimilation algorithm. In particular, we find that the climate model produces a more reliable ensemble (as characterised by the ranked probability histogram) which better detects the transitions between slow metastable states. This leads to a far superior analysis in these regimes than that produced by the full deterministic model.

Additionally in this chapter we have discussed the general problem of predictability in multiscale systems containing both fast and slow regimes, using the FSLE for this model. Interestingly we found that the metastability of the fast dynamics produces regions of rapid divergence of trajectories of the slow degrees of freedom, indicating poor predictability at those scales. The associated loss in predictability is found to be far greater than that defined by the maximal Lyapunov exponent. Nonmonotonous behaviour of the FSLE has been previously reported in (Torcini et al., 1995; Cencini and Torcini, 2002). Discrete maps involving singular derivatives such as the circle map are simple examples where finite size perturbations can grow faster than infinitesimal perturbations. Here we discussed the occurrence of narrow well-defined peaks which are caused by the fast dynamics switching between regimes. For our toy model a simple analytical argument can be employed to calculate the predictability times and the critical perturbation sizes. We also found that minima seen in the large scale part of the FSLE spectrum are indicative of the presence of slow metastable states. Detection of peaks and troughs may serve as a useful guide for interpreting FSLE spectra and determining the presence of fast and slow regimes respectively in the system.

SUMMARY AND OUTLOOK

In this thesis we have developed methods for incorporating climatological information into data assimilation algorithms. We found that incorporating climatological data can help address problems arising from finite size effects, such as covariance underestimation and overestimation as well as filter divergence. In particular, we have focused on two ways in which statistical data can be used to augment the existing ensemble Kalman filter (EnKF) algorithm: firstly through the analysis step in Chapters 3 and 4 and secondly through the forecast model itself in Chapter 5.

In Chapter 3 we looked at the problem of data assimilation in sparse observational networks, and proposed a variance limiting Kalman filter (VLKF) to incorporate climatological data into the analysis step of the EnKF. We did this through the addition of a weak constraint to the cost function and by requiring that the posterior analysis ensemble have covariance no larger than the climatic variance. We applied this filter to a linear toy model where we showed analytically that for small observation intervals the climatological information provided through the weak constraint can improve the analysis skill, particularly in multiscale systems with large time scale separation. A further direction of research would be to apply the VLKF to nonlinear multiscale systems to see if the results for the toy model hold for more complex dynamics. In keeping with the use of Lorenz models as testbeds for data assimilation experiments, one possibility would be the Lorenz-05 models (Lorenz, 2005). Instead of incorporating just the mean and variance for a particular variable into data assimilation, the VLKF method could be generalised to weak constraints which incorporate integrated quantities for the system. One application of this could be to keep the fast energy bounded in Hamil-

tonian systems. It would be particularly interesting to apply the VLKF to multiscale Hamiltonian systems such as the Lorenz-86 model (Lorenz, 1986) for which slow balance manifolds exist (Bokhove and Shepherd, 1996; Wirosuetisno and Shepherd, 2000). For this model the weak constraint in the VLKF could be generalised to a nonlinear weak constraint on balance, which may help produce a balanced analysis. These ideas have previously been explored in the context of 4D-VAR; weak constraints on integrated quantities have been used by Watkinson et al. (2005, 2007) for assimilating Hamiltonian systems, while Dixon and Roulstone (2003) have looked at how to control imbalance in the shallow-water equations using a weak constraint. Neef et al. (2006) studied the problem of balance in both the extended Kalman filter and EnKF using the Lorenz-86 model. A challenge in modern high-resolution numerical weather prediction is how to incorporate imbalance into the analysis. In the VLKF framework one could imagine using some technique such as potential vorticity (Wlasak et al., 2006) to separate out the balanced and unbalanced part of the flow, and then using the VLKF only on the relevant subsystem to help create a balanced portion of the analysis, while retaining some unbalanced components. This could find application in convective-scale forecasting and data assimilation.

We also applied the VLKF to a sparsely observed Lorenz-96 model in Chapter 3. Numerical simulations showed that for small observation intervals the incorporation of climatological data can improve the analysis skill for sparse or noisy observations, and can lessen the instances of catastrophic filter divergence for small observational noise. Further research is required to understand how filter divergence can be avoided: is there a way to construct an ensemble such that it explores enough of the state space to avoid collapse during the forecast step? Some interesting research in this direction has been done recently by Trevisan and Palatella (2011) on extended Kalman filtering in the unstable subspace, as well as by Ng et al. (2011) for divergence in the EnKF. These ideas may provide some insight on choosing an optimal subspace on which to project the error covariances.

In Chapter 4 we looked at a problem motivated by the use of artificial damping to control small-scale numerical instabilities. We applied the VLKF to a data assimilation problem using a forecast model with less damping. The VLKF was found to control the covariance overestimation resulting from the use of the underdamped forecast model. Unlike the perfect model results from Chapter 3, in the presence of large model error

we obtained skill improvements over standard EnKF techniques for large observation intervals. There was however found to be a trade-off between skill improvement over the EnKF and skill improvement over the observations/climatology.

In Chapter 5 we turned to the question of how to incorporate climatological information into data assimilation through the forecast step. We used stochastic homogenisation to obtain a reduced climate model for a multiscale dynamical system, which we employed in the forecasting step of an EnKF. We found that while the characteristic time scales of the reduced model were sensitive to the inaccuracies in the estimation of the drift and diffusion terms, forecasting with the reduced model was still beneficial in an ensemble filter. Indeed, the reduced model with sufficiently large diffusion was better at detecting transitions between metastable regimes than the full model was. This was linked to the increased ensemble spread when using a reduced model, in a comparable way to covariance inflation or larger ensemble size. Because the optimal level of forecast model diffusion differed depending on whether the slow variable was in or between metastable regimes (see Figure 5.23), it would be beneficial to vary the forecast model diffusion parameter depending on what state the slow variable was in. This would require the prediction of regime changes in the system. There is work in the literature on early warning signs for critical transitions, such as through an increase in the autocorrelation function (Scheffer et al., 2009) or spikes in the growth rate of bred vectors (Evans et al., 2004; Yang et al., 2006). If applicable, such methods could be used to further improve the performance of reduced forecast models in data assimilation. In Chapter 5 we also explained an interesting phenomenon of nonmonotonous finite size Lyapunov exponent (FLSE) spectra with troughs indicating enhanced predictability and peaks indicating a significant loss of predictability for perturbations of a well defined size. We found that peaks and troughs in the FSLE spectrum could be used as an indicator of fast and slow regimes in the system.

APPENDIX AND BIBLIOGRAPHY

A

APPENDIX: ANALYTICAL EXPRESSIONS FOR VLKF APPLIED TO TOY LINEAR MODEL

The analytical expression we obtain for $\mathbf{P}_f(t_{i+1})$ from *Mathematica* is

$$\mathbf{P}_f(t_{i+1}) = \begin{pmatrix} P_{11} & 0 & P_{13} & P_{14} \\ 0 & P_{11} & -P_{14} & 0 \\ P_{13} & -P_{14} & P_{33} & 0 \\ P_{14} & P_{13} & 0 & P_{33} \end{pmatrix} \quad (\text{A.1})$$

where

$$\begin{aligned} P_{11} &= \frac{(\gamma_y \sigma_x^2 (\gamma_x + \gamma_y) + \lambda^2 \sigma_y^2)}{2\gamma_x \gamma_y (\gamma_x + \gamma_y)} \\ &\quad + \frac{\Delta t_{\text{obs}} \left(\lambda^2 \sigma_y^2 \left[\gamma_x^2 (2\gamma_y^2 + 1) + 4\gamma_x \gamma_y + \gamma_y^2 \right] + (\gamma_x^2 + 1) \gamma_y^2 \sigma_x^2 (\gamma_x + \gamma_y)^2 \right)}{4\gamma_x^2 \gamma_y^2 (\gamma_x + \gamma_y)^2} \\ P_{13} &= \frac{\lambda \sigma_y^2}{2\gamma_y (\gamma_x + \gamma_y)} - \frac{\lambda \sigma_y^2 \Delta t_{\text{obs}} (\gamma_y^3 - \gamma_x \gamma_y^2 - \gamma_x - 3\gamma_y)}{4\gamma_y^2 (\gamma_x + \gamma_y)^2} \\ P_{14} &= \frac{\lambda \sigma_y^2 \Delta t_{\text{obs}}}{(\gamma_x + \gamma_y)^2} \\ P_{33} &= \frac{\sigma_y^2}{2\gamma_y} + \Delta t_{\text{obs}} \left(\frac{\sigma_y^2}{4\gamma_y^2} + \frac{\sigma_y^2}{4} \right). \end{aligned}$$

Using (A.1) and the assumptions from Section 3.3, we can obtain the following expression for the skill $\mathcal{S}$ from *Mathematica*, where we have Taylor expanded in terms of Δt_{obs} and weak coupling λ , and retained only terms up to first order in Δt_{obs} and second order in λ :

$$\mathcal{S} = \frac{\delta (S_1 - 1) \lambda^2}{4\gamma_x \gamma_y} + \delta \quad (\text{A.2})$$

$$+ \Delta t_{\text{obs}} \left(\frac{S_2}{2\gamma_x \gamma_y^2 (\gamma_x + \gamma_y)^3 (\delta \sigma_x^2 + 2R_{\text{obs}} \gamma_x)^4} + 2\gamma_y (1 - \delta) \right)$$

where

$$S_1 = \frac{-4\gamma_x^2 (\delta^3 \sigma_x^4 + 4R_{\text{obs}} \gamma_x \delta^2 \sigma_x^2 - 4R_{\text{obs}}^2 \gamma_x^2 (1 - 2\delta)) \sigma_y^2}{(\gamma_x + \gamma_y)^2 (\delta \sigma_x^2 + 2R_{\text{obs}} \gamma_x)^3}$$

$$S_2 = \left(\gamma_y \delta^3 \left(\gamma_y (\gamma_x + \gamma_y)^3 \delta (2\delta - 1) \sigma_x^2 - 4\gamma_x^2 (\gamma_x + \gamma_y) \left(\gamma_x (\delta - 1)^2 - \gamma_y \delta \right) \sigma_y^2 \right) \sigma_x^6 \right.$$

$$+ 4R_{\text{obs}} \gamma_x \delta^2 \left(2\gamma_y^2 \delta (2\delta - 1) \sigma_x^2 (\gamma_x + \gamma_y)^3 \right.$$

$$+ \gamma_x \left[-\gamma_y (\delta - 1) (7\delta - 4) \gamma_x^3 + \left(((15 - 4\delta)\delta - 4) \gamma_y^2 + \delta^2 \right) \gamma_x^2 \right.$$

$$+ \gamma_y \delta \left((\delta + 6) \gamma_y^2 + 2\delta - 1 \right) \gamma_x - \gamma_y^2 (\delta - 1) \delta \left. \right] \sigma_y^2 \sigma_x^4$$

$$+ 8R_{\text{obs}}^2 \gamma_x^2 \delta \left(3\gamma_y^2 \delta (2\delta - 1) \sigma_x^2 (\gamma_x + \gamma_y)^3 \right.$$

$$+ \gamma_x^2 \left[(\delta (3\delta + 2) + 2) \gamma_y^3 - 2\gamma_x^2 (\delta - 1) (2\delta - 1) \gamma_y \right.$$

$$+ \delta (3\delta - 2) \gamma_y + \gamma_x \delta \left((\delta + 6) \gamma_y^2 + \delta \right) \left. \right] \sigma_y^2 \sigma_x^2$$

$$+ 32R_{\text{obs}}^3 \gamma_x^3 \gamma_y^2 (\gamma_x + \gamma_y) \left((\gamma_x + \gamma_y)^2 \delta (2\delta - 1) \sigma_x^2 + \gamma_x^2 ((\delta - 1)\delta + 1) \sigma_y^2 \right)$$

$$+ 16R_{\text{obs}}^4 \gamma_x^4 \gamma_y^2 (\gamma_x + \gamma_y)^3 (2\delta - 1) \left. \right) \lambda^2.$$

In the limits of large and small noise respectively, the skill $\mathcal{S}$ can be simplified to the more compact form

$$\lim_{R_o \rightarrow \infty} \mathcal{S} = \frac{\delta^2 (4\gamma_x \gamma_y - \lambda^2)}{4\gamma_x \gamma_y \delta + 2\gamma_y \Delta t_{\text{obs}} (4\gamma_x \gamma_y (\delta - 1) - \delta \lambda^2)}$$

and

$$\lim_{R_o \rightarrow 0} \mathcal{S} = \frac{N}{D}$$

where

$$\begin{aligned}
N &= \frac{\delta^2 ((\gamma_x + \gamma_y)^2 \sigma_x^2 + \lambda^2 \sigma_y^2)}{(\gamma_x + \gamma_y)} \\
D &= \frac{4\gamma_x \delta (\gamma_y (\gamma_x + \gamma_y) \sigma_x^2 + \lambda^2 \sigma_y^2)}{4\gamma_x \gamma_y - \lambda^2} + \frac{D_1}{D_2} \\
D_1 &= 2\Delta t_{\text{obs}} \left((\gamma_x + \gamma_y)^2 (4\gamma_x \gamma_y (\delta - 1) - \delta \lambda^2) \sigma_x^2 - \lambda^2 (\delta \lambda^2 + 4\gamma_x (\gamma_x + \gamma_y - \gamma_y \delta)) \sigma_y^2 \right) \\
&\quad \times \left[\gamma_y (\gamma_x + \gamma_y)^3 \delta (4\gamma_x \gamma_y - \lambda^2) \sigma_x^4 \right. \\
&\quad + (\gamma_x + \gamma_y) \lambda^2 \left(4\gamma_x ((\delta - 1) \gamma_x^2 + 2\gamma_y \delta \gamma_x + 2\gamma_y^2 \delta) - (\gamma_x + 2\gamma_y) \delta \lambda^2 \right) \sigma_y^2 \sigma_x^2 \\
&\quad \left. + \delta \lambda^4 (4\gamma_x (\gamma_x + \gamma_y) - \lambda^2) \sigma_y^4 \right] \\
D_2 &= \delta (\lambda^2 - 4\gamma_x \gamma_y)^2 ((\gamma_x + \gamma_y)^2 \sigma_x^2 + \lambda^2 \sigma_y^2)^2.
\end{aligned}$$

As $\Delta t_{\text{obs}} \rightarrow 0$, we have

$$\mathcal{S} \rightarrow \delta \frac{(4\gamma_x \gamma_y - \lambda^2) ((\gamma_x + \gamma_y)^2 \sigma_x^2 + \lambda^2 \sigma_y^2)}{4\gamma_x (\gamma_x + \gamma_y) ((\gamma_x + \gamma_y) \gamma_y \sigma_x^2 + \lambda^2 \sigma_y^2)}, \quad (\text{A.3})$$

which is linear in the inflation factor δ .

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